



Fishing boats moored at Carey's Bay, Otago Peninsula (*Raewyn Peart*)



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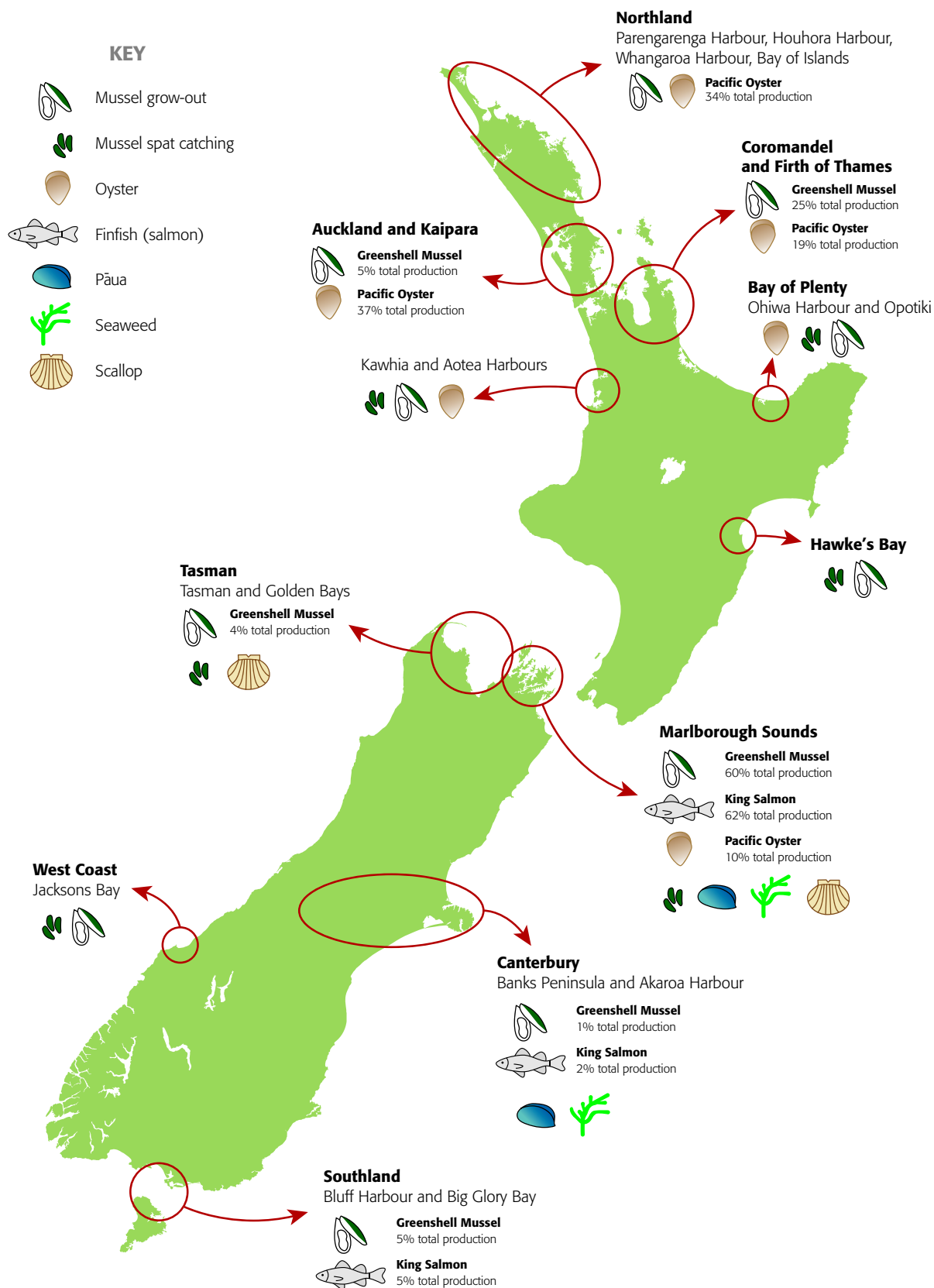


Figure 10.2 Location of aquaculture in New Zealand (*Aquaculture New Zealand*)

Offshore extensive farms hold some interest for the aquaculture industry. These can be several miles offshore (although still within the coastal marine area) and several thousand hectares in size. Due to the more exposed conditions offshore, the mussel lines must be spaced further apart to avoid entanglement. Such farms are still very much at the experimental stage, are untested commercially, and rely on substantial infrastructure investment before becoming viable.

Offshore mussel farming

In 2006, Eastern Sea Farms Limited, which is majority-owned by the Whakatōhea Māori Trust Board, was granted a licence for a 3,800 hectare marine farm to be established off the coast of Ōpōtiki.¹³ The farm is sited three nautical miles offshore where the water depth and currents are suitable for mussel growing. The site has been chosen to minimise visual, navigational and environmental effects. It is away from sensitive habitats and important fishing areas. The planned marine farm will be the largest aquaculture venture in New Zealand, with the potential to produce approximately 20,000 tonnes of mussels per annum by the year 2025.¹⁴ There is also potential for further aquaculture development off the Ōpōtiki coastline.

Oysters

Pacific oysters (*Crassostrea giga*) have been farmed in New Zealand since the 1960s. It is unclear as to when the Pacific oyster species arrived in New Zealand from Asia, but it is thought to have been in the late 1950s.¹⁵ They soon started displacing the New Zealand native rock oyster and so farmers adopted the new arrival as their principal commercial species.

Today Pacific oysters are the main farm-raised oyster in New Zealand. They are grown on wooden racks and baskets in sheltered and shallow intertidal bays around the northern North Island coast and in the Marlborough Sounds. The seed used to stock most farms is caught from the wild, predominantly from the Kaipara Harbour. Oyster larvae drifting in the water settle on bundles of wooden sticks which are placed in the water column. The sticks are then transported to marine farms, where the larvae are transferred to racks for on-growing the baby oysters. Sometimes, the oysters are transferred to mesh bags, hanging baskets or wire racks for growing. Using baskets suspended on wires can be an effective method of farming, allowing oyster farms to be located in more exposed locations because in rough weather the baskets swing and thereby resist the damage that would be caused to the traditional fixed racks. Pacific oysters grow to marketable size within 12 to 18 months.¹⁶

A growing proportion of oyster spat is obtained from hatcheries where it has been selectively bred. Cawthron's Pacific oyster breeding programme has seen growth rates more than 20 per cent higher than those normally achieved from a crop of wild spat.¹⁷ Hatchery-bred oysters have become much more important to the industry during the past couple of years due to the devastating



Oysters can be raised in baskets in shallow intertidal bays such as those shown here in Te Matuku Bay, Waiheke Island, Hauraki Gulf (Raewyn Peart)

Indigenous finfish

The farming of indigenous finfish in New Zealand, such as hāpuku (*Polyprion oxygeneios*) and yellowtail kingfish (*Seriola lalandi*), has yet to occur on a commercial scale. Interest in the development of finfish aquaculture in the Firth of Thames and southern Hauraki Gulf led the Government to legislate a zone of approximately 390 hectares for the culture of yellowtail kingfish and hāpuku in the Waikato regional coastal plan.²³ A resource consent is still required before such farming can proceed.

Currently there is limited information available about the specific finfish farming methods which might be adopted. However, assessments of the anticipated impacts of such farms have assumed that the two finfish species would be grown in floating sea-pens, with juvenile stock sourced from land-based seawater hatcheries. Plastic circular pens (typically 28 to 51 metres in diameter) are anticipated because the wave energy is expected to be too great for the square metal pens used at the South Island salmon farms.²⁴

Hāpuku

Hāpuku is a member of the *Polprionidae* or wreckfish, which is comprised of nine species of large fish that live and feed on or near the bottom of the sea. Hāpuku is large and thick-bodied, reaches 150 centimetres in length, and can vary in colour from silver and blue, to a pink, brown and silver combination. Most often found in deep waters over a depth range of 50 to 854 metres, this species is found throughout southern waters, and is common in the waters around New Zealand and Australia. Where stocks are still abundant (such as around the Chatham Islands) the species is known to frequent shallow waters. Unlike kingfish, hāpuku is a winter-spawning species producing eggs in late July to early September.²⁵ Scientists at NIWA have been undertaking detailed research on growing hāpuku in controlled environments and these early trials indicate that the species could potentially be farmed in New Zealand.

Yellowtail kingfish

Yellowtail kingfish are widely distributed throughout the warm, temperate waters of the Southern Hemisphere. New Zealand kingfish, also known as kingi or yellowtail, are found from the Kermadec Islands to Banks Peninsula during the summer months. In the wild they can reach 1.7 metres in length and weigh 56 kilogrammes. Wild populations have been significantly reduced due to fishing pressure and commercial catches of New Zealand kingfish are small, seasonal and unpredictable. Closely-related species are commonly farmed overseas, in countries such as Japan and Australia.

Research undertaken by NIWA indicates that kingfish could be an attractive aquaculture species for a number of reasons including:²⁶

- It is highly valuable (earning up to \$17 per kilogramme on the European market)
- It has a rapid growth rate (reaching marketable size of three kilogrammes in 12–15 months)
- It is amenable to aquaculture conditions
- It has excellent flesh quality for a range of product options (such as whole fillets, sushi and the highly valued sashimi)
- It has significant domestic and international market opportunities

Around 500,000 kingfish fingerlings per year could be produced from NIWA's facility in Northland, which would meet the needs of the early stages of an industry in New Zealand. Research is now focused on selective breeding and the on-growing technology, including the setting up and running of sea pens to grow the hatchery-produced fingerlings to marketable size.²⁷



Kingfish and hāpuku are being bred and raised at the NIWA hatchery in Bream Bay, Whangarei (Raewyn Peart)

Biosecurity

Biosecurity pathways are human activities that, intentionally or unintentionally, may move a harmful organism from one place in New Zealand to another.³² Figure 10.3 below highlights some of the pathways related to marine farming.

	Transfer mechanism for both finfish and shellfish
Farm structures and activities	Provide a vector for marine pathogens and diseases Facilitate the movement of species attached to equipment around the coast, including: • Fouled sea pens or harvesting gear • Fouled hulls or contaminated bilge water Can create environmental conditions that facilitate the establishment of pest species
Wider impacts	Can spread disease to wild populations Can breed with wild fish, altering the wild stock. Can out-compete wild stock Can act as a reservoir from which marine pests or disease agents spread to the environment, potentially leading to irreversible regional-scale effects

Figure 10.3 Marine pest pathways related to aquaculture

The fixed structures found in marine farms can provide ideal habitats for some marine pest species, particularly biofouling species such as barnacles, tube worms and seaweed that grow on submerged structures and on the shellfish themselves. A marine farm provides a substrate on which these organisms can settle and multiply when there might not naturally have been such an opportunity, hence increasing the chance of establishment in an area.

Such biofouling can include invasive species such as the Japanese kelp (*Undaria pinnatifida*) which has been associated with mussel farms in the Hauraki Gulf. The nitrogen discharges produced by finfish farming may increase the growth of such unwanted species.³³

As well as potentially increasing biosecurity risks, biofouling is of direct concern to marine farmers, as it can lead to lower yields through poor growth of the shellfish. Removal of biofouling organisms – which is usually done by hand – can be a very expensive and labour-intensive job for marine farmers. Biofouling can also result in higher maintenance costs, as a result of equipment becoming damaged or unusable because of the growths. When biofouling organisms grow on finfish nets, they can prevent water flow through the nets and increase their weight, so that they can be pulled from their moorings. Nets must therefore be changed or cleaned regularly.

The transportation of spat, gear, stock and vessels around the country has the potential to spread pests and diseases. It is therefore important to ensure that all stock and spat comes from healthy sources and that all equipment is cleaned properly before transportation. International transfers of finfish feed and used aquaculture farm equipment are controlled by import health standards that outline stringent quarantine and control measures to minimise the risk of disease transfer.

Most marine farmers follow a code of practice in order to minimise the risk of spread of pests and diseases. The industry has also become involved in joint biosecurity efforts, including the long-term management plan for the invasive sea squirt *Styela clava* (clubbed tunicate). In addition, marine farmers play an important role in biosecurity monitoring, due to their frequent surveillance of marine stock and equipment.

Marine biosecurity is discussed further in Chapter 5.



It is important that aquaculture equipment is kept clean to reduce the risk of spread of invasive species. These salmon nets at Ruakākā Bay, Marlborough Sounds, are being cleaned by hand (Raewyn Peart)

Seabed habitats

The effects on the benthic community from marine farms are well understood and can be summarised as:⁴²

- Enrichment of the organic content of the seabed in the immediate area beneath the farm
- Smothering of benthic organisms
- Debris changing the composition of the benthic community
- Biofouling drop-off
- Shading of the seabed which may impact on algal production

These potential impacts are described in more detail in the sections below addressing specific farmed species.

Landscape, natural character and amenity

Locations for siting marine farms are typically restricted by suitable environmental parameters. For example, shellfish filter feed, taking particulates of food (plankton) from the water column, and can concentrate toxins from the water in their flesh.⁴³ Thus for successful production of key New Zealand aquaculture species, unpolluted and sheltered waters are required, away from centres of high population density. This is to ensure that runoff from land-based activities does not affect the marine environment where the farm is located.

There are relatively few places which are appropriate sites for aquaculture, and these tend to be places which are also valued for other activities, and for their unspoilt landscape values and natural character.

Shellfish farms usually have elements that are visible from the shore. Mussel farms have lines of buoys holding up the mussel lines which are visible as horizontal structures on the water plane. Some lines are deliberately suspended below the surface to avoid swells. Although not permanent fixtures, barges working the farms may be perceived as having an impact on amenity values when stationary. In addition, navigational lighting marking the farms is typically visible from 2 nautical miles away at night time. In the case of oyster farms, rows of wooden racks can be visible in the intertidal zone during low tide.



Finfish farms can impact on landscape values because of infrastructure such as nets and sheds constructed above the water surface as shown here with the salmon farm in Ruakākā Bay, Marlborough Sounds (Raewyn Peart)

Finfish farms are usually visible as vertical structures on the surface of the water. They can be particularly noticeable as they typically use high nets to keep predators out. Similar to mussel farms, service structures and craft working on the farm will also be visible.

In terms of location, several key variables affect the ability of the marine environment to visually absorb marine farming activities. These are the visual scale of the affected coastal environment, the degree of visual interest in the view, and the elevation and distance from which the farm is likely to be viewed.⁴⁴

Where an area has outstanding natural character, or is an outstanding natural landscape, Policies 13 and 15 of the NZCPS require any adverse effects to be avoided. In practice, this means that consent should not be granted for marine farms which have adverse effects on the landscape values of these areas.⁴⁵ This issue is discussed further in the decision-making framework section below.

Restoration and enhancement

Aquaculture can play a role in fisheries enhancement. For example, the salmon industry provides smolt to recreational fisheries and marine farmers provide pāua to enhance recreational, customary and commercial pāua fisheries. A case study on the way the aquaculture industry is helping to replenish mussel beds is provided at the end of this chapter.

of the lines relative to the predominant direction of the currents. The effects can be minimised through good design, with larger, denser and poorly orientated farms having greater effects. Research suggests that dispersed development of marine farms in New Zealand is unlikely to significantly impact on the hydrodynamics over a wider area.⁴⁸

Limits and trigger points for phytoplankton depletion

Work has been underway for a number of years looking at the pelagic impacts from marine farms in the Firth of Thames. An adaptive management framework was recommended in 2005 whereby Limits of Acceptable Change would be developed and agreed upon. The limits agreed to were that phytoplankton depletion shall not exceed 20 per cent over 10 per cent or more of the Firth of Thames and phytoplankton depletion shall not exceed 25 per cent over an area twice the size of the Aquaculture Management Area.^{49,50} 'Trigger points' were used to determine when limits have been reached. These can then be reviewed and modified as necessary. In addition, these trigger points enable marine farm managers to respond with a range of management actions.⁵¹

Wild species

Material falling from mussel farms can attract small fish and encourage the development of reef-type communities including sponges, crabs, starfish and tube worms. Juvenile fish commonly use artificial structures placed in the sea for shelter, which can lead to the creation of new communities and increased biodiversity.

The overall effect of marine farms on fish abundance is not well understood. Research on farms in the Marlborough Sounds indicates that there is more abundance and diversity of fish adjacent to the farms than in surrounding areas. Anecdotal evidence from recreational fishers indicates that species such as snapper and kingfish can be found in higher numbers around marine farms, attracted to the food supply provided by the mussels and juvenile fish, especially during harvesting. An increasingly popular charter fishing industry is developing around the Wilson Bay mussel farms in the Firth of Thames.

Shellfish farms can also change the community structure in their vicinity because of the way they change the seabed substrate, increasing the amount of shell and nutrients. This results in a shift towards species better adapted to the new conditions. For example, studies of farms in the Marlborough Sounds found that heart urchins and brittle stars declined in the locality of mussel farms and were replaced by other species.⁵²

Spat for mussel farms is predominantly sourced from Ninety Mile Beach in Northland and then moved to the rest of New Zealand to seed the farms. There is currently no evidence that wild green-lipped mussels are declining in population fitness due to reliance on spat from one region, although mixing with farmed mussels has probably already occurred. This could be due to the long-lived dispersive larval stage of green-lipped mussels, which facilitates wide genetic mixing.

Pacific oyster farming in areas where they don't already exist could result in the establishment of populations of a non-native oyster in the wild. However, wild populations of Pacific oysters are already widespread around the coastline, and are present in areas where there are no marine farms.

Increasingly, selective breeding is being used to supply spat for oyster farms and, with new research underway, is likely to become more common for mussel farms. Trials have started where selectively bred hatchery-produced spat mussels have been transferred to a marine farm.⁵³ There is little information available about the impact of selective breeding on wild populations of oysters or mussels.

Farm structures

Abandonment of marine farms can result in degrading structures being left in the marine area. The structures required for finfish and mussel farming are generally of a high enough value to incentivise farmers to remove them, whereas oyster racks are not. If oysters are left behind on the racks they can outgrow the structures and cause the farms to collapse into the mud and build up on the seafloor. The abandoned structures can be very visible for many years. This problem can be addressed by conditions of resource consents and bonds to ensure structures are removed on the cessation of farming, or by the requirement for reasonable assurance (such as a fidelity fund).

Siltation and alteration of natural currents

Oyster farms can cause siltation on the seabed by disturbing the natural flow of water in the area, and changing the height of the seabed beneath and around the farm through scouring or silt build-up. These effects can be minimised by ensuring that oyster racks are oriented parallel with, rather than at right angles to, currents and wave action. The growing of oysters in baskets, which is becoming more common, minimises the trapping of sediment.

Currently in New Zealand, feed for farmed salmon does not contain antibiotics, vaccines, steroids or colourants. However the feed contains zinc, necessary to prevent cataract formation and other health problems, which can accumulate in sediments beneath salmon farms at levels potentially toxic to benthic organisms. The zinc now used in salmon feed is organic which has reduced impacts on the benthic environment. Any effects are thought to be limited to the footprint of the farm.⁵⁶

Therapeutants for treating parasites and diseases have not been required in New Zealand salmon farms, due to the relatively low density and small scale at which they are farmed and absence of diseases. However, the expansion of finfish farming into new areas and other use of types of fish may require new methods of disease prevention. Kingfish trials undertaken in the Marlborough Sounds indicate that there will be a need for disease treatments because kingfish are unusually susceptible to some parasites. It is proposed that these parasites will be managed by low-concentration in-water baths of hydrogen peroxide.

Finfish farmers, have in the past, used antifouling paint to prevent the build-up of biofouling organisms on structures. These paints gradually wash off the structures and can accumulate on the seabed under the farms. Historically, many of these paints contained copper, which in high concentrations is toxic. The extent to which such concentrations of copper may impact on marine life is not fully understood, although it may be toxic to organisms living in the sediment.⁵⁷ The effects are normally confined to the boundary of the farm.

Although manual removal of fouling is possible for farm nets, there are few alternatives to putting antifouling paint on items such as predator nets which must be kept clear of debris to allow water to flow through. Where the use of antifouling paints cannot be avoided, copper-free antifouling paints are now available, and there has been a significant reduction in the use of copper in antifoul paints by the aquaculture industry.

Decision-making process

Aquaculture reforms, which took place in 2004, placed all aquaculture planning and consent processing under the RMA. Previously marine farmers had to obtain separate consents from the regional council and the former Ministry of Fisheries. The reforms were intended to address the 'gold-rush' situation which had developed in some areas, where numerous consent applications for marine farms were lodged in an attempt to occupy areas of the seabed in a 'first-in, first-served' system. The reforms placed a moratorium on the processing of such applications to give councils time to undertake proper planning for the activity through the creation of aquaculture management areas.

Further aquaculture reforms came into effect on 1 October 2011 designed to free up marine space for farming, as a result of slow progress in creating operative aquaculture management areas. These removed the requirement for such an area to be in place before a resource consent application could be lodged. As a result, the normal RMA decision-making framework now applies to marine farming, subject to the following unique provisions:

- Permitted activity status is not allowed – regional coastal plans are not able to classify aquaculture activities as permitted activities.⁵⁸ This is because the 'Undue Adverse Effects' test (described below) is carried out alongside the consent process.
- A 20 year minimum consent term – coastal permits for aquaculture are required to have a minimum term of 20 years, unless a shorter term is requested by the applicant or a shorter term is required to manage the effects of the activity⁵⁹
- A three-year lapse period – generally resource consents must be exercised within five years of being granted otherwise they will lapse. The reforms reduce the lapse period for aquaculture consents to three years.⁶⁰

The 2011 RMA reforms now give the Minister responsible for aquaculture the power to recommend regulations to amend provisions in a regional coastal plan related to the management of aquaculture activities.⁶¹ This process allows changes to be made directly to regional coastal plans without following the normal plan-making process set out in Schedule 1 of the RMA.

Allocation of space

New allocation mechanisms enable councils to better manage a 'gold rush' situation, should it arise under the 2011 rules. Figure 10.6 lists the tools that are now in place.

- The likely effect of the proposed marine farm on fishing, including the proportion of any fishery that would be affected
- The degree that the proposed marine farm would exclude fishing
- The extent that fishing for a species in the proposed marine farm area could occur in other areas
- The extent that occupation of the coastal marine area by the proposed marine farm would increase the cost of fishing
- The cumulative effect on fishing of any authorised aquaculture, including any structures authorised before the introduction of any relevant stock to the QMS

The test is limited to the effects on the practicalities of catching, taking and harvesting fish.

New Zealand Coastal Policy Statement

The NZCPS contains a number of policies relevant to aquaculture developments and operations. These are described more fully in Chapter 2: Marine management. The policies attempt to strike an effective balance between providing for such activities in the coastal environment and ensuring that important natural resources and community and cultural values are protected in doing so.⁶⁴ The Department of Conservation has provided some guidance to planners in interpreting the application of the NZCPS policies to aquaculture planning and decision-making, including a checklist of important matters to consider.⁶⁵

Policy 8 recognises the significant existing and potential contribution of aquaculture to the social, economic and cultural well-being of people and communities. This is done through directing that aquaculture activities are provided for in appropriate places within regional planning documents. It recognises that aquaculture activities require high water quality and land-based facilities. In addition, the policy directs that development in the coastal environment does not make water quality unfit for aquaculture activities in areas approved for that purpose.

NZCPS, Policy 8: Aquaculture

Recognise the significant existing and potential contribution of aquaculture to the social, economic and cultural well-being of people and communities by:

- a. including in regional policy statements and regional coastal plans provision for aquaculture activities in appropriate places in the coastal environment, recognising that relevant considerations may include:
 - i. the need for high water quality for aquaculture activities; and
 - ii. the need for land-based facilities associated with marine farming;
- b. taking account of the social and economic benefits of aquaculture, including any available assessments of national and regional economic benefits; and
- c. ensuring that development in the coastal environment does not make water quality unfit for aquaculture activities in areas approved for that purpose.

Protecting outstanding natural landscapes and natural character

Policies 13 and 15 of the NZCPS require adverse effects on outstanding natural features and landscapes and areas of the coastal marine area with outstanding natural character to be avoided. This effectively means that marine farms will not be able to proceed in sensitive areas of the coastal marine area and should instead be directed to localities where the natural character or landscape values of the area have already been compromised or are not outstanding. A recent Supreme Court decision on a salmon farm proposal in the Marlborough Sounds confirmed that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes.⁶⁶

Case law on marine farms in outstanding natural landscapes

New Zealand King Salmon Company Limited applied for plan changes and resource consents for salmon farms at nine sites in the Marlborough Sounds. The Board of Inquiry found that an industrial-scale salmon farm within Port Gore site would adversely affect an outstanding natural landscape, but the farm was still approved. The Supreme Court overturned the decision, making it clear that a plan could not provide for the establishment of a marine farm in a location where it results in adverse effects on an outstanding natural landscape.⁶⁷

Existing marine farming rights

Existing aquaculture consent holders have two key protections, which apply even if the consent holder is applying to establish a different aquaculture activity, although it must be within the same space:

- When a resource consent is due to expire the consent holder can apply for a new consent for an aquaculture activity in the same space (at least six months before the expiry of the existing consent) and can continue to exercise the existing consent until a decision is made to approve or decline the application
- If a consent is due to expire and an application has been lodged for an aquaculture activity in the same place, the existing consent holder is notified and thus has an opportunity to lodge a competing application which is given priority in terms of processing

It is important to note that these provisions do not amount to a right of renewal, or even a presumption that the existing consent will be renewed.

Case law on renewal of aquaculture coastal permits in an outstanding natural landscape

Sanford Limited and Port Gore Marine Farms applied to renew coastal permits for three mussel farms in Port Gore, in the Marlborough Sounds. The Marlborough District Council declined one application and granted the other two applications for limited terms. The applicants appealed the decision to decline/grant for limited terms. The marine farms were within a zone where new farms were prohibited. An appendix to the plan allowed existing farms to apply for consent as a discretionary activity. The Court determined that the landscape of the inner Port Gore is an outstanding natural landscape. Policy 15(a) of the NZCPS requires adverse effects on outstanding natural landscape areas to be avoided. The Court considered the existing environment of Port Gore as if the existing farms were not present (because the coastal permits had expired). The Court determined that each farm would, in a more than minor way, diminish the natural character of the locality. The Court refused the appeals by Sanford and Port Gore Marine Farms.⁷⁰

Elements of best practice

Some of the risks from aquaculture activities discussed in the previous sections can be minimised through good siting of farms and good design. These include marine mammal interactions, adverse impacts on landscape and natural character, and alterations of water flows. Other risks, such as those associated with marine biosecurity, can be managed through good on-farm practices. Water-quality impacts and cumulative risks are harder to predict and are therefore best managed adaptively, by phasing development within initially precautionary principles. Figure 10.7 below provides guidance on best practice responses to potential impacts.

Effects	Best practice response
General aquaculture activities	
Transporting spat and machinery used on marine farms, around the country, can spread diseases and pests	Good management, including careful washing and drying of equipment before transportation, should be adopted to prevent spread of disease and pests
Fixed structures used by a marine farm can attract biofouling species which can spread to the surrounding environment	Fixed structures should be regularly cleaned to prevent the build-up of biofouling species
Copper from antifouling paints can build up on the seabed to toxic levels	Marine farms should either manually defoul equipment or use paints which are free of copper
Alteration of the behaviour of marine mammals as a result of marine farms obstructing access to, or use of, feeding grounds and other areas of their habitat	Marine farms should be carefully located in order to ensure that access by marine mammals to habitual feeding, haul out, migratory routes and other areas is not obstructed Marine farms should not be located in areas which are used by dolphins for co-operative hunting
Potential for marine mammals to become caught in longline or finfish nets used on the marine farm	Marine farms should use the best available technology and techniques to minimise the risk that marine mammals will become trapped in nets (such as adequate net tension, enclosing anti-predator nets at the bottom, eliminating food waste and reducing the mesh size of nets) ⁷¹

Effects	Best practice response
Deposition of faeces and uneaten food causing changes to the physical, chemical and biological nature of the seabed and water quality	High-density stocking rates should be avoided, high-quality feed used to reduce the amount required, and automated feeding systems deployed which stop dispensing when they detect uneaten food in the water Farms should be located in areas which flush well Adopt integrated co-culture/multi-trophic aquaculture where possible (where finfish, mussels, seaweed and other species are grown in an integrated farm where one species counteracts the negative effects of another)
Effects on local biodiversity as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or of ecological importance
Escape of farmed fish and spread of diseases to wild fish	Fish farms should be designed to minimise escapees
Breeding of escaped fish with wild fish that adversely affects the genetic robustness of the wild population	All reasonable efforts should be made to prevent the escape of farmed fish, including utilising new technology where available
Harm to the marine environment and to humans consuming the products from chemicals used to treat the fish	Finfish should be farmed at a density which limits the likelihood of disease and therefore does not require the use of antibiotics and other therapeutants, as has been the case up to this point in New Zealand

Figure 10.7 Best practice considerations for aquaculture activities

The aquaculture industry has developed Environmental Codes of Practice designed to promote the adoption of best industry practices, throughout the growing and harvesting cycles, in order to minimise potential effects on the environment.⁷² These Codes have been independently benchmarked against similar international standards. They are specific to three species:

- Greenshell™ Mussel Industry Environmental Code of Practice, 1999 (Revised 2007)
- Oyster Industry Code of Practice, 2007
- Finfish Aquaculture Environmental Code of Practice, 2012

Aquaculture New Zealand is currently redrafting the Environmental Codes of Practice to reflect current industry practices, changes to the regulatory framework and the latest market conditions, as well as to incorporate any new global standards. Key performance indicators will be set for each Code. An online sustainability assessment and reporting tool is also being considered. This is intended for use by marine farm operators, third-party auditors and the industry body to help rate and consequently improve environmental performance against the standards. One of the aims of this work is that individual operators can use the environmental management system as a pathway to become fully accredited with an international certification.



Aquaculture New Zealand has developed codes of practice for marine farms, such as the salmon farm shown here in the Marlborough Sounds, in order to promote the adoption of best industry practice (Raewyn Peart)

Revive Our Gulf

Subtidal green-lipped mussels once covered over 500 square kilometres of reefs throughout much of the Firth of Thames and Tāmaki Strait, forming one of New Zealand's most productive shallow-water ecosystems.⁷⁶ From the early 1940s, these mussel beds were dredged for around five decades, until the commercial fishery collapsed. The damaged mussel beds never recovered, resulting in the loss of habitat and alteration of food webs as they gradually turned to less productive mud.⁷⁷

Mussels are powerful filter feeders. A single mussel is capable of passing 350 litres or a bathtub of water a day through its gills, secreting nutrients that support other marine life at the same time.⁷⁸ In their original numbers, the waters of the Firth would have been filtered daily, but with the few mussels left, filtering is anticipated to take two years. Without the filtering abilities of the mussels, the waters of the Hauraki Gulf have become murkier and more vulnerable to the effects of increasing nutrient run off.

In an effort to restore the mussel beds of the Hauraki Gulf, the Mussel Reef Restoration Trust set up the 'Revive Our Gulf' initiative in 2012. In December 2013 the first seven tonnes of mussels, stock donated by North Island Mussels Limited, were deposited in the Gulf. They were used to create 7 'living room'-size trial plots in a sheltered bay off eastern Waiheke Island.⁷⁹ The plots are being monitored by a student from the University of Auckland as part of work towards a doctorate. Within the first six months, there have been reports of spat starting to settle, an early signal of a self-sustaining reef.⁸⁰ Snapper and starfish have been attracted to the plots as the mussels provide a valuable source of food. In addition, new habitat has been provided to invertebrates and fish which have started to colonise the beds.

With the assistance of New Zealand's aquaculture industry, the Hauraki Gulf's mussel reefs are being re-introduced and re-established. It is hoped that water quality can be improved, and fish life will once again flourish amidst the newly-created habitat. Revive Our Gulf Chairman John Laurence said the group planned partnerships with mussel farms to utilise reject mussels for further field trials and initiatives to build awareness about the project's potential: 'Our vision is to prove a technology and then provide it to communities around the Gulf to enhance environmental quality'. This project highlights how valuable industry support can be to delivering marine restoration projects.



The Hauraki Gulf's mussel reefs are being re-introduced and re-established and it is hoped that water quality can be improved and fish life will once again flourish amidst the newly-created habitat (*Shane Kelly and Shaun Lee*)

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In more recent years, quartz-rich silica sand has been mined from sandbars off Mangawhai Heads and Pākiri Beach. The sand found at Parengarenga is very pure silica. Titanomagnetite iron sand (sand that contains both titanium and iron oxides) is found along the west coast of the North Island between Kaipara Harbour and Whanganui. This sand has been mined from the beach, from the adjacent coastal dunes, and from the near-shore marine area.² New Zealand Steel mines up to 1.2 million tonnes of sand each year from the north head at Port Waikato.³ The Taharoa iron sand mining site, located just south of the Kawhia Harbour, has the largest deposits of iron sand in New Zealand with an estimated 300 million tonnes of concentrate available at commencement of mining.⁴ There are also the rutile and ilmenite sand deposits of Westland, and the platinum, uranium and thorium-bearing sands of the South Island.

Inshore sand mining is regulated under the Crown Minerals Act for allocation and under the RMA for managing environmental effects, which is discussed in the section below on decision-making framework and in Chapter 2: Marine management.

Offshore sand mining

Offshore deposits of sand, associated with beaches formed during previous ice ages when the sea level was lower, occur off the west and east coasts, including Taranaki and the Bay of Plenty. Iron sand deposits are found in the northern and southern Taranaki Bights with localised concentrations of greater than 10 per cent titanomagnetite.⁵ An iron sand mining project to excavate up to 50 million tonnes of seabed material per year from the South Taranaki Bight was recently declined consent (this is discussed further in the case study at the end of this chapter).

Sand mining from offshore areas is regulated under the Crown Minerals Act for allocation and the EEZ Act for managing environmental effects, which is discussed in the section below on decision-making framework and in Chapter 3: EEZ legislation.

Mining methods

Both inshore and offshore sand mining follow a staged process from prospecting through to active mining, as shown in Figure 11.1. During each phase, different methods are used to assess or extract minerals from the seabed or land adjacent to the coast. Quantifying potential resources through prospecting and exploration typically takes around seven years before the active mining phase can begin.⁶

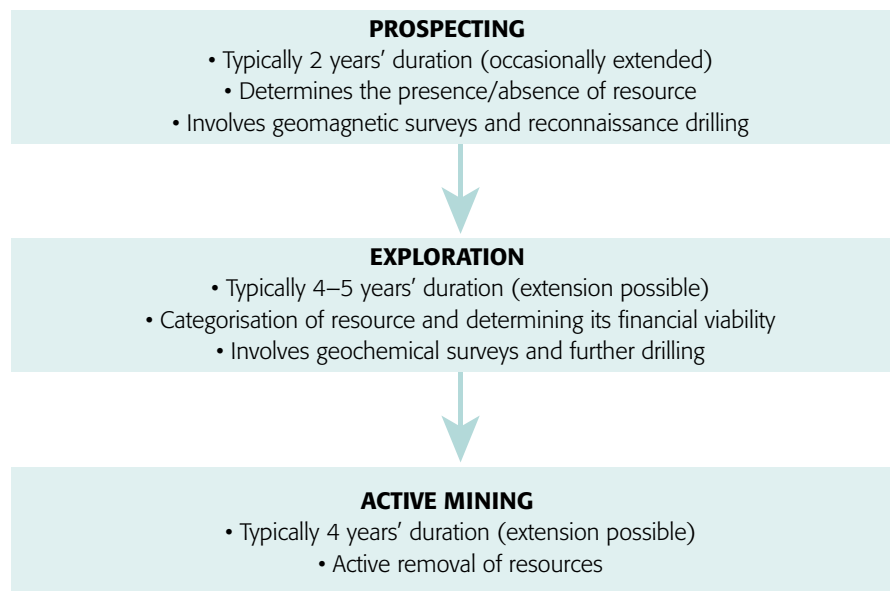


Figure 11.1 Phases of mineral extraction (Thompson K, 2012)

Prospecting and exploration

Prospecting generally involves reconnaissance drilling to recover sediment samples for analysis onshore, airborne geomagnetic studies and seismic surveys, as summarised in Figure 11.2 below.⁷ A limit on the volume of sediment which can be removed, and on the total number of core samples which can be taken, is normally set in the prospecting permit. For example, the Trans-Tasman

Active mining

Active mining of the seabed is generally undertaken using suction pipes to pump sand that has been mixed with seawater from the seabed. Iron ore is magnetically separated from the sand whilst other minerals are extracted by sieving, before returning the residue back to the sea.¹⁰ The sand mining methods used will vary, depending on the depth of the seafloor where the sand is being extracted from, and the distance from shore.

Offshore sand mining methods

Figure 11.3 shows the process used for mining offshore sand resources.

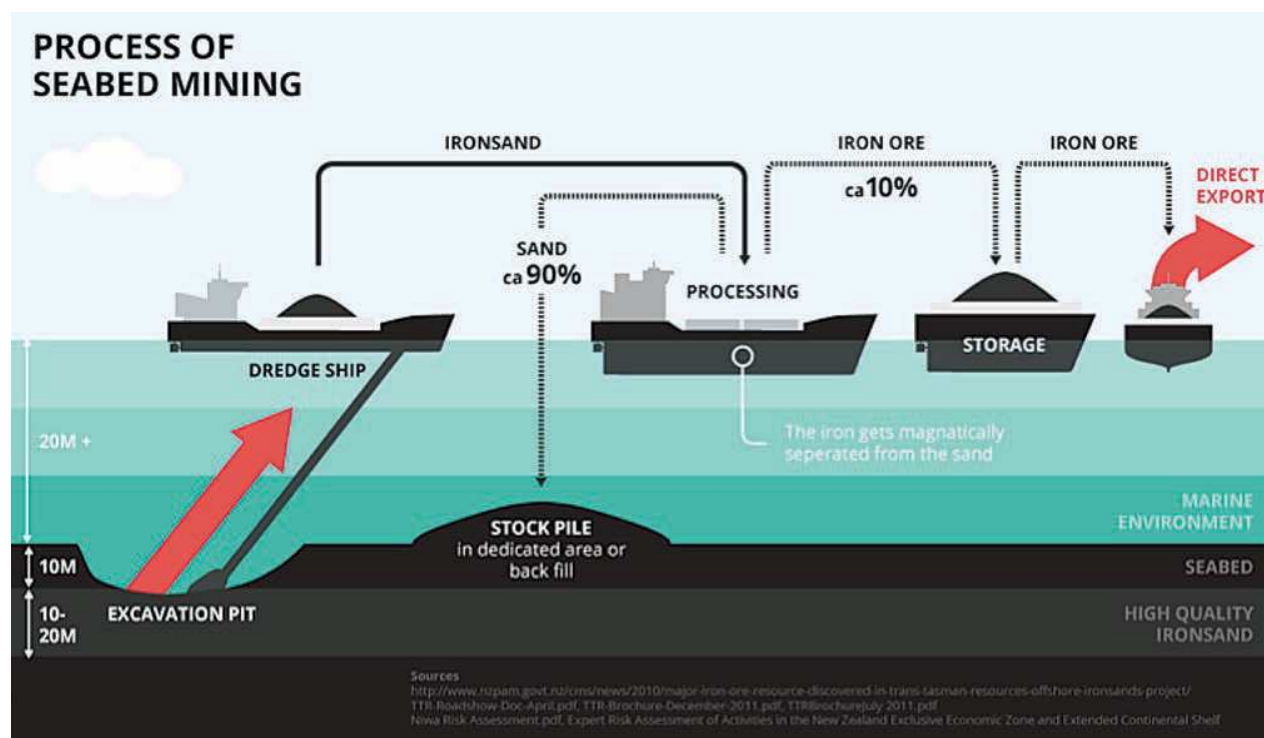


Figure 11.3 Process used for offshore sand mining (KASM¹¹)

Process proposed for offshore iron sand mining

Trans-Tasman Resources Limited applied to mine iron sands in the South Taranaki Bight using a subsea sediment extraction device called a 'Crawler'. This would create a mining cut approximately 12 metres wide and up to 11 metres deep. The seabed material would then be pumped by a slurry delivery pipe to a 'Floating Processing Storage and Offloading Vessel' where it would be processed to separate the iron ore (approximately 10 per cent yield). Processing would involve screening to remove material greater than two millimetres in size, and grinding and magnetic separation. The de-ored sand would have been redeposited into areas which had been mined.¹² The iron ore concentrate would be transferred to a Floating, Storage and Offloading Vessel where dewatering and storage would occur. It would then be transferred to bulk carrier vessels for export.

Inshore sand mining methods

Different methods are used for inshore mining, depending on the location of the sand resource. For example, at the Waikato North Head mine site, the iron sand is processed at an on-site concentration plant, where a magnetic concentrate is extracted through a series of separation processes producing a slurry. The slurry is pumped



Seabed habitat

Any sampling of the seafloor during prospecting and exploration will disturb the benthic community to some extent. In addition, research has suggested that during drilling or suction sampling a 'sand plume' has the potential to be formed, risking the smothering of benthic life as suspended material settles.²⁰ Sand from a plume quickly drops onto the seabed due to its weight, but fine sediment (silt and particularly clay) stays suspended for longer, and therefore can travel further afield before it settles, causing wider impacts.

Seabed mining causes considerable disruption to aquatic habitats on the seabed as large amounts of material are removed, filtered and then returned, once the mineral has been extracted.²¹ Within the dredged area itself, benthic communities will be significantly disturbed and may be destroyed. These can include shellfish beds and other benthic fauna living within the sand being extracted.

Bulk extraction of sand and redeposition of tailings on the seabed cause large sediment plumes which can travel many kilometres. The significance of their impacts will depend on the communities affected and whether they are adapted to regular disruption. For example, they would be greater if the plume travels outside the mobile sandy habitat and settles onto rocky or biogenic reefs. These impacts will move as the site is progressively mined and new areas disturbed, and are likely to be cumulative over time and space.²²

Some coastal and continental shelf areas are used by fish as nursery grounds and feeding areas. Changes to the benthic fauna following large-scale removal of sand may result in cumulative impacts higher up the trophic levels, including changes to feeding patterns of fish.²³ For example, research has shown that juvenile red snapper use sandy shoal areas to feed on small crustaceans commonly found in such material.²⁴ In addition, destruction of benthic communities in one area may have implications for those elsewhere. A 2008 report on the Kaipara Harbour raised concerns about the impact that mining may be having on tuatua populations there. In particular, it was thought that the tuatua beds located in and around the sand extraction area may be an important brood-stock for other populations found in the harbour.²⁵

The recovery of benthic habitats post-exploration and mining will depend on the extent of impact and the species affected, but in general those living in mobile sand environments are adapted to frequent disruptions and have the ability to rapidly repopulate areas.



Sand mining can change coastal dynamics, thereby affecting surf breaks such as this one shown at Wainui Beach, Gisborne (Raewyn Peart)

Cessation of sand mining due to erosion

The extraction of sand from the Pouto shoreline in the Northland Region has occurred for many decades, but has now been abandoned due to possible effects of the mining on shoreline erosion.²⁸ Several companies extracted sand from Pouto, particularly during the late 1970s. In 1992, Mt Rex was granted the sole resource consent for the continued extraction of sand, and this expired in 2004 and was not renewed. The resource consent allowed for a maximum of 60,000 cubic metres to be extracted per year. Monitoring of the Pouto shoreline began in 1990, at the request of the Department of Conservation. The monitoring confirmed that the shoreline was eroding and therefore that it was not appropriate to continue to extract large quantities of sand immediately adjacent to the shoreline.²⁹



Sand mining was stopped at Poutu in the Kaipara Harbour due to concerns about the eroding shoreline (Raewyn Peart)

Marine mammals, fish and seabirds

Seabed sand mining, through the introduction of equipment, increased underwater noise levels and potential ecological changes to the marine environment, can have a range of impacts on marine mammals, fish and seabirds. Small coastal cetaceans, such as Hector's and Maui's dolphins, are at particular risk from human activities such as sand mining, due to their restricted ranges and the cumulative impacts of many activities affecting one location.³⁰ Figure 11.5 provides some examples of the potential impacts of various sand mining activities on marine mammals.

Mining activity	Example of potential impact	Potential level of impact
Boat-towed surveying equipment	• Behavioural changes	Low – for example, no behavioural changes were observed during research work in the Bay of Plenty and Hauraki Gulf ³¹
Reconnaissance drilling and sampling	• Behavioural changes	Proximity dependent

areas which are undeveloped and normally little used by vessels. However, the significance of these impacts is directly related to the location of the sand mining operation, and the level of 'naturalness' of the background environment.

Findings on effects of sand mining on natural character and amenity

In 2006 there were appeals against the former Auckland Regional Council's decision to refuse consent for the continuing extraction of sand from near-shore areas of Mangawhai-Pākirī Bay by Sea-Tow Limited. Sea-Tow had applied for consents to enable extraction of 27,000 cubic metres of sand per year for 20 years. McCallum Bros applied for consents to enable extraction of 49,000 cubic metres of sand per year for 20 years. Both extractions were to continue activities authorised under previous expired consents and were for discretionary activities.⁴⁰ Parties within the meaning of section 274 of the RMA variously claimed that extraction of sand from a closed system such as this would result in degradation of the natural character of the beach. The Court held that the Hauraki Gulf is not a wilderness waterway. It is frequented by pleasure boats, shipping and other craft. Although a barge and tug passing regularly may not be an attractive view to residents or leisure users of the beach, their presence at sea could not be deemed markedly out of place. It held there were no more than minor adverse visual effects. The Court also held that there would not be any significant adverse effect on natural character. This was largely related to the expectation that the resource would not be depleted because the system was not closed.⁴¹

Other

Equipment and vessels brought into New Zealand from overseas can act as vectors for non-indigenous species.⁴² These impacts are discussed in detail in Chapter 5: Marine biosecurity.

Mining may negatively impact on areas and resources of importance to Māori. These could include kaimoana areas, wāhi tapu or urupa sites which lie on or under the seabed. This may be through the direct impacts of the mining activity on the seabed, through accelerated coastal erosion caused by the removal of sand from the system, or through the wider impacts of the sediment plume.

Consideration of impacts of sand mining on Māori cultural and spiritual values

The Environment Court heard appeals against the former Auckland Regional Council's decision to grant replacement consents for sand extraction in the Kaipara Harbour in 2006. The Oruawharo Marae Trust opposed the applications on the grounds that continued sand extraction would have a negative adverse effect on Māori cultural and spiritual values, and that the extraction site was wāhi tapu, among other matters. The Court held that the concerns raised by the Trust did not of themselves provide a basis for declining or recommending against the granting of consents, and that monitoring and review conditions were satisfactory.⁴³



The Environment Court found that mining from near-shore areas of Mangawhai-Pākirī Bay, shown here, would not have more than minor visual effects and would not have significant effects on natural character (Raewyn Peart)

The NZCPS contains a number of policies that are relevant to applications for minerals activities, including:

- It is a national objective to recognise that some uses and developments which depend upon the use of natural and physical resources in the coastal environment are important to the social, economic and cultural well-being of people and communities⁴⁶
- It is recognised that the extraction of minerals is an activity which is important to the social, economic and cultural well-being of people and communities⁴⁷
- A precautionary approach is to be adopted towards proposed activities whose effects on the coastal environment are uncertain, unknown, or little understood, but potentially significantly adverse⁴⁸
- Adverse effects on certain indigenous biodiversity, such as taxa which are threatened or at risk or naturally rare, must be avoided⁴⁹
- Adverse effects must be avoided in areas with outstanding natural character⁵⁰
- Adverse effects must be avoided in outstanding natural landscapes⁵¹
- Requiring that subdivision, use or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water⁵²
- Protecting surf breaks of national significance by ensuring that activities in the coastal environment do not adversely affect the surf breaks⁵³
- In managing discharges to water in the coastal environment have particular regard to avoiding significant adverse effects on ecosystems and habitats after reasonable mixing⁵⁴

Figure 11.6 NZCPS policies relevant to sand mining applications

Often the scientific evidence supporting an application contains considerable uncertainties, because the marine environment is a dynamic system subject to constant change. Obtaining scientific data can be expensive and challenging because of the difficult environment within which the research is being undertaken. A lack of certain scientific information means that there is often disagreement about how coastal sand systems work.

Adaptive management is an approach which has developed to address scientific uncertainty. It is about informed trial and error, essentially learning as the mining progresses. There is an emphasis on monitoring and measurement (against pre-defined thresholds), learning what works best, and then adapting management processes accordingly. Monitoring needs to be long-term, and is a major commitment.⁵⁵ Increasingly, consent authorities deal with the lack of information by granting consent accompanied by strict conditions providing that a programme of adaptive management and monitoring must be followed. Conditions can provide that reviews of the consent must be undertaken once particular extraction quantities have been reached. This approach is discussed further in Chapter 3: EEZ legislation.

Case law on need for adaptive management and monitoring

In 2006, the Royal Forest and Bird Society sought the imposition of a scientifically-credible monitoring regime to protect conservation values, when replacement consents were granted for sand extraction in the Kaipara Harbour. This included an 'adaptive management regime' involving the undertaking of baseline studies, particularly in relation to Maui's dolphin, as a pre-condition of increases in extraction rates.⁵⁶ In relation to marine mammals, the Environment Court concluded that there was no evidence that harm had arisen from the current high level of operations, and that the risk of adverse effects was less than minor and probably negligible. The Environment Court also rejected a sophisticated monitoring programme proposed by a submitter in relation to shorebirds on the basis that the effects would be less than minor. Essentially, this means there is a need to demonstrate a more than minor risk of harm before the Court is likely to require an adaptive management regime.

Management under the EEZ Act

Under the EEZ Act, prospecting and exploration for sand are permitted, subject to conditions. Mining is a discretionary activity. This means that a marine consent is required for sand mining, and the application for consent is notified, allowing the public to make submissions and participate in the hearing. To date, no minerals activities have been classified as non-notified discretionary. Further details about requirements for sand mining under the EEZ Act are outlined in Chapter 3: EEZ legislation, and in the case study at the end of this chapter.

- Thresholds are set to trigger remedial action before the effects become overly damaging
- Effects that might arise can be remedied before they become irreversible

In addition to the areas to be avoided (discussed above) the following best practice considerations outlined in Figure 11.7 can be applied to sand mining activities, both offshore and inshore.

Area	Detail
Destruction of benthic communities	• Mine in a chequerboard or strip pattern to allow recolonisation from adjacent populations • Ensure that sufficient research has been undertaken to identify what species are likely to be present in the area to be mined and their distribution • Minimise the area which is directly affected by mining activity • Assess options for biodiversity offsetting as appropriate to protect or restore seabed • Adopt mining methods which minimise the amount of sediment which is released into the water column
Disruption of sedimentation and coastal processes	• Monitor changes in shoreline position • Complete beach profiles prior to mining commencing • Minimise the area which is directly affected by mining activity
Disruption of marine mammals, fish and seabirds	• Ensure a comprehensive understanding of the presence, distribution, abundance and density of marine mammals in the area both pre-operation and during operation to assist to determine whether and to what extent mining operations and associated activities may be impacting on marine mammals • Make sure all personnel on board project vessels receive training in regards to monitoring and management of any impacts on marine mammals and seabirds, such as identification and behaviour observation • Limit unnecessary lighting and ensure that lights that are required are directed vertically onto work surfaces • Monitor bird strike and review mitigation measures in response to bird mortality above an identified level • Minimise underwater noise levels • Reduce the speeds of vessels in the vicinity of marine mammals
Changes to natural character and amenity	• Assess options for biodiversity offsetting as appropriate to protect or restore inshore area
Other	• Regularly check, clean and apply antifouling paints (preferably copper-free) to infrastructure that is brought into New Zealand's marine area from overseas. Biosecurity New Zealand has prepared a craft risk management standard – discussed in Chapter 5: Marine biosecurity. • Adopt an adaptive management approach, where the impacts of a pilot operation are monitored and assessed before considering whether expanding to full production can be allowed or whether amendments are required • Undertake rigorous baseline monitoring prior to lodging application so that sufficient information is available during the hearing and so that ongoing monitoring can be compared to the baseline • Develop and undertake a robust monitoring regime – this should include biological and geomorphological monitoring as well as baseline monitoring, so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner

Figure 11.7 Best practice considerations for sand mining activities

Area	Detail
Biosecurity	There is a risk that non-indigenous organisms could be introduced to New Zealand by ships transiting into New Zealand to support the proposed activities. The Committee accepted that these would be managed under the Biosecurity Act.
Human health	The Committee considered four issues: air discharges from the combustion of heavy fuel oil in the engines on the vessels, contamination to seafood from the proposed operation, risk to other users of the South Taranaki Bight (e.g. recreational) and health and safety of workers. The Committee concluded that the effects on human health would have been manageable through consent conditions.
Physical environment	The Committee considered four issues: seabed bathymetry, waves and surf, coastal stability and the sediment budget and freshwater springs. It concluded that there would not be significant adverse effects on the physical environment. There was no evidence that changes to the seabed bathymetry would cause more than minor effects on waves, surf, coastal stability or the sediment budget.
Visual, natural features, natural landscapes and natural character	The mining activity, and the sediment plume created, had the potential to cause adverse visual effects. The Committee found that visual effects from the mining activity would be minor and visual effects from the plume would be moderate. The impact on natural features and landscapes, outside the mining area itself, would be minor. The effects on natural character would be minor to moderate at the mining site and minor in coastal waters.
Archaeology	The Committee considered the potential for shipwrecks or other archaeological artefacts to be present in the proposed mining site and for these to be destroyed by the mining operations. It was satisfied that a proposed 'Discovery Protocol for Shipwreck Finds' would have appropriately addressed archaeological effects.

Figure 11.8 Summary of decision on application to extract sand from the South Taranaki Bight

The Decision-Making Committee also considered effects on existing interests:

- Iwi – iwi were concerned that they had not received adequate information to allow them to make an informed response to the proposal. The Committee found that iwi's existing interests would potentially be adversely affected in more than a minor way by the project. It acknowledged that the proposed consent conditions sought to involve iwi in a substantive way. However, iwi were not engaged in drafting those consent conditions and had not agreed to them.
- Fishing – the Committee heard evidence regarding potential effects on commercial, recreational and customary fishing. It concluded that the 'effects [on commercial and recreational fisheries] are at best uncertain'. It also concluded that there was 'considerable uncertainty' with respect to customary fishing.
- Recreation and tourism – the Committee found that it was unlikely that the proposal would harm New Zealand's international reputation
- Origin Energy Resources – the proposed mining area is adjacent (1.14 kilometres) to the existing Kupe platform. The Committee found that Origin Energy Resources' interests could be satisfactorily addressed, predominately through existing Crown Minerals Permit conditions.
- Marine traffic – the proposed mining area is removed from regular marine traffic routes and therefore no conflict was considered likely



The Decision-making Committee considered that the effects on recreational users of the Taranaki Bight of the sand mining proposal would have been manageable through consent conditions (Raewyn Peart)

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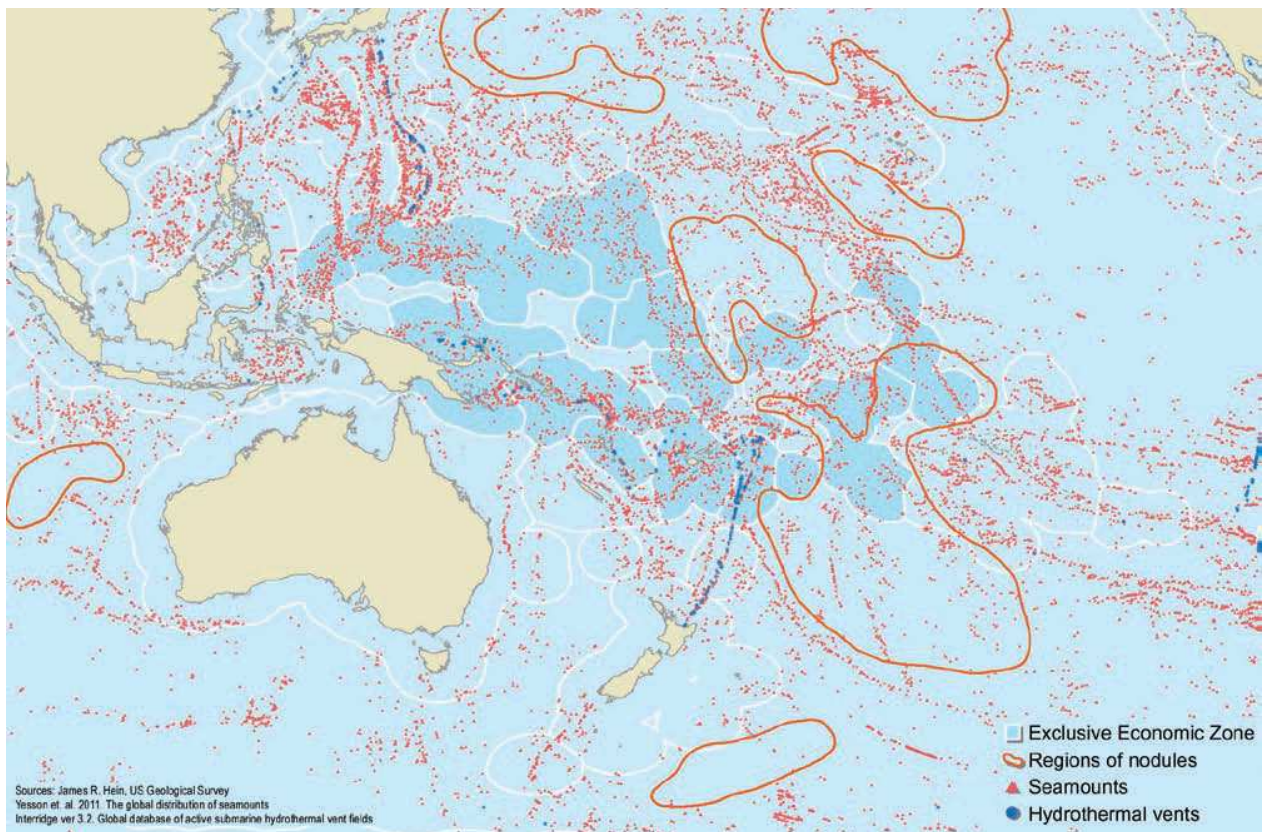


Figure 12.1 Distribution of sedimentary basins and some mineral resources, including those within New Zealand's marine jurisdiction (NIWA, 2012)

Mineral	Detail
Phosphate nodules	• They are formed in highly-productive areas of upwelling • The nodules are found within a 1 metre deep seabed layer consisting of fine sandy silt³ • The nodules that are economically viable to recover range in size from 2 to 150 millimetres • The phosphorite nodules are ground up, turning them into rock phosphate • Phosphate nodules are found on the Chatham Rise east of Christchurch at water depths of about 400 metres⁴ • Phosphate is an essential ingredient in fertilisers, and is therefore potentially commercially valuable, particularly as the majority of New Zealand's fertiliser is currently imported • Seafloor mining for phosphorite nodules is unprecedented at the moment, thus New Zealand could be the first country to permit this type of mining activity
Seafloor massive sulphides	• These are deep undersea deposits of high-grade polymetallic sulphides rich in copper, zinc and lead, and with a high gold and silver content • The largest of these deposits form by mineralisation of fragmented rocks • Chimney-like deposits are particularly rich • They are produced in underwater volcanic regions, usually on seamounts, by the deposition of metals from hydrothermal fluids • Hydrothermal vents, which contain these minerals, have been discovered on the Kermadec Ridge off the north-east coast of the North Island of New Zealand. The locations of these hydrothermal vents in the Kermadec Arc are known and mapped by researchers.⁵ • Both inactive and active vents with a low level of hydrothermal activity have the potential to be mined

- Exploration – the purpose of this stage is to evaluate the feasibility of mining identified deposits. This stage can include any drilling, dredging or excavations needed to assess the nature and size of a mineral deposit.¹⁵ The exploration phase may involve core sampling at higher concentrations (than undertaken in the prospecting phase) in small areas, test pit excavation, test drilling, trials of extraction methods and bulk sampling. This exploration could involve the removal of larger volumes of deposits.
- Production – the purpose of this stage is to actively extract and process the mineral resource. Techniques for mining minerals in the marine environment may involve seafloor suction dredging, seafloor slurry pipes, use of tracked vessels on the seafloor and seafloor cutting/fragmentation. Following extraction and processing, tailings (unwanted material) are deposited back onto the seafloor. Some production operations involve a processing vessel that moors to ‘permanent’ anchor blocks or anchor moorings.

Due to the high variation in the geological processes governing resource formation, and concentration associated with the different mineral deposits, there is no single, universal method which can be deployed. A variety of methods will be required to mine these deposits. Some of the tools used are shown in Figure 12.3. Technology that can be used to mine minerals from the seabed efficiently in deeper waters is only just being developed, and work is still ongoing to determine what the most effective methods of mining may be, both in terms of resource recovery and for minimising environmental impacts.

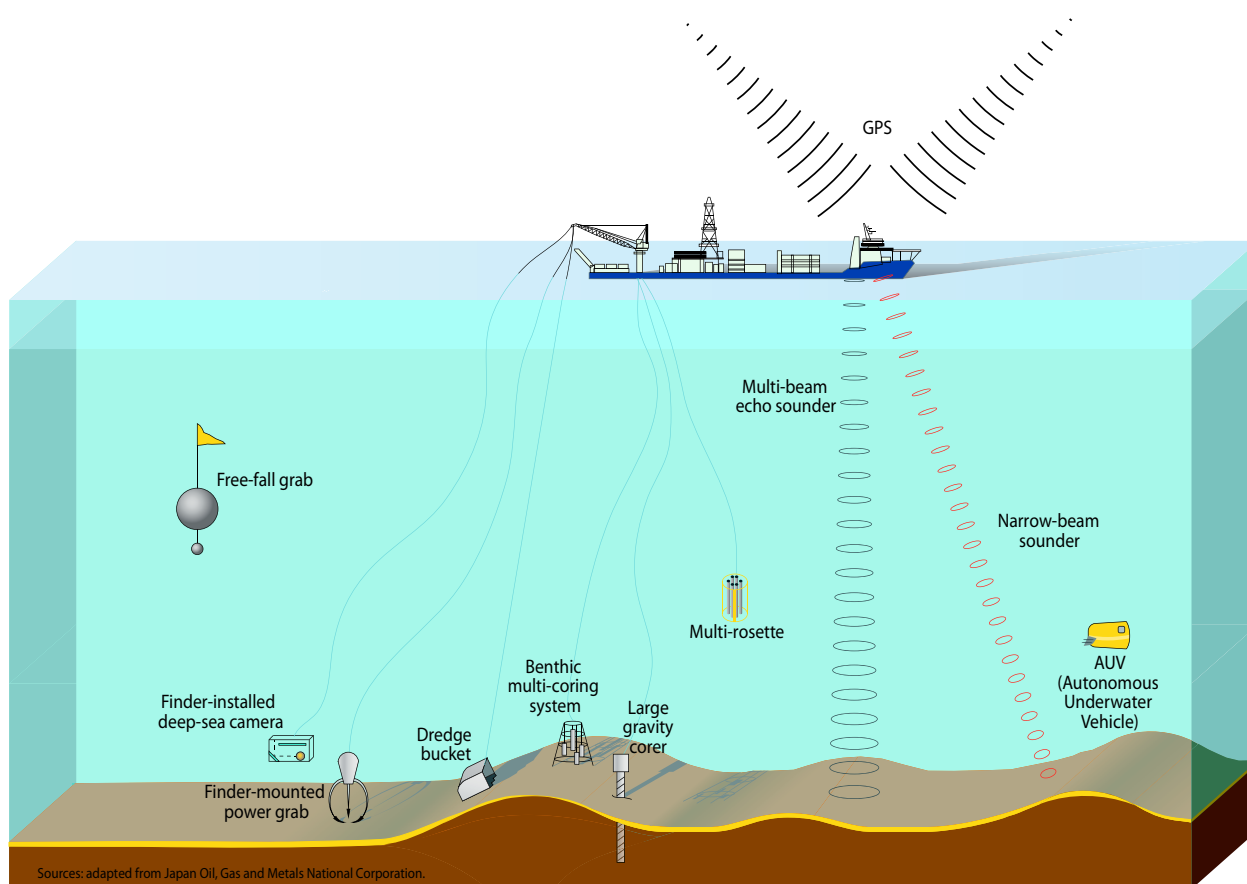


Figure 12.3 Examples of deep-sea minerals exploration tools (*Secretariat of the Pacific Community, 2013a*)

Phosphates

Mining for phosphate generally requires a mining vessel that deploys a trailing suction draghead that moves along the seabed. Water jets and possibly cutting teeth are used to loosen the top centimetres of seabed sediment, and the resulting slurry of sediment and seawater is pumped to the mining vessel. The material will be mechanically processed on board the ship to separate the coarse phosphatic material from the finer non-phosphatic sediments. The fine non-phosphatic sediment and seawater will be discharged to the seafloor or above the seafloor.¹⁶

- Transport of a slurry (ore and seawater) vertically from the seafloor to a vessel or platform on the sea surface
- If substrate rock is collected, separation of ore and gangue (the material in which ore is found that has no economic value) using froth flotation on board a vessel
- Dewatering of the ore on board a vessel or platform
- Transfer of the ore from the vessel to a transport barge or bulk carrier/storage facility
- Disposal of the separated seawater

Technologically, the mining of cobalt crusts is considerably more complex than that of manganese nodules, as the crusts form a hard substrate. Ideally, a mining operation would only remove the crust, and not the underlying volcanic rocks. In addition, the slopes of the underwater volcanoes where the crusts are located are very rugged, which can make the use of excavation equipment more difficult. Cobalt crust mining is only at the conceptual stage at present. It is believed that the greatest barrier to mining both manganese and cobalt resources is the processing stage, with viable processes to extract the minerals from the ore once it has been removed from the seabed yet to be developed.



Manganese nodule collector (COMRA (Secretariat of the Pacific Community, 2013b))

Methane hydrates

Mining for methane hydrates is still in its early stages. In March 2013 a Japanese company became the first to extract gas from offshore methane hydrates. It is aiming to commence commercial production by early 2019.²³

In 2008, a joint Canadian-Japanese expedition in Mallik in the Northwest Territories, Canada, established that methane hydrates could be collected by using a water pump to depressurise a well already drilled. This process involved lowering the pressure by pumping out the water that naturally accumulates in the well. Standard oil and gas drilling equipment was used to re-enter the existing well drilled to a depth of just over 1,000 metres and then to 'refurbish' it by casing the entire well with lengths of steel tubing cemented into place to prevent it from collapsing.²⁴

Subsequently, gas hydrate resources on the Alaskan North Slope have been classified as technically recoverable, and the first production tests offshore in Japan suggest production of gas from offshore hydrates is now also technically feasible.²⁵

Exploitation in New Zealand remains a major challenge for a number of factors:²⁶

Mining may also change the topography of the seabed, and associated hydrodynamics of the area, as well as the composition of the substrate. As a consequence, the original community may be unable to re-establish after mining activity ceases, and be replaced by different species better adapted to the new environment. The significance of this change is dependent on whether the affected species and communities are unusual, rare and/or play a particularly important role in the food web and broader ecosystem functioning.



Crab shown on the flange of a hydrothermal area associated with a deep-sea vent
(Secretariat of the Pacific Community, 2013a, photo courtesy of Chuck Fisher)

Deposition of sediment tailings into the marine environment is likely to extend the environmental footprint of the mining activity over a wider area than just the active mine site. Smaller particles may become suspended in a plume, which can travel some distance with ocean currents before settling out. Water-borne sediment is known to be a particular problem for suspension-feeding organisms, such as sponges and corals, as the fine particles can clog their filtering apparatus. It can also affect the gills of fish. In addition, such sediment can affect primary productivity by reducing light levels and thereby the ability of phytoplankton to grow.

When sediment settles, it can smother benthic organisms, and also change the nature of the substrate making it unsuitable for attachment by invertebrates such as corals. The distance and direction the sediment plume will travel depends on its composition (i.e. grain size) and on oceanographic conditions (e.g. direction, frequency and velocity of currents).

Mining activity has the potential to impact on fish stocks in the longer term, if it disrupts food supplies or spawning and nursery areas on which the fishery is dependent.

These impacts will be sequential during the long production phase, as adjacent areas are progressively mined. Recovery potential, once mining has ceased, will depend on a range of factors. These include whether the affected species are reliant on the specific features of the habitat which have been removed and whether there are remaining communities in adjacent areas that provide a source for recolonisation. As many deep-sea species are slow-growing and long-lived, recovery of the benthic communities could take considerable time; in some cases recovery may never occur.

The major challenge for deep-sea mining is the lack of baseline data on what marine species are associated with prospective mining areas. There is even less understanding of the interactions between different species and trophic levels, the ecological significance of any impacts, and recovery and recolonisation potential.

Marine mammals and seabirds

Underwater sound may have acoustic impacts on marine mammals as well as other fauna.²⁷ Vessel and mining equipment, including underwater pumps, riser and sinker activity and the vessel-based processing plant, produce varying levels of underwater noise. Generally, such activities are of a short duration. The possible consequences vary from none (affected animals may leave the area), to acute injury (ear drum damage), to serious death, which depends on the noise level encountered, species and habitat.²⁸ There may also be cumulative effects from repeated exposure. In addition, the use of mining vessels can increase the risk of ship strike, and the use of lights at night can attract seabirds, causing them to collide with ships.²⁹

Biosecurity

Vessels and mining equipment brought into New Zealand from overseas can act as vectors for non-indigenous pest species. Non-indigenous species are unlikely to survive independently of the vessel in deep offshore waters, but may be transferred to and invade shallow coastal habitats.³⁰ (This topic is discussed in more detail in Chapter 5: Marine biosecurity.)

Active deep-sea vents, where seafloor massive sulphides are found, are associated with highly unusual and productive communities that rely on sulphur compounds as an energy source. These communities were only discovered in 1977³³ and remain poorly understood. The environments around active vents, and the communities they support, are dynamic and subject to irregular flows of hot fluid.

Mining will leave permanently altered areas at the mine site, with surrounding areas potentially impacted by the sediment plume. The spatial scale of the impacted zone is dependent on the volumes of mined material, and current regime near the seafloor.³⁴ In areas of active hydrothermal venting, species may be adapted to natural changes in the environment, and therefore be able to recolonise more rapidly than other deep-sea communities, dependent upon there being a viable source of recruits nearby.³⁵

A major minerals research programme in New Zealand, led by GNS Science, aims to gain a better understanding about how these minerals are formed. The programme is also considering the development of effective exploration and extraction practices. This research is currently examining seafloor massive sulphide mineralisation along the Kermadec Arc and in Frontier regions, including the Colville Ridge and Kermadec Ridge.³⁶ The work should assist with building a deeper understanding of how these minerals form and the environments surrounding them.

Manganese nodules

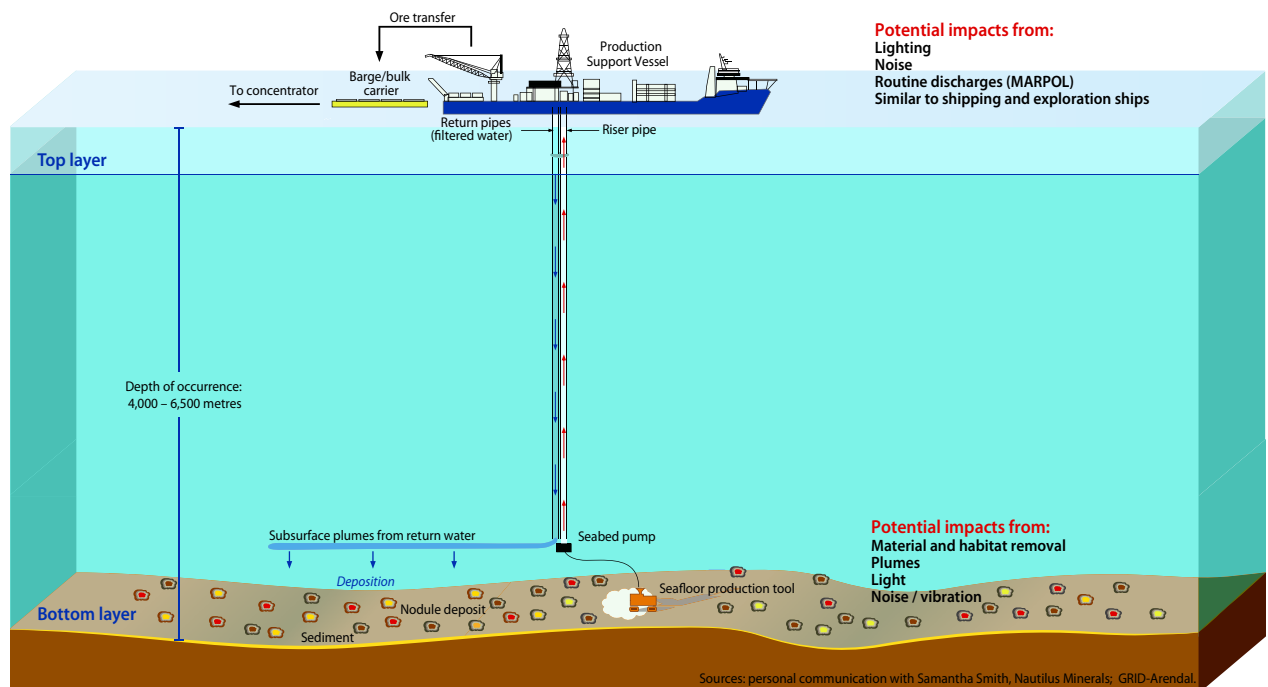


Figure 12.6 Example of seafloor manganese nodule mining system and potential environmental impacts (*Secretariat of the Pacific Community, 2013b*)

The excavation of manganese nodules would disturb parts of the seabed. Potentially large amounts of sediment, water and many organisms would be dug up with the nodules (depending on the method proposed), and the destruction of the deep-sea habitat could be substantial.³⁷ Nodules take millions of years to form, so these habitats would be permanently lost from large areas of seabed following mining.³⁸

The potential environmental impacts from exploration and mining for manganese nodules are highlighted in Figure 12.6 and summarised as:³⁹

- Destruction of the benthic community in the mining area including organisms living in the crevices of the nodules which would be permanently lost
- A sediment plain left remaining after the removal of the nodules which is suitable only for a subset of the original benthic community. Nodules will not reform for millions of years.
- Mobilisation of sediment into the water column, which is potentially a major concern for the affected ecosystem as currents at that depth are often very slow, as is the settling velocity of the disturbed material

Decision-making framework

Allocation of mining rights

Applications for prospecting and mining licences are processed by New Zealand Petroleum and Minerals. The 'Minerals Programme for Minerals (Excluding Petroleum) 2013' sets out the policies and procedures to be applied to the allocation of mineral resources (including consultation with iwi and hapū by the Crown prior to allocation). The requirements to be met by permit holders are set out in regulations.

The Crown Minerals Act separates activities into Tier 1 and Tier 2. Tier 1 includes high-return, high-risk projects, and most, if not all, marine minerals operations will fall within this category. The Act requires a high-level preliminary assessment of the applicant's health, safety and environmental policies, and technical and financial capabilities to be undertaken for all applications for Tier 1 activities. The purpose is to ensure that the applicant is able to undertake the work proposed, and that the applicant does have a satisfactory history and hence will not likely cause health, safety or environmental issues in New Zealand.

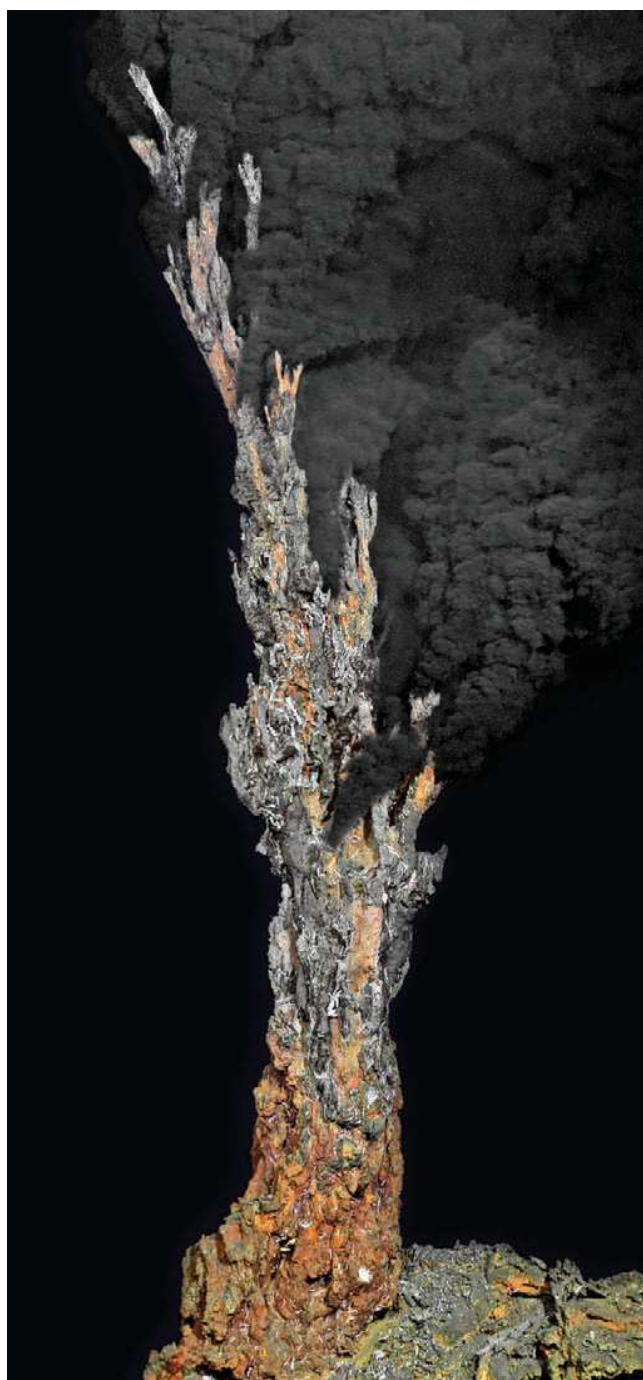
Before granting a permit for Tier 1 exploration or mining, the Minister of Energy and Resources must be satisfied that the proposed permit operator has, or is likely to have by the time the relevant work in any granted permit is undertaken, the capability and systems that are likely to be required to meet the environmental requirements of all specified Acts for the types of activities proposed under the permit. The Minister may seek the views of other regulatory agencies, including local authorities, the EPA and Maritime New Zealand, in satisfying himself or herself of these matters.⁴⁴

Environmental management under the EEZ Act

The EEZ Act regulates the environmental effects, and effects on existing interests, of minerals mining (and other seabed activities) beyond the territorial sea. Under the EEZ Permitted Activity Regulations prospecting and exploration for minerals are permitted activities, subject to conditions.

Minerals production mining in the EEZ is a discretionary activity. This means a marine consent is required, and the application is notified, allowing the public to make submissions and participate in the hearing. To date, no minerals activities have been classified as non-notified discretionary.

Discharges and dumping from seabed mining vessels and installations are also regulated under the EEZ Act. At the time of writing the relevant regulations were still being developed. The provisions of the EEZ Act which apply to mining activities are described in more detail in Chapter 3: EEZ legislation.



Seafloor massive sulphides are rich in copper, zinc and lead and also have a high gold and silver content (GEOMAR)

Seafloor massive sulphide mining operations	
Area	Action
Seafloor	• Setting aside part of the habitat, with representative animals, to serve as reference areas and sources of animals to assist recolonisation of the impacted site • Relocating some of the representative animals • Installing artificial substrates to help recolonisation • Considering location, positioning and method of any subsea discharge to limit its impact • Employing technologies/methodologies so that sediment resuspension is minimised during cutting • Limiting the number and/or size of sites
Mid-water column	• Developing and implementing emergency response procedures in the event of accidents leading to spills that might affect water quality • Using fully-enclosed ore delivery systems
Surface	• Releasing discharge and water from dewatering plant where possible, at least below the top 200 metres of the water column (i.e. below the light zone) • Developing and implementing environmental management plans that specifically cover waste minimisation and loss prevention, in order to reduce impacts on water quality • Developing and implementing emergency response procedures in the event of accidents leading to spills that could affect water quality • Adopting effective mitigation measures to minimise the risk of injury to marine animals from ship strike or collision • Installing approved sewage treatment plants, certified through relevant international standards, to handle such normal ship discharges as treated sewage • Maintaining surface deck lights at the lowest levels needed to ensure safe working conditions • Developing safety, health and environmental policies and plans for all offshore operations
Manganese nodule and cobalt crust mining operations	
Area	Action
Operational	• Developing and implementing environmental management plans that cover waste minimisation and loss prevention to minimise impacts on water quality – a working example of a management plan was prepared by the International Seabed Authority in 2012 • Developing and implementing emergency response procedures in the event of accidents leading to spills to the environment • Using effective mitigation measures to minimise the risk of injury to marine animals from ship strike or collision • Installing an approved sewage treatment plant, certified to meet relevant international standards and/or other relevant regulations • Developing and implementing safety, health and environmental policies and plans for all offshore operations
Spatial	• Setting areas aside for conservation, possibly within mine site/mine lease areas. Depending on the site, it may not be possible to find an appropriate site within the mine lease that will not be impacted by the mining (e.g. by sediment plumes), in which case a similar site nearby may be nominated as the conservation area instead. Attention should be given to the representativeness, adequacy, resilience and connectivity of a network of areas. • Establishing marine management areas, which involve zoning of different areas for different uses or intensities of use. Such an approach may designate areas that are acceptable for total mining, areas that can only be partially mined, or areas set aside for conservation. • Evaluating other generic measures, if feasible, including installation of artificial substrates to encourage more rapid colonisation by fauna impacted by mining, and animal relocation if there are populations of rare, endemic or highly endangered species • Evaluating the location of discharges to ensure minimal impact on ecosystems • Considering the use of diffusers to aid dispersal • Considering animal relocation, if there are populations of rare, endemic or highly endangered species • Employing reserve networks, such as marine protected areas
Temporal	• The time scales of nodule and crust growth and faunal recovery in the abyssal deep sea almost certainly make consideration of short time-scale measures impractical • Measures to rehabilitate degraded areas or encourage longer-term faunal recruitment are much less likely to be effective than spatial management approaches
Figure 12.9 Possible mitigation and management strategies specific to seafloor massive sulphide, manganese nodule and cobalt crust mining operations (<i>Secretariat of the Pacific Community, 2013a, b and c</i>)	

	Potential effect
Water quality	The mining activity will result in the formation of a suspended sediment plume reducing water clarity. The disturbance of sediment will result in the release of chemical elements (nutrients such as nitrogen and phosphorus and trace elements such as arsenic, nickel and uranium) found in the sediments into the water column. Chatham Rock Phosphate states that the elements released are likely to result in the 99 per cent ANZECC (Australian and New Zealand Guidelines for Fresh and Marine Water Quality) trigger value being exceeded, but that following dilution, concentrations will not exceed the trigger value. In the case of uranium, there will be a near-field concentration increase of 10 per cent.
Sediment quality	There will be discharge of organic matter after mining due to the incorporation of benthic biota in the mined sediment. This discharge is likely to increase oxygen demand. If organic enrichment occurs, then the oxygen demand and consumption by bacteria decomposing the organic matter will increase, possibly leading to hypoxia or anoxia. Chatham Rock Phosphate states that it will not result in anoxia at the water–sediment interface.
Food web	Chatham Rock Phosphate has commissioned the development of a trophic model of the food web on the Chatham Rise. The model indicates that much of the Chatham Rise fisheries resource is reliant on primary production through phytoplankton and the buildup of food resource in the zone between 200 and 1,000 metres below sea level. It suggests that there is limited food dependency between fisheries and benthic fauna because the proposed mining is below the photic zone. Chatham Rock Phosphate concludes that the trophic impacts of removing benthic fauna in the mining area will be minor in the broader context.
Fish	More than 200 species of fish have been identified on the Chatham Rise. Key commercial species include hoki, hake and ling. Fish may be impacted by increased suspended sediment, changes in water quality and noise.
Marine mammals	Based on data sets of opportunistic sightings, Chatham Rock Phosphate states that 12 species of whales or dolphin and one species group (beaked whales) have been observed in Chatham Rise waters between 1981 and 2007. These sightings include two threatened species (killer whale and the southern right whale) and three International Union for Conservation of Nature endangered species (sei, blue and humpback whales). Marine mammals may be impacted by noise created by mining activities, interactions with the mining vessel (e.g. ship strike), disruption of food sources, and introduction of pollutants into the environment. Chatham Rock Phosphate states that noise effects may occur within 2 kilometres of the vessel for some marine mammal species. It states that the sound levels produced by an operating dredger (similar to the equipment to be used on the mining vessel) are comparable to the sound levels produced by a similar-sized ship transiting across the Chatham Rise. It proposes consent conditions to avoid any impacts of noise on marine mammals.
Seabirds	Chatham Rise is one of the most important areas for seabirds in New Zealand, with the majority of seabirds being present over the course of a year. The high marine productivity in this area attracts foraging birds. The company has identified that approximately 28 species are ‘key members of the Chatham Rise avifauna’, most of which are threatened or at risk. Seabirds may be impacted by interaction with the mining vessel (particularly lighting impacts), disruption of food sources, and the introduction of pollutants into the environment.
Cumulative impacts	Currently, the only human use of natural resources on the Chatham Rise is fishing, which results in damage to benthic environments and bycatch of marine mammals, seabirds and other non-target species. Chatham Rock Phosphate has identified that approximately 92,000 square kilometres of the Chatham Rise (under 1,000 metres depth) was trawled between 1989 and 2011. The company states that the impacts of mining on the crest of the Chatham Rise (450 square kilometres over 15 years) will be proportionally very small compared to the area on the flanks of the Chatham Rise that has been and continues to be impacted by commercial fishing.

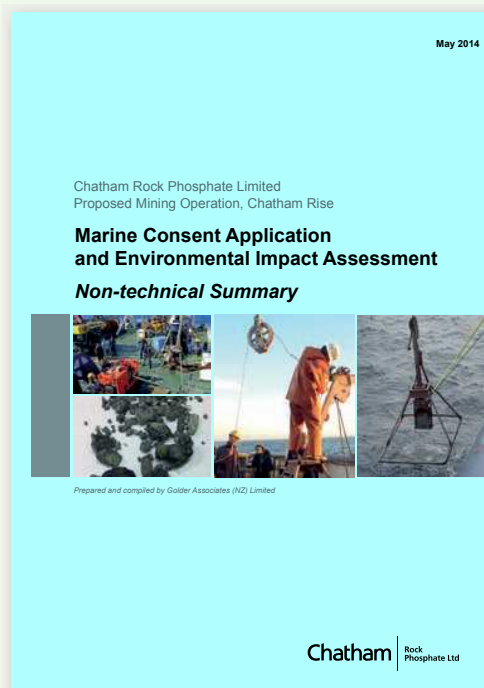
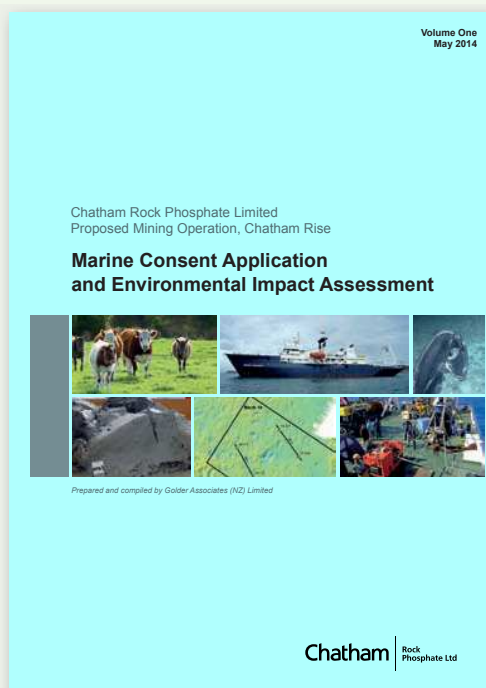
Figure 12.10 Potential environmental effects of the Chatham Rise phosphate mining
(Adapted from the Chatham Rock Phosphate marine consent application)

- Where total suspended solids concentrations exceed 50 milligrams per litre at a point 5 kilometres or greater away from the mining operations, or at a point 50 metres or greater above the seabed, the applicant had:
 - Undertaken additional monitoring to confirm that the threshold was exceeded
 - If the threshold was exceeded, advised the EPA
 - Undertaken investigations to identify the point at which total suspended solids reach 50 milligrams per litre and the point at which they reach 'background' levels
 - Within three months of advising the EPA, completed an assessment of responses which would avoid, remedy, or minimise total suspended solid levels. If a solution was identified, the applicant must identify timeframes for implementation.
 - If an adaptive management approach was to be implemented, it had to be implemented in accordance with the timeframe identified

The Committee found that the adaptive approach proposed by the applicant was inadequate because it did not provide for the validation of the modelling undertaken regarding, *inter alia*, the existence of equivalent habitat and benthic ecology outside the mining area and the extent of sediment dispersal. The Committee considered that this would require a three-stage adaptive management strategy. This would begin with a no-mining stage to allow for the collection of sufficient data regarding coral communities. If the uniqueness of the communities in the mining area was disproven, a trial mining stage to validate the sedimentation plume model over a small area could be authorised. Only then (provided necessary adjustments were made) would fully operational mining be permitted. The first two stages would contain a 'go/no go trigger', which could terminate the consent if the field evidence was sufficiently contrary to the applicant's prediction.

The staged adaptive management approach was rejected by Chatham Rock Phosphate on the basis that it would render the project unviable. As a result, it could not be imposed without frustrating the consent.

The Committee concluded that the proposal would create significant and permanent adverse effects which were incapable of being avoided, remedied or mitigated and this aspect of sustainable management was of paramount concern in this instance. With a staged adaptive management approach requiring considerable pre-mining research being dismissed as unviable by the applicant, the Committee declined consent.



A sample of the detailed documentation supporting the Chatham Rock Phosphate Limited's marine consent application

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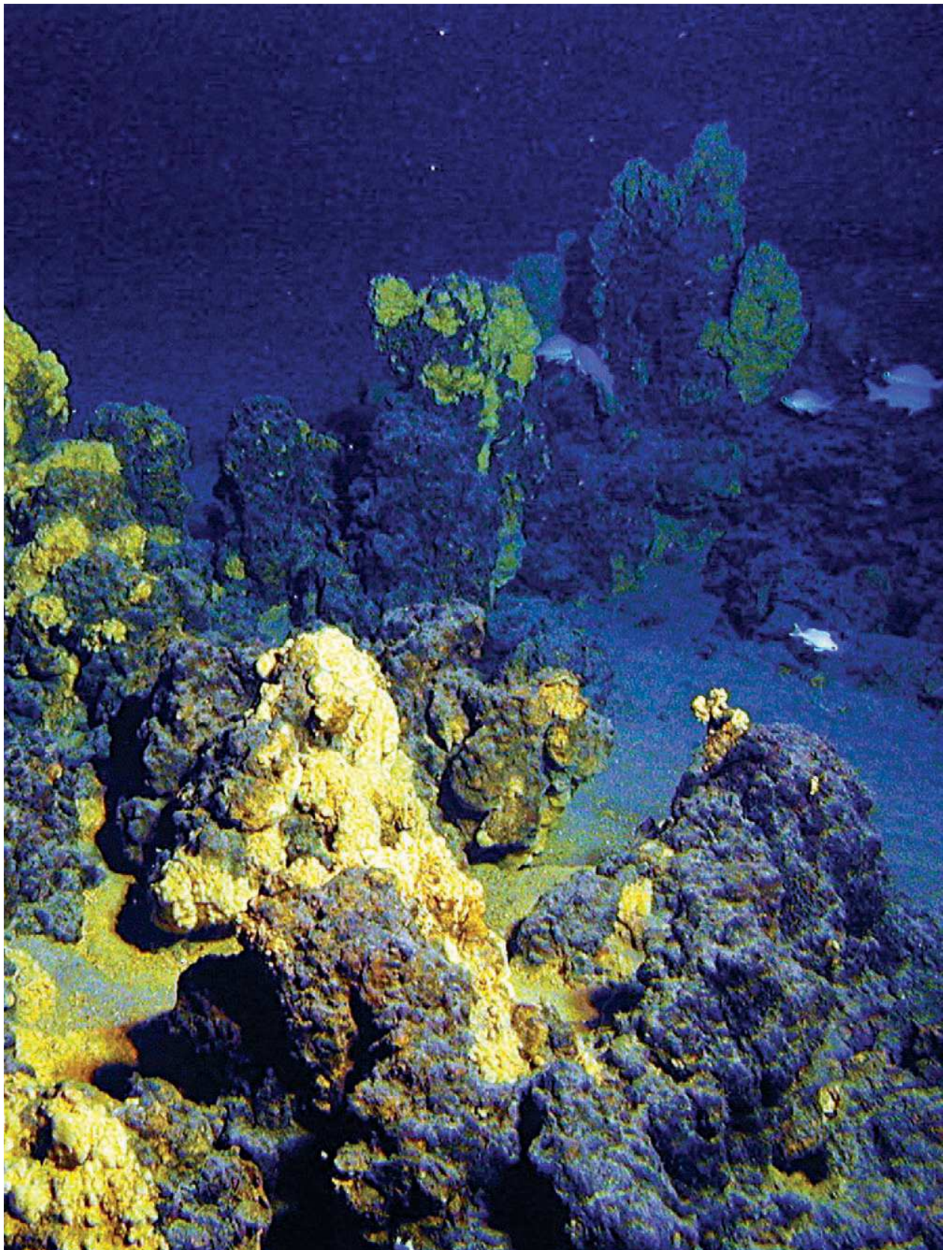
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Iron and silica rich chimneys on the summit of Giggenbach volcano, Kermadec Arc
(New Zealand-American Submarine Ring of Fire 2005 Exploration; NOAA Vents Program)

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NEW ZEALAND
PETROLEUM & MINERALS

New Zealand Petroleum Basins

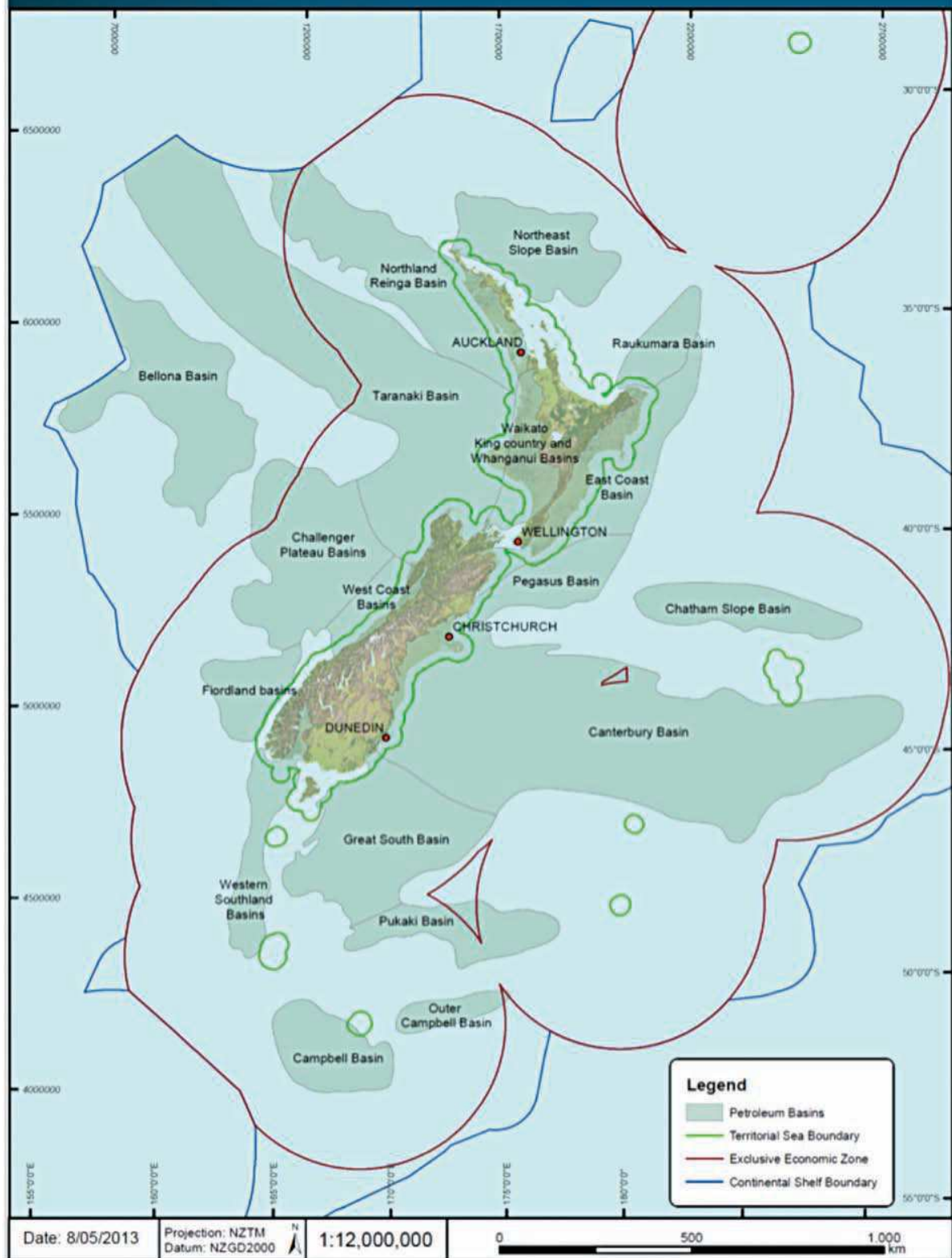


Figure 13.1 Locations of New Zealand's petroleum basins (Ministry for the Environment, 2013)

Exploration

If the initial phase delivers some positive results, then well drilling to discover and define potential oil and gas deposits may take place. Exploration typically focuses on a smaller spatial area of interest than prospecting, which then becomes the focus of more intense investigation. Activities authorised by an exploration permit can also include further more detailed seismic surveying (usually three-dimensional).

A mobile offshore drilling unit is used to drill a well in order to assess whether oil and/or gas is located in a specific geological zone, and in the specific location that has been identified as a 'prospect' based on seismic surveying and other analysis. The type of rig used will depend on the depth of water, ocean conditions and other factors. The three main types of rigs used for exploration are:¹³

- Jack-up rigs – used in shallower water up to around 120 metres deep. The rig is towed to the location, and its legs are lowered to the bottom to enable the rig to be raised above the water surface.
- Semi-submersibles – used in depths of 100 to 3,000 metres. These consist of a floating, column-stabilised vessel, with a submerged hull. The rigs can be towed and anchored, or are able to maintain their own position with thrusters.
- Drillships – used in depths of up to 3,000 metres. These are purpose-built ships with drilling and other equipment installed on board. They function as a ship until they reach the drill site, where they are either anchored or maintain their own position with thrusters while undertaking drilling operations.

Drilling a single well may take weeks to a few months to reach the target depth, depending on various factors including the depth of the target, the rock types encountered and weather conditions. One or more wells may be drilled under a single exploration permit. Exploration permits require a certain number of wells to be drilled within a specified time frame.

In order to create an offshore well, an oil rig drills through the seabed and into the subsurface reservoir. There are a number of steps that are needed for drilling a well which are described below:¹⁴

1. Spudding – a metal tube is forced into the ground (this tube can be up to 100 metres long by 92 centimetres in diameter). This provides a casing which is the backbone of the well. The casing is lowered onto the seafloor by a 'drill pipe'. More drill pipe is added over time to force the casing into the seabed until it reaches the targeted petroleum prospect. Some rigs use large hammers to push the casing deeper. Once the right depth is reached the drill pipes are pulled back to the surface.
2. Drilling a hole for a second string of casing – a drill bit is lowered inside the casing. This is generally drilled between 300 and 1,000 metres into the seabed. High-pressure drilling fluids called 'mud' is pumped down the centre as the drill is rotated. This fluid helps to keep the drill cool, provides lubrication, and carries chipped rock pieces away from the well. This mud is also the primary barrier for keeping hydrocarbons in the well during drilling. A second smaller casing (often 56 centimetres in diameter) is lowered down the hole.
3. Cementing the casings in place – cement is pumped down the drill pipe and back up between the two casings. This helps to cement the casing to the rock in the seafloor and is a vertical barrier on the outside of the casing.
4. Connecting the blowout preventer and marine riser – a large underwater control valve known as the 'blowout preventer' is placed on top of the wellhead. It is a critical piece of equipment, that is generally fixed directly on top of the wellhead on the ocean floor, and functions as the secondary barrier to controlling hydrocarbon pressures. The blowout preventer is connected by a marine riser to the drilling rig or drillship. Once these are installed, drill pipe is lowered through the marine riser and the blowout preventer. The marine riser allows the mud and rock cuttings to be pumped back to the surface and onto the drilling rig for processing.
5. Drilling the remaining sections of the well – steps 2 and 3 can be repeated to drill the remaining well sections. The rock cuttings and mud are circulated back up to the drilling rig.
6. Conducting test logs – once the correct depth has been reached, tests are conducted to see whether oil and gas is present, and if so the nature and extent of it. The information is logged. In some cases hydrocarbons will be brought to the surface to flow test the well.
7. Once an exploration well is completed, it will usually be plugged with a cement plug and abandoned. In some situations, it may be temporarily abandoned so that it can be subsequently tested or produced from. Once the well is abandoned, the drilling rig will leave the drill site.

involves plugging the well with cement and dismantling removing of equipment, plant and machinery. All New Zealand's offshore fields remain in production so decommissioning is yet to take place here.

Prior to decommissioning, surveys are usually undertaken to map the location and quantity of debris, pipelines, power cables and natural marine environments. Post-decommissioning surveys can identify any debris left behind during the removal process and confirm any environmental damage. Remote underwater vehicles and divers can be used to remove any debris and to ensure the area is free from any of the operational equipment.

Abandonment of oil rig infrastructure has become a major issue in the North Sea where there are many fields reaching the end of their producing phase. Best practice is to remove as much of the infrastructure as safely as possible. Another option being investigated is the storage of CO₂, where some of the existing infrastructure of depleted offshore oil and gas fields might be used for CO₂ sequestration after adaptation.¹⁷

Potential environmental effects

Blowouts and oil spills

The risk and consequences of oil spills are probably the greatest environmental concern associated with oil and gas operations. While most oil enters the oceans unseen, massive oil spills highlight the catastrophic effects that oil can have on marine wildlife and human communities.¹⁸ There has been only one major deepwater blowout, which was the Deepwater Horizon well in the Gulf of Mexico, which blew catastrophically in 2010. So far, there have been no major well blowouts in New Zealand, but the risks may increase as oil and gas activity expands. Because this country is so far away from any offshore oil and gas servicing ports, it would take a long time for international aid to respond.¹⁹

Formations deep in the ground, where hydrocarbons are located, are under intense pressure. Controlling these pressures is fundamental to successfully drilling wells. Various 'barriers' are employed to ensure pressures are controlled throughout the life of the well, including the initial drilling phase. The key principle is that there are always two tested barriers in place ('the two barrier rule').

The high pressure drilling fluids called 'mud', which are pumped down the centre of the drill string during drilling, create a primary barrier which exerts hydrostatic pressure to stop hydrocarbons coming to the surface. Mud weights are adjusted during drilling to ensure balance is maintained. The blowout preventer generally acts as the second barrier if the mud fails when drilling an exploration well.²⁰

Well blowouts occur when the pressure in the reservoir being drilled overwhelms the primary mud barrier, and the blowout preventer and other secondary barriers are unable to stop the flow. This can be a result of faults in the safety device which is intended to seal the well, poor well design, and/or human error. Often it is the result of a combination of these factors. Blowouts can sometimes be brought under control quickly, but in other situations they can last for a prolonged period. In 1979, when the Ixtoc 1 drilling rig sank as the result of a blowout, it took engineers nearly a year to stem the flow of oil into the Gulf of Mexico.²¹

Deepwater Horizon blowout

A report by the Chemical Safety Board, an independent investigative arm of the USA federal government, concluded that the 2010 Deepwater Horizon disaster was caused by the blowout preventer failing to properly close the well when a surge of natural gas from the deep formation caused a blowout. The blowout preventer itself punctured the riser pipe, allowing oil to start leaking into the Gulf of Mexico close to the seabed. This triggered the worst offshore oil spill in history. About five million barrels of oil flowed into the Gulf before the well was closed off, 87 days later.²²

The impact of such oil spills can vary from minimal to large-scale mortality of marine life, depending on the sensitivity of the environment and its ability to recover. The Ixtoc 1 disaster contaminated the Gulf of Mexico waters with millions of barrels of crude. Fisheries had to close, birds and other wildlife perished, and vast lengths of coastline were soiled.²³

Figure 13.5 highlights some of the key potential environmental impacts resulting from oil spills caused by oil and gas drilling, including well blowouts. Oil spills from vessels and their impact on the environment are discussed in Chapter 15: Vessels.

Activity	Potential impact
Ship movements	- Noise, lights and the physical presence of ships affecting the movement of sensitive species - Dredging of navigation channels if necessary, disturbing and modifying the seafloor
Seismic surveys	Underwater noise generated by air-based energy sources in seismic arrays disturbing marine animals during the duration of their deployment, especially marine mammals
Drilling rig placement	Localised physical damage to the seafloor where the rig is placed
Anchoring	Localised physical damage to the seafloor where anchors are lodged
Drilling	- Localised smothering and toxic impacts on marine life from drill cuttings if they are discharged overboard - Release of toxic substances into the marine environment from clay-based drilling fluids which include metals such as barium and a range of additives, depending on the type of rock being drilled - Reduction of light in the water column due to suspended sediment which may reduce plant growth until the particulate material settles
Platform placement	- Localised physical damage to the seafloor where the platform is placed - Enhanced marine habitat through the creation of artificial reefs - Provision of important resting places for seabirds and haul outs for seals and/or sea lions
Produced formation water	- Localised contamination of seawater through the discharge of low levels of hydrocarbons contained in the produced formation water. Long exposures to low levels of particular hydrocarbons can result in cumulative and/or sublethal impacts. - Localised warming of seawater from the discharge of warmer produced formation water - Localised changes in salinity levels as produced formation water may have a salt content greater or less than seawater
Greenhouse gas emissions	Contributes to global greenhouse concentrations through flaring and use of fuel for vessels and power generation

Figure 13.6 Potential environmental impacts from offshore oil and gas exploration and mining

(Adapted from the World Petroleum Organization)²⁴

Fisheries

Permanent oil and gas production activities may affect fisheries through direct effects on fished species. Fish may move away from the area of influence due to increased levels of activity.²⁵ In addition, the new structures will alienate a small area of seafloor, so that it no longer provides foraging habitat for demersal fishes (which live and feed on or near the seafloor). However, these same structures may provide new artificial reef-like surfaces and increase reef habitat in the area.²⁶

There may be impacts on fisheries management as production platforms and pipelines invariably have a zone of restricted access around them that restricts fishing access in these areas. Unless commercial fish quota is bought by the petroleum company, and retired for the duration of the petroleum activities, commercial fishing will be displaced into the remainder of the quota management area, increasing pressures on fish populations there. The impact of a single production facility is unlikely to be great, but the cumulative impact could be significant if multiple production facilities and pipelines are located within a single quota management area.²⁷ There are fisheries exclusion zones around rigs and pipelines in the North Sea. However, it has been estimated that despite more than 500 platforms and thousands of kilometres of pipelines being in place there, the loss of fishing area was only around one per cent.²⁸



Fisheries may be impacted by oil and gas production activities (Daniel Sharp)

Biosecurity

Oil drilling platforms create biosecurity risks as they are large structures which lie stationary in coastal waters for long periods of time, which enables organisms to grow on the immersed parts. Drillships and platforms brought in from overseas can potentially bring with them invasive marine pests. In the past rigs have been towed to New Zealand, but recently, they have been transported on heavy lift vessels, so biosecurity can be well managed.

The South African brown mussel (*Perna perna*) was detected in Tasman Bay following the cleaning of a semi-submersible oil rig's submersed surfaces that had arrived from Australia in November 2007.³⁴ In total, 10 non-indigenous species were found on the rig. This resulted in a clean-up operation to remove the risk of the exotic pests establishing in New Zealand. Transported species are unlikely to survive independently of the rig in deep offshore waters – the risk is that the platform acts as a stepping stone for non-indigenous species to invade shallow coastal habitats. The risk can be lowered by cleaning and/or fresh antifouling if necessary.³⁵

Biosecurity risks and impacts are discussed further in Chapter 5: Marine biosecurity.

Amenity

Near-shore oil and gas activities can potentially have the following amenity effects:

- Noise from the operation could affect residents and visitors to the area, if operations are close to the shoreline
- Vessels passing regularly through the marine area can create an unwelcome visual impact, particularly in areas which are normally little used by vessels
- The presence of oil rigs in the sea can industrialise the natural ocean landscape

Other impacts

Near-shore oil and gas activities may negatively impact on areas and resources of importance to iwi and hapū. These could include kaimoana areas and wāhi tapu or urupa sites which lie on or under the seabed.

There may be other ecological impacts resulting from increased light and noise in the deep-sea environment. These are poorly understood but could impact on fish and marine mammals that are dependent on sound for locating prey, navigating and communicating.³⁶ There is also the risk of accidents leading to spillages and loss of equipment within the marine environment.



Māui B platform off the Taranaki coast (Shell Todd Oil Services Limited)

No prohibited activities are identified in the regulations, so a marine consent can be sought to authorise petroleum drilling and mining activity anywhere within New Zealand's EEZ and extended continental shelf.

The EEZ Act will also regulate discharges and dumping from offshore installations, as well as discharges from ships if it is a mining-related discharge, once regulations are made. Discharges and dumping from vessels continues to be regulated under the Maritime Transport Act. The Exclusive Economic Zone and Continental Shelf (Environmental Effects – Discharge and Dumping) Regulations 2014 classify discharge and dumping activities.⁴⁴ At the time of writing these were still in draft format and had not yet been promulgated.

Figure 13.7 sets out the activity status of discharge and dumping activities that specifically relate to oil and gas activities. These were not final at the time of writing.

Activity	Activity status
Discharge of harmful substances from petroleum exploration drilling	Non-notified discretionary
Discharge of harmful substances from petroleum production drilling	Discretionary
Discharge of oily waste from machinery waste	Permitted, subject to conditions Prohibited, if the oil content exceeds 15 parts per million
Discharge of contaminated seawater from operations	Permitted, subject to conditions Prohibited, if the oil content exceeds 15 parts per million
Discharge of drilling fluids	Non-notified discretionary
Discharge of food waste	Permitted, subject to conditions
Discharge of garbage	Prohibited, unless necessary for safety reasons or accidental
Dumping of fish waste, sewage sludge, organic material or mineral exploration structures	Non-notified discretionary
Dumping of dredged material or vessels in authorised dumping grounds	Non-notified discretionary
Dumping of other dredged material, other vessels, other structures, CO ₂ streams from CO ₂ capture processes or inert inorganic geological material	Discretionary
Dumping of waste or other material not otherwise classified	Prohibited

Figure 13.7 Proposed activity status of discharge and dumping activities that specifically relate to oil and gas activities

The marine consent process is described in full in Chapter 3: EEZ legislation.

To date the EPA has considered two non-notified applications for marine consent for oil and gas exploration wells from Shell Todd Oil Services Limited and OMV New Zealand Limited. Both applicants commenced drilling under the transitional provisions of the EEZ Act but were required to seek a marine consent because drilling was not able to be completed before 28 June 2014. The OMV New Zealand application was to drill two exploration wells near the Maari platform in the South Taranaki Bight. The wells will be drilled in a water depth of 98 metres and to a total depth of 2,839 metres.⁴⁵ The Shell Todd application was to drill two exploration wells near the Māui field in the South Taranaki Bight. The wells will be drilled in a water depth of 105–108 metres and to a total depth of 3,370–5,183 metres.⁴⁶ Both applications were granted.

Marine Mammals Protection Act

Marine mammal sanctuaries can contain restrictions on extractive industry-related activities. For example, the West Coast North Island Marine Mammal Sanctuary restricts seabed mining and puts stringent conditions on seismic surveying.⁴⁷

The Department of Conservation's 2013 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations' was developed in conjunction with international and domestic stakeholders representing industry, operators, observers and marine scientists. Its purpose is to provide effective and practical mitigation measures for minimising acoustic

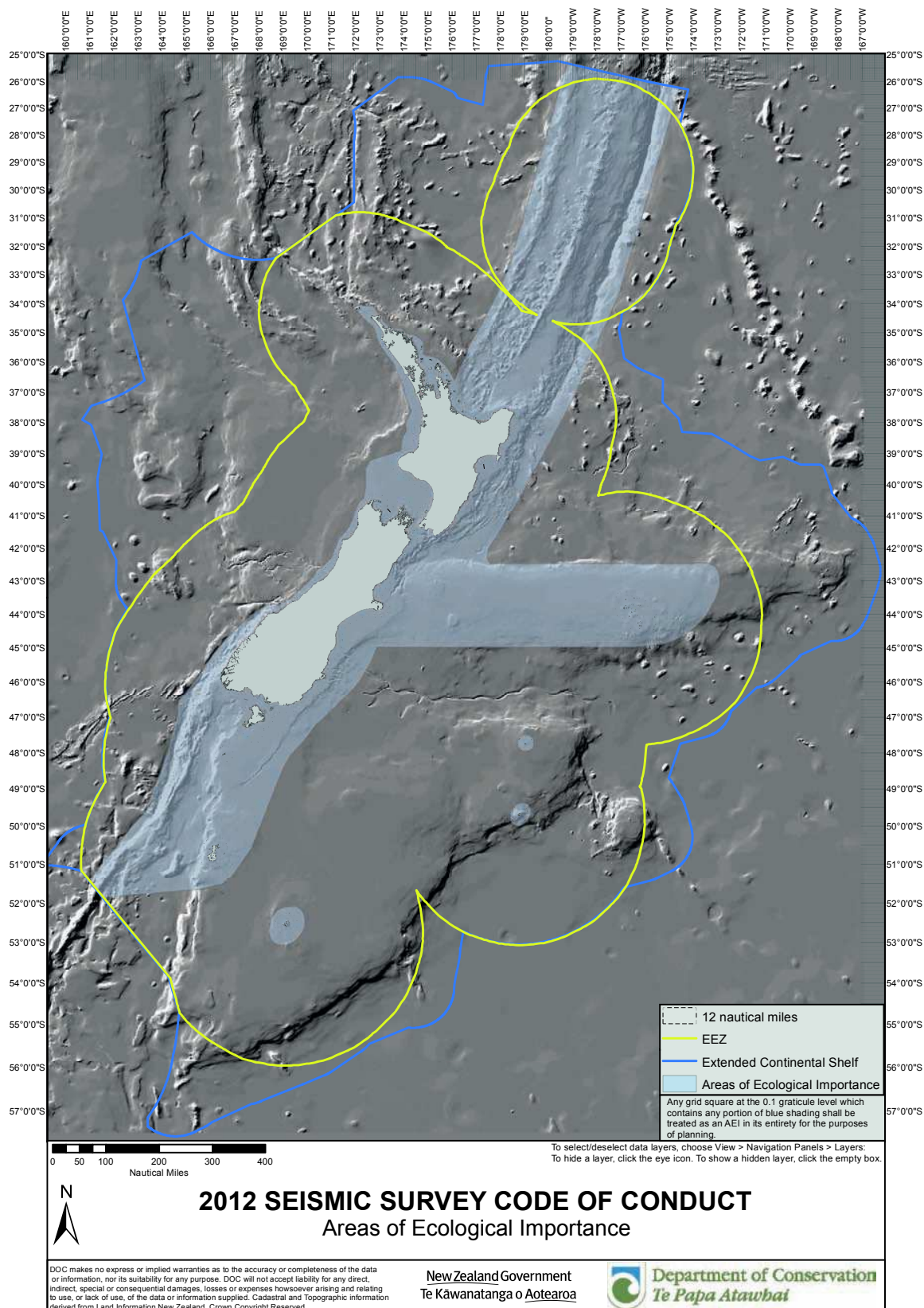


Figure 13.8 Map of Areas of Ecological Importance in relation to seismic surveying (Department of Conservation⁵⁰)

- Disposal on land, into air and in water

Marine disposal is considered the last resort where no other viable option exists, and in the specific circumstances of decommissioning this implies that all practically recoverable material will be removed.⁵² In 2010, Maritime New Zealand began a process intended to develop a national policy on offshore decommissioning in consultation with the industry. This is still underway.

Health and safety management

Although health and safety regulations are directed at protecting employees from hazards, they can provide some environmental protection. This is because avoiding hazards to people, and avoiding environmental damage such as by minimising the risk of oil spill, often involve the same technical steps (e.g. keeping the hydrocarbons controlled). For example, the 2014 Ministry for the Environment Guidelines for onshore petroleum development states, 'Well design and integrity are important matters to consider for managing risk to both health and safety and the environment. The regulation of well design and integrity to manage risk to health and safety is the responsibility of the High Hazards Unit (part of WorkSafe New Zealand)'.⁵³

The High Hazards Unit within WorkSafe New Zealand (a Crown Agent with an independent Board appointed by the Minister of Labour) was established to enforce health and safety legislation in New Zealand's mining, petroleum and geothermal sectors, recognising the inherent and significant hazards in these industries. However, WorkSafe New Zealand has no legal mandate to protect the environment.⁵⁴

The Health and Safety in Employment Act 1992 is the overarching legislation relevant to workplace health and safety. The object of the Act is to promote the prevention of harm to all persons at work or in a place of work. It requires employers to take all practicable steps to ensure the safety of employees while at work.

The Health and Safety in Employment (Petroleum Exploration and Extraction) Regulations 2013 establish requirements specific to the oil and gas industry. The regulations contain a general duty to take all practicable steps to ensure that the installation is safe for any person on or near it and that all work and other activities carried out on the installation are carried out in a manner that is safe for any person on or near the installation. There are also specific duties pertaining to safe operations.

These regulations also require:

- Each offshore installation to have a safety case approved by the High Hazards Unit
- Each offshore installation and its key systems to be annually certified by an independent certification agency
- An independent well examiner to examine every exploration or production well over its full lifecycle (i.e. from design to abandonment)

An emergency response plan for the installation must be prepared. This is a plan for responding to emergencies that occur while petroleum workers are working on an installation. The emergency response plan must identify:

- People authorised to set emergency procedures in motion
- Person in charge of co-ordinating the emergency response
- Foreseeable conditions or events that could bring about a major accident
- Actions that could control those conditions or events or limit their consequences
- Arrangements for limiting the risks to persons on or near the installation
- Arrangements for training petroleum workers in the duties they will be expected to perform in an emergency

Oil spill response

The Maritime Transport Act requires Maritime New Zealand and regional councils to respond to a marine oil spill (Part 23). New Zealand has a three-tiered approach to marine oil spill preparation and response. Industry (Tier 1), regional councils (Tier 2) and Maritime New Zealand (Tier 3) have clear roles and responsibilities under the Act.

Tier 1 plans are prepared by industry and are site-specific. Tier 2 plans are the regional marine oil spill contingency plans that regional councils must produce. Regional councils must respond to marine oil spills within their jurisdiction that exceed the clean-up capability of the Tier 1 response. Regional marine oil spill contingency plans are approved and audited by Maritime

Area	Options for mitigation
Exploration	Ensure that sufficient research has been undertaken to identify what species are likely to be present in the area and their distribution Avoid areas frequented by threatened or engendered marine species such as the Māui's dolphin and New Zealand sea lion Follow the Department of Conservation's Code of Conduct for minimising acoustic disturbance to marine mammals from seismic survey operations Adapt or schedule seismic surveys to avoid seasonal migrations or key breeding locations Avoid anchoring on seabed structures of environmental or other significance Check, clean and apply non-toxic antifouling paints to infrastructure that is brought in from overseas Undertake sufficient monitoring so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner For any exploratory well drilling, consider the options outlined for production well drilling
Production	Choose a site for platform placement after an environmental impact assessment has been undertaken Locate drilling platforms as far away as possible from sensitive species or habitats, avoiding areas of high ecological value including areas of high endemism Avoid areas with outstanding biodiversity, natural character or landscape values Avoid significant shellfish beds, fish spawning areas and areas of high fish density Avoid areas of recreational significance, including nationally important surf breaks Ensure well integrity is regulated for the purpose of protecting the environment Purchase and retire appropriate commercial fish quota for the duration of the production phase Limit unnecessary lighting and ensure that lights that are required are directed vertically onto work surfaces Reduce levels of toxic components in fluids by removing or recycling them before discharge Ensure any dredge spoil is only dumped in locations where impacts will be minimal In sensitive environments, collect all waste from drilling operations, and store on board for disposal on land Re-inject gas to decrease the volume of gas that might be otherwise flared (reducing emissions of greenhouse gases) Deploy robust safety features, such as automatic shut-down valves, to reduce the risk of a significant oil spill Ensure adequate systems and training are in place to minimise oil spill risk Treat sewage prior to discharge Check, clean and apply non-toxic antifouling paints to infrastructure that is brought in from overseas Comply with the requirements of the craft risk management standards Minimise greenhouse gas emissions through careful management of sources, including power generating equipment, flaring and fugitive fuel emissions Undertake sufficient monitoring so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner Involve tangata whenua and stakeholders in overseeing monitoring programmes Make monitoring information publicly available in a timely fashion
Decommissioning	Leave a clean seafloor following decommission Ensure that the well is decommissioned properly and formations are sealed off permanently

Figure 13.10 Possible mitigation options for managing the potential environmental impacts of oil and gas exploration
(Adapted from the World Petroleum Organization)⁵⁸

Response: Oil spill preparedness and response recommendations

- Creation of a joint industry project to improve co-ordination between key stakeholders (oil spill response organisation, governments and companies) internationally and with all responsible representative bodies
- Establish principles on dispersant use with key stakeholders
 - document the effectiveness and the value of the use of surface, aerial and, in particular, subsea dispersants
 - recommend how best to deploy dispersants
 - obtain regulatory pre-approval for their use
- Promote research to advance understanding and response methodologies and risk assessment models
- Enhance existing recommended practices for *in situ* burning (controlled burning of oil that has spilled)

Final conclusions

- Action on prevention will yield the most effective outcomes
- Implementation of the following recommendations by member companies will improve industry prevention and intervention capabilities
 - Rigorous adherence to standards and practices is central to prevention
 - With specific enhancements, current standards will be effective if applied and observed rigorously. We strive to ensure greater alignment between standards internationally
 - Verification of adherence to standards and practices is required
 - Transparency in sharing and expediting learning from well incident analyses is required
- An ongoing commitment from the industry is required to work together for continuous improvement in skills development, competency and a culture of compliance
- Implementation of proposed response research and development activities will lead to the development of new response technologies and techniques, as well as a better understanding of and more effective application of existing technologies and techniques
- Active co-ordination and collaboration with international initiatives and national oil industry associations is essential

Figure 13.11 Key recommendations of the Global Industry Response Group (*International Association of Oil and Gas Producers, 2011*)



The 2011 report by the Global Industry Response Group provided important recommendations for the oil and gas industry (*International Association of Oil and Gas Producers*)

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Components of major marine developments

Large fixed structures

A wide variety of structures have been placed in the coastal marine area. These include relatively small objects such as moorings and navigational aids, as well as larger fixed structures that are rigidly attached to the seabed, such as wharves, breakwaters, seawalls, causeways and groynes. Some developments, such as ports and marinas, require the construction of a large number of structures within a confined location of the coastal marine area together with extensive development above mean high water springs.

A marina development is likely to include numerous piles and floating pontoons to create marina berths and fuelling facilities, breakwaters to protect the boats from wave damage, slipways, haul-out facilities and washdown areas for boat maintenance, boat ramps, and onshore facilities such as hard stand areas, administration and toilet blocks, shops, cafes and car parks.

Canal developments typically include seawalls around the edges of the canals, piles and floating pontoons to provide berths adjacent to the canal-side homes and may include bridges or causeways to provide pedestrian and/or vehicle access across the canals. Onshore structures typically include houses, apartments and commercial developments. A marina may be included as an integral part of a canal development.

Tidal energy can be exploited by building semi-permeable barrages across estuaries or by harnessing offshore tidal streams.⁵ In 2011, a 10 year consent was granted to construct a marine tidal turbine power station in the mouth of the Kaipara Harbour. The consent provided for 200 completely submerged marine tidal turbines with a maximum generating capacity of around 200MW to be located underwater, but it has yet to be constructed.

Dredging and disposal

Dredging of the seabed is commonly undertaken to increase, maintain or restore navigable water depths in the coastal marine area, particularly in association with ports, marinas and canal developments. There are two kinds of dredging: capital dredging which aims to physically change the depth of the seabed permanently; and maintenance dredging which aims to maintain a depth of the seabed. Figure 14.1 shows the different activities for which dredging consents have been granted in the Auckland region between 1994 and 2012.

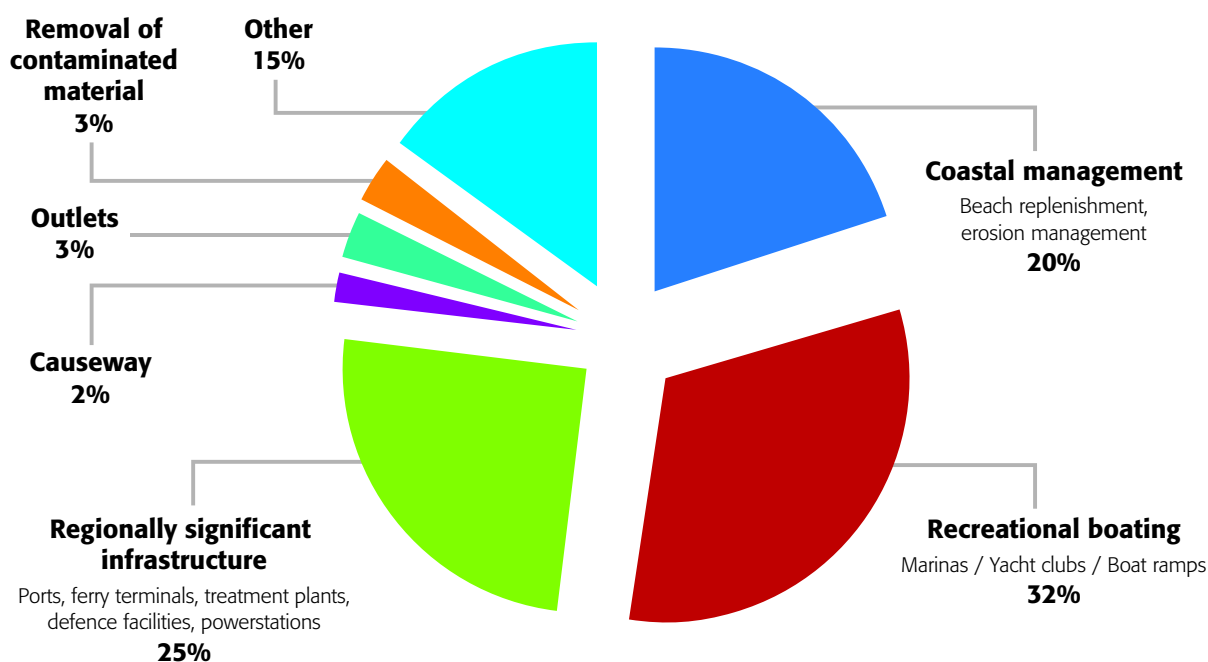


Figure 14.1 Categories of dredging in the Auckland region 1994 to 2012 (Kelly S and S Faire, 2012)

Potential environmental effects

Major marine infrastructure can create positive economic and social benefits, through enabling transport and trade, and facilitating public access to, and enjoyment of, the marine environment. At the same time, because of the large scale of many of these projects, there can be significant adverse environmental effects on the marine environment which need to be addressed. These are summarised in Figure 14.2 and discussed in further detail below. Best practice ways of reducing these effects are discussed later in the chapter.

Area	Potential negative impact
Degradation of natural coastal landscapes	• Significant visual intrusion • Replacement of natural shapes and forms with hard lines and edges • Disruption of natural coastal patterns and processes • Domestication of the natural coastal landscape
Damage to marine ecosystems	• Direct loss of coastal habitat • Disruption of sensitive and/or ecologically productive ecosystems and transition zones • Damage to habitats of birds and intertidal fish and animals • Reduction in natural flushing, potentially leading to eutrophication • Shading areas of water resulting in the longer-term loss of marine plants • Removal of all marine life from parts of the seabed • Permanent loss of slow-growing, sensitive species which are unable to recover • Release of sediment into seawater which reduces light penetration, disrupts juvenile species and filter feeders, and smothers benthic communities over a wide area • Release of organic-rich sediments into seawater which can exacerbate algal blooms • Release of metals, chemicals, dioxins, organochlorines and polychlorinated biphenyls into the marine environment which can impact on the health and viability of marine species (such as where dredging of contaminated sediments is undertaken)
Changes to coastal processes	• Disruption of wave energy, currents and tidal flows • Alteration of sediment transport regimes causing coastal erosion or sediment accumulation • Alteration of seabed bathymetry • Alteration of particle sizes entering the marine environment, making an area more muddy than previously, and therefore changing the habitat
Degradation of water and sediment quality	• Discharge of sediment or pollutant-laden water which can cause sedimentation and contamination of the wider marine environment • Increased levels of rubbish, oil and antifoul paints entering the water through increased use of an area
Disrupting the relationship of tangata whenua with the marine environment	• Contamination of water and sediment which can render kaimoana unsafe to eat and reduce the size of harvestable populations • Loss of intertidal areas, which reduces the areas where kaimoana can be harvested • Loss of access to, or destruction of, culturally important sites
Degradation of marine heritage	• Damage to the integrity of archaeological sites, wāhi tapu and other sites of historical significance
Loss of public access and amenity	• Restriction or exclusion of public access to parts of the coastal environment • Loss of amenity through increased noise, light and traffic

Figure 14.2 Potential adverse environmental effects of major developments

such shading will depend on the habitat affected, but most plants such as mangroves, seagrasses and seaweeds will deteriorate or die if shaded for significant periods.

One of the main concerns associated with carbon sequestration is the potential for leakage of CO₂ both during transportation and after storage. These leakages could be either abrupt or gradual. Their environmental impacts are not well understood, but is believed to be dependent upon the scale of the sequestration, and the location, geographical extent and prevailing conditions within the local marine environment.⁹



The causeway across the Moanaanuanu Estuary at Whangamatā has reduced the speed of water flowing out of the harbour, resulting in increased levels of sediment being deposited within the harbour, which in turn have supported the expansion of mangroves (Raewyn Peart)

Dredging and marine disposal

The main environmental impacts from dredging and marine disposal are related to the:¹⁰

- Direct disturbance and physical changes to the seabed
- Suspension of sediment and associated changes in water quality
- Release and remobilisation of contaminants on or in the seabed
- Changes in local hydrodynamics and settlement patterns of sediment

Capital dredging necessarily removes flora and fauna from the seabed and increases its depth. With the exception of deep-burrowing animals, the activity will completely remove all life from the seafloor for a period of time. The recovery rate of the benthic communities will depend on the particular nature of the ecosystem: some ecosystems dominated by opportunistic species may recover in only a few months, whereas those populated by slow-growing, sensitive species may take years, or may never recover.

Maintenance dredging usually involves frequent removal of material and the amount removed differs considerably depending on the site and type of development. For example, Pine Harbour maintains the channel by dredging 3,000 cubic metres per year, Westpark Marina dredges up to 10,000 cubic metres per year and the Lyttelton Port maintenance dredging has varied between 290,000 and 800,000 cubic metres per year over the past 10 years.

Dredging and disposal can also discharge sediment into the water column:

- As the seabed is disturbed by the dredge
- When sediment overflows from hoppers and barges carrying the dredged material
- As the sediment is dumped at a marine disposal site
- By resuspension of material at the disposal site

Some dredged material can be re-used, such as for beach renourishment or to fill a new reclamation. For example, Ports of Auckland disposes of dredgings into the Fergusson reclamation where the dredged material is mixed with cement to make mudcrete.¹¹ Where such options are not available, land disposal is typically costly and logistically difficult. For this reason, marine disposal is often the option of choice.

One of the significant risks of marine disposal is that the dredgings will be resuspended due to wave action and will be redistributed over a wide area. Where sediment is released into the water column, it can affect the clarity and turbidity of the water, potentially reducing light penetration. The suspended material may eventually settle on areas around the dredged site, blanketing the benthic animals and plants that live there. The effects of this will vary, depending on the resilience of the communities and the conditions that they are used to, but it may cause stress, reduced rates of growth or fatalities. Suspended sediment from maintenance dredging is likely to have less impact than that from capital dredging, due to the already disturbed nature of the site.¹²

Coastal processes

Structures such as jetties, breakwaters, seawalls, reclamations and groynes may alter wave energy, currents and tidal flows. This, in turn, has the potential to change the transport of sediments along the coast, resulting in coastal erosion in some areas and higher levels of sedimentation and accretion in others. Reclamation inside harbours and inlets also reduces water space, and consequently the volume of water that moves in and out of the area on each tide, thereby affecting tidal flows and flushing. This is sometimes referred to as the tidal compartment.

The size, shape and orientation of structures relative to the shoreline and natural water flows will determine their hydrological and sedimentation effects.



Concerns have been raised that dredging the access channel to the Whangamatā Marina has changed the coastal dynamics of the area and degraded the Whangamatā surf break (Raewyn Peart)

Impact of Timaru breakwater

A notable example of how a structure can change sedimentation flows is the breakwater that was constructed as part of the Timaru Port in the late 1870s. This resulted in major erosion at Washdyke but also in the creation of a new beach at Caroline Bay. Before works to construct Timaru Port began, Caroline Bay was moderately deep, with narrow mixed sand. Now approximately 30,000 cubic metres of sand has accumulated annually in the shelter of the North Mole, and the shoreline has advanced seaward by more than 650 metres.¹⁷

Dredging can also alter the processes that govern the movement of sediments in the coastal marine area. Dredging for capital works has the potential to cause a significant change to the equilibrium of the system, depending on the circumstances. In some situations this can alter the bathymetry of the seabed, impact on the flow of seawater and have an effect on wave patterns, with consequential effects on areas such as surf breaks (this is an issue that was raised in connection with the marina at Whangamatā).

There are concerns that the construction and operation of marine tidal turbines could impact on local tides. For example, it was estimated that in the Bay of Fundy, tidal power plants could decrease local tides by 15 centimetres. However, the environmental impact assessments that have been completed indicate that impacts are site-specific and depend greatly upon local geography.¹⁸

Water quality

During the construction phase of a marine development, developers are generally required, as part of their resource consent, to use best practice preventative measures including silt curtains, settlement ponds, temporary retaining walls, temporary dry docking and new technologies. These measures are intended to reduce the risk of sediment or any potential contaminants being introduced into the wider marine environment. Compliance with these consent conditions is managed by unitary and regional councils.

After construction has been completed, the potential for degradation of water quality surrounding the structure will depend on what it is used for. Wharves, jetties and marinas typically attract large numbers of people and boats that, in turn, have the potential to create discharges into the marine area – including fuel spillage, sewage from boat toilets and rubbish. To help address this, Westhaven Marina has a ‘no discharge’ policy within the marina, which is discussed further in the first case study at the end of this chapter.

The parking and boat washing areas found at marinas may also be a source of contaminated runoff into the sea if they are not appropriately managed. In 2012 a project was undertaken by NIWA for the EPA which looked at the levels of copper leaching from antifoul paints. This research indicated that in many of New Zealand’s marinas copper concentrations could be above guidelines for the protection of marine life.¹⁹ Since then, the EPA has carried out a reassessment of antifouling paints²⁰ and the Ministry for Primary Industries has prepared draft antifouling and in-water cleaning guidelines.²¹

The location of wastewater treatment plants in the coastal environment has caused much concern among coastal Māori. A Cultural Impact Assessment of the Matata Wastewater Scheme prepared in 2014 identified significant concerns for Tangihia Hapū. This included the disposal of partially treated wastewater in a wāhi tapu area where the bones of tupuna were buried, and the location of the treatment plant in proximity of the Rangitihi Marae, both of which were considered to be culturally unacceptable.²⁹



The construction of seawalls, such as the one shown here at Waihi, can result in the loss of intertidal areas and places where kaimoana can be harvested (Raewyn Peart)

Public access

Structures built along and near the coastal edge can enhance public access (through boat ramps and jetties, as well as walkways around buildings) and also obstruct it (through restrictions to access both to the marine area and to areas within it). Port developments typically exclude public access to the coastal edge for safety and security reasons. However, some ports provide organised educational tours for school groups and interest groups. Marina and canal developments vary in the extent to which they provide for public access.

Auckland waterfront promenade

The waterfront walkway and cycleway development in Auckland will create 6 kilometres of continuous walkway and cycleway across the waterfront to encourage walking and cycling, increase access to the coastline, and provide sightlines to the water (e.g. Daldy Linear Park) while protecting, where appropriate, the commercial use of the water's edge.³⁰



Waterfront Auckland is currently developing a water's edge promenade and cycleway around Westhaven Marina (Waterfront Auckland)

Public access to marina berths is often excluded, particularly at night, for security reasons. The marina berths themselves are occupied by licencees, and are not freely available to other boat users, although some may be available to visiting boats for short-term periods. Some marinas incorporate public walkways, boat ramps, parking, picnic areas and cafes as part of the complex, and this enables members of the public to access and enjoy the coastal environment.

Public access is typically not available around the coastal edge of canal developments as privately-owned residential properties extend right down to the coastal edge. The waterways themselves are usually in public ownership and open to all users. Some canal developments also provide small public parks alongside sections of the canals.³¹

There has been considerable controversy over the extent to which members of the public should have access to private wharves and jetties. The general legal principle is that unless a coastal permit expressly limits public access, and such a limitation is necessary to give effect to the permit, the permit holder cannot exclude the public from the coastal marine area.³² The situation

Decision-making framework

Resource Management Act

Under section 12 of the RMA, a range of activities in the coastal marine area are restricted. No person may undertake the following activities unless they are allowed by a national environmental standard, a regional coastal plan as well as a proposed regional coastal plan (if there is one), or a resource consent:³⁶

- Reclamation or drainage of the foreshore or seabed
- Construction, alteration or demolition of a structure in the coastal marine area
- Disturbance of the foreshore or seabed (e.g. by excavation or drilling) in a manner that is likely to have an adverse effect (except for lawfully harvesting a plant or animal)
- Deposition of substances that are likely to have an adverse effect on the foreshore or seabed
- Introduction of any exotic or introduced plant in, on, or under the foreshore or seabed
- Generally destroying, disturbing or damaging any part of the foreshore or seabed which is likely to have an adverse effect on historic heritage
- The occupation and use of space in the common marine and coastal area
- Carrying out activities within the coastal marine area

The regional coastal plan, required to be prepared by regional councils and approved by the Minister of Conservation, determines whether a major marine development in the coastal marine area will require consent, and what kind of consent will be required. In practice, all major developments in the marine area are likely to require resource consent, because of the potential significance of their effects. When considering a resource consent application, the consent authority must have regard to the actual and potential effects on the environment of allowing the activity (section 104) including the provisions of the NZCPS and applicable regional policy statement and regional coastal plan.



All major marine developments, including the construction of reclamations, canals, breakwaters and jetties as shown here in the Whangarei Harbour, will likely require a resource consent under the RMA (Raewyn Peart)

Port zones in district plans and regional coastal plans are one mechanism that can be used to address the needs of a port (and where, how and when to provide for the port) as part of an integrated and strategic approach to the management of the coastal environment.⁴⁰ The Port Zone provided in the Bay of Plenty Proposed Regional Coastal Plan 2014 provides for the current operational needs of the Port of Tauranga as a matter of priority while 'limiting the effects of those activities on cultural values and the environment'. It also provides for the 'future expansion and operational needs of the Port of Tauranga and its shipping channels' in 'appropriate locations, having regard to the potential adverse effects on the environment'.⁴¹

Reclamation

Reclamation is generally discouraged by Policy 10 of the NZCPS unless it results in regional benefits. The NZCPS directs that reclamation be avoided in the coastal marine area unless land is not available, the activity can only occur in the coastal marine area, there are no practicable alternative methods to provide for the activity, and the reclamation will provide significant regional or national benefit.⁴² The Bay of Plenty Proposed Regional Coastal Plan recognises that reclamation, which is identified in the 'Outline Development Plan Port of Tauranga', is appropriate in terms of specific Port Zone policies and Policy 10 of the NZCPS, 'provided that any adverse effects are appropriately managed, including by use of off-site mitigation'.⁴³

Particular care will be needed to determine which parts of a marine development need to be located at the land–water interface, and what the safety and efficiency effects would be of alternative locations.⁴⁴ Where a reclamation is considered suitable, the design and form of the structure should have regard to the effects of climate change and sea-level rise, and avoid the use of contaminated materials. De-reclamation of redundant land is encouraged only where it would help restore the natural character and provide for more public space.

NZCPS, Policy 10: Reclamation and de-reclamation

1. Avoid reclamation of land in the coastal marine area, unless:
 - a. land outside the coastal marine area is not available for the proposed activity;
 - b. the activity which requires reclamation can only occur in or adjacent to the coastal marine area;
 - c. there are no practicable alternative methods of providing the activity; and
 - d. the reclamation will provide significant regional or national benefit.
3. In considering proposed reclamations, have particular regard to the extent to which the reclamation and intended purpose would provide for the efficient operation of infrastructure, including ports, airports, coastal roads, pipelines, electricity transmission, railways and ferry terminals, and of marinas and electricity generation.
4. De-reclamation of redundant reclaimed land is encouraged where it would:
 - a. restore the natural character and resources of the coastal marine area; and
 - b. provide for more public open space.



The NZCPS recognises the importance of ports, such as Port Chalmers shown here, and the need to provide for their safe and efficient operation as well as future development (Raewyn Peart)

NZCPS, Policy 23: Discharge of contaminants

1. In managing discharges to water in the coastal environment, have particular regard to:
 - a. the sensitivity of the receiving environment;
 - b. the nature of the contaminants to be discharged, the particular concentration of contaminants needed to achieve the required water quality in the receiving environment, and the risks if that concentration of contaminants is exceeded; and
 - c. the capacity of the receiving environment to assimilate the contaminants; and
 - d. avoid significant adverse effects on ecosystems and habitats after reasonable mixing;
 - e. use the smallest mixing zone necessary to achieve the required water quality in the receiving environment; and
 - f. minimise adverse effects on the life-supporting capacity of water within a mixing zone.
4. In managing discharges of stormwater take steps to avoid adverse effects of stormwater discharge to water in the coastal environment, on a catchment by catchment basis, by:
 - a. avoiding where practicable and otherwise remedying cross contamination of sewage and stormwater systems;
 - b. reducing contaminant and sediment loadings in stormwater at source, through contaminant treatment and by controls on land use activities;
 - c. promoting integrated management of catchments and stormwater networks; and
 - d. promoting design options that reduce flows to stormwater reticulation systems at source.
5. In managing discharges from ports and other marine facilities:
 - a. require operators of ports and other marine facilities to take all practicable steps to avoid contamination of coastal waters, substrate, ecosystems and habitats that is more than minor;
 - b. require that the disturbance or relocation of contaminated seabed material, other than by the movement of vessels, and the dumping or storage of dredged material does not result in significant adverse effects on water quality or the seabed, substrate, ecosystems or habitats;
 - c. require operators of ports, marinas and other relevant marine facilities to provide for the collection of sewage and waste from vessels, and for residues from vessel maintenance to be safely contained and disposed of; and
 - d. consider the need for facilities for the collection of sewage and other wastes for recreational and commercial boating.

Coastal hazards

The extent to which a major marine development will affect and be affected by coastal processes, including natural hazards, erosion and inundation, will also be considered by a consent authority. The NZCPS requires that applications avoid land use changes or development which would increase the risk of adverse effects from coastal hazards.

Managing public access

The NZCPS directs decision-makers to maintain and enhance public walking access to, along and adjacent to the coastal marine area, including by avoiding, remedying or mitigating any loss of public walking access resulting from subdivision, use or development.⁴⁹ As a result, applicants will need to carefully consider how public access to the coastal marine area can be provided for despite the development. There are specified circumstances when imposing restrictions is appropriate. Within some parts of a port, restricting public access along the waterfront may be necessary to



The impact of several smaller structures in the coastal marine area taken together can have the most effect on natural character values of the area. This is shown here at Mangonui where a coastal road and seawall, reclamation, wharf and several buildings overhanging the water collectively significantly reduce the natural character of the coastal fringe (Craig Potton)

Potential adverse environmental effects	Best practice response
Degradation of natural coastal landscapes	• Locate reclamations and structures within the coastal marine area only if they are absolutely necessary and there is no practical alternative • Locate reclamations and structures where natural character and landscape values have already been compromised • Seek opportunities for restoring or enhancing areas that have been compromised • Design reclamations and structures to visually 'fit in' with the natural coastal environment – by minimising their size and where possible adopting natural shapes
Damage to marine ecosystems	• Locate marine developments away from ecologically important marine habitats such as salt marshes, shell banks, seagrasses, shellfish beds, mangroves, fish spawning areas and bird roosting, feeding and nesting sites • Locate marine developments away from any threatened, at-risk or rare species or ecosystems • Design marine developments to minimise impacts on natural coastal processes including sediment transport, tidal flows and local currents • Minimise shading of marine habitat in areas where mangroves, seagrasses or seaweeds are present • Locate and design marine developments to minimise the amount of capital and maintenance dredging required • Time dredging operations to minimise disturbance to sensitive marine animals • Follow the principles of disposal in the London Convention 1972 Annex III⁵² – including considering carefully the characteristics of the material being disposed, the disposal site and method (such as rate of disposal, dilution and dispersal characteristics, other materials already dumped in the area and water characteristics) and the possible effects of the disposal (on marine life, other uses of the sea and options for land-based disposal) • Dispose of dredgings on land as a first preference • Use clean sand dredgings to replenish local beaches • If there is no alternative to disposing of dredgings at sea, ensure that the disposal site is deep enough to avoid any resuspension of sediment in the coastal environment, preferably over the edge of the continental shelf • Minimise the number of dumping sites, by identifying a suitable site which can be used for dumping from a range of dredging activities within a region • Undertake frequent monitoring of the site to ensure that sediment is not dispersing outside the site boundaries
Changes to coastal processes	• Avoid reclamations unless there is no practicable alternative and the reclamation will provide significant regional or national benefit • Where reclamations occur, minimise their size and consider the shape of the reclamation and materials used to ensure they are compatible with the coastal environment • Carefully design reclamations and coastal structures in terms of size, shape and orientation to minimise impacts on natural seawater flows • Locate and design marine developments to minimise the amount of capital and maintenance dredging required • Where possible, accommodate dredged material within the reclamation, to avoid the need to dispose at sea

Westhaven Marina development

Westhaven Marina, located in the centre of Auckland by the Harbour Bridge, is the largest marina in the southern hemisphere. St Mary's Bay became a popular place for yachting during the late 1800s and shortly after the turn of the century the Ponsonby Cruising Club established premises on the shores of the bay. A solid breakwater was constructed along the western end of the bay during the 1930s to provide better shelter for the yachts moored there, which were being damaged in storms. Several yacht clubs established premises on what became known as the Western Reclamation. By the 1980s the northern side of the marina was full and building started on the southern side to cater for ever increasing demand. Today there are more than 2,000 boats moored at the marina with berths provided from 10 to 30 metres long. Four yacht clubs and 50 commercial charter operators are also based there.⁵⁵

Waterfront Auckland has recently developed a plan for how Westhaven Marina is to be improved for the future. This plan comprehensively covers the land- and water-based development in the marina that is planned to occur over the next 30 years.⁵⁶ There are two drivers for the plan, being a desire to improve use of the public edge and also to ensure the best use of water space.

The vision for Westhaven Marina is 'to be the most successful marina in the Southern Hemisphere and home to a globally renowned marine industry. An iconic Auckland destination that is vibrant, accessible and attracts people to the sea'. The Westhaven Plan provides for a range of multiple uses, including boat parking, public access and improved environment. It is based on a number of design principles, as shown in Figure 14.4, and a range of measures to manage environmental impacts (discussed below).

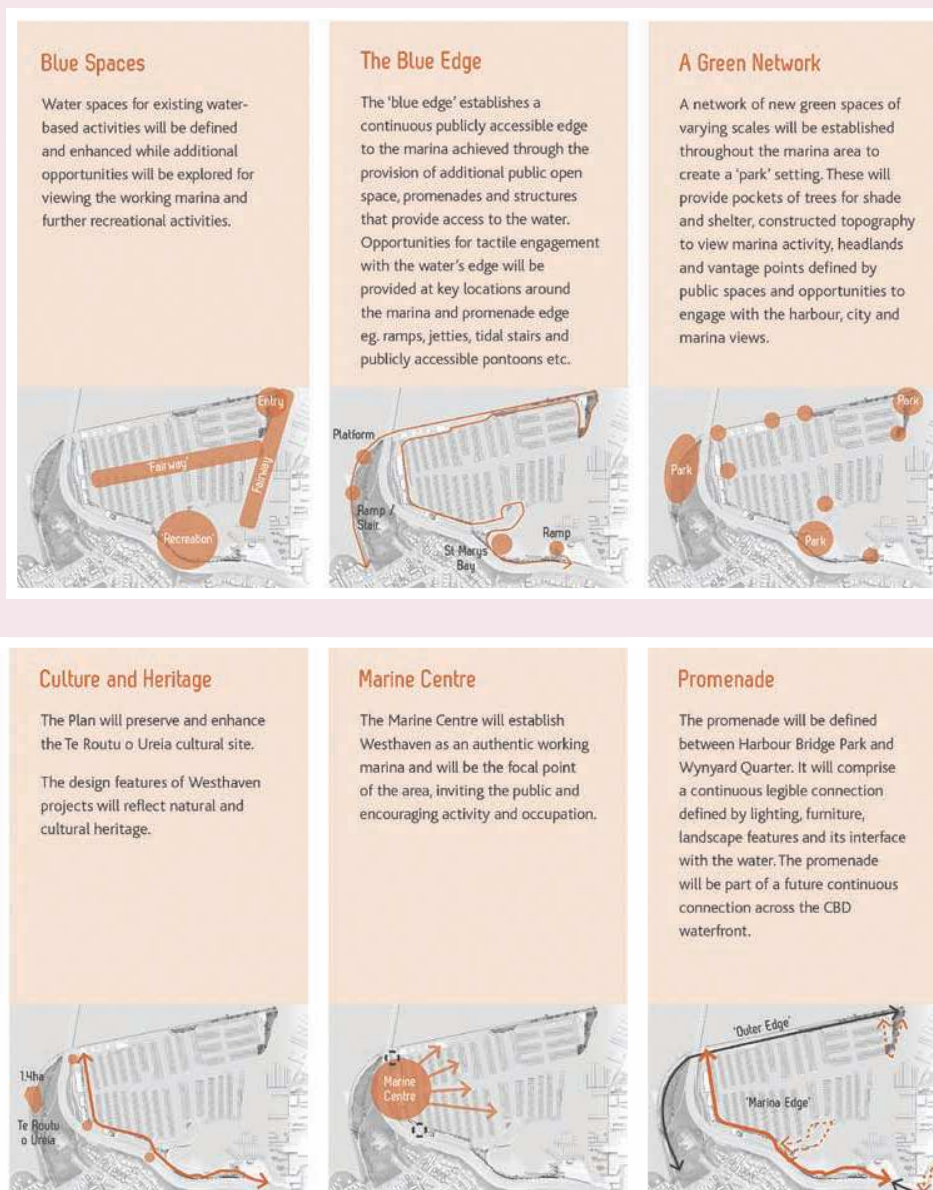


Figure 14.4 Westhaven Marina design principles (*Waterfront Auckland, 2013*)

Mangere sewage ponds restoration

The Mangere Wastewater Treatment Plant on Auckland's Manukau Harbour once supported the largest oxidation ponds in the world, covering more than 500 hectares of intertidal foreshore, and processing all domestic and industrial waste produced within a 40 kilometre radius.⁵⁷ With increasing volumes of waste from the fast-growing Auckland urban area, stress was placed on the plant, and this resulted in the need for a substantial upgrade.

With growing public awareness, and a need to future-proof the facility against the region's inevitable growth, Watercare Services Limited embarked upon a period of extensive public consultation. This resulted in Project Manukau, New Zealand's largest marine restoration effort.⁵⁸ The project, supported by the public, ran from 1998 till 2005 when the foreshore was re-opened to the public as a coastal walkway.

The primary objective of the project was the removal of the oxidation ponds and restoration of 13 kilometres of coastline. Oruarangi Creek and Mangere Lagoon, which had been closed off when the old sewage site was built, were reopened allowing them to flow into the sea for the first time in over 40 years.⁵⁹ The length of foreshore was reconnected to the sea, restoring it back to being part of the intertidal zone. Extensive areas of white shell beaches were constructed, extending along more than 13 kilometres of coastline, along with more than 270,000 native shrubs and trees which were planted with support from local community groups.

Water quality in the area has improved considerably, with a notable decline in bacteria levels, to the point where it is now deemed to be of swimming quality. Odours and midges (which had plagued the area whilst the ponds were in operation) disappeared.⁶⁰ Pipi beds re-established themselves as the estuarine ecosystems were restored. Previously destroyed fishing grounds have gradually replenished as aquatic life once again becomes established in the area. The native bush has provided habitat and food to an increasing number of wildlife and birds, particularly waders, which have been drawn back to the area by the expanse of beaches.

To further support the returning birdlife, purpose-built roosting sites were placed in the harbour. These have attracted native and migrating birds, turning the area into an internationally-recognised roosting and feeding area. The area is now inhabited by 80 bird species, including a breeding flock of 12 New Zealand dotterels of which there are only 2,000 left in the country. In peak season, the number of birds swell to an estimated 20,000 individuals, with half the world's wrybill population using Mangere Bridge to roost.⁶¹ An extensive network of paths has been created for the public to enjoy the natural surroundings and rich diversity of life that has been restored.



The restoration of the Mangere sewage ponds included the construction of extensive areas of shell beaches (Raewyn Peart)

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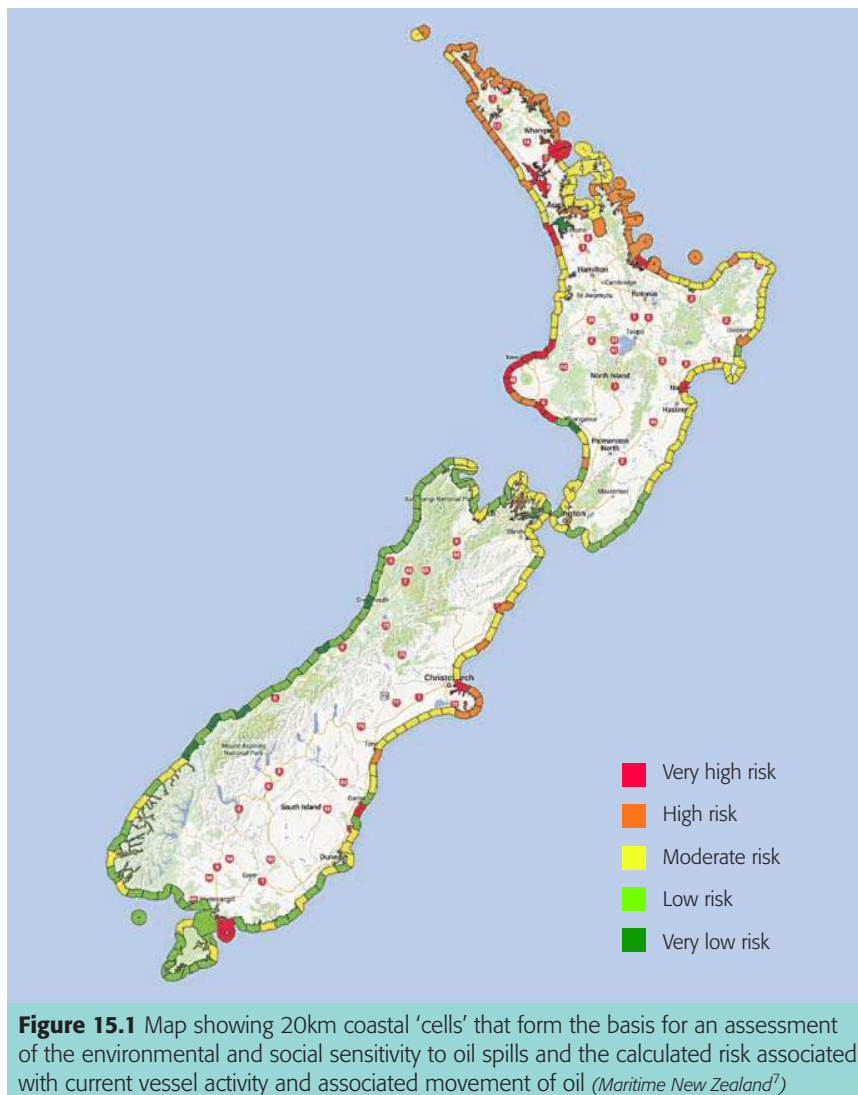
Pollution

Oil and chemical spills

Every year there are spills of diesel, petrol, oil and other toxic chemicals into the sea. For example, in 2006, over 80 spills were reported. Major oil spills can occur when vessels collide or run aground and hole their tanks, although the risk is reduced if vessels have double hulls. Spillages can also occur when oil cargoes are transferred. Recreational vessels can also release oil into the marine environment when oily water is pumped from bilges and when fuel tanks overflow during refuelling.

There have been several significant spills in New Zealand waters over the last two decades:³

- 1998: *Don Wong 529* in Stewart Island (400 tonnes of automotive gas oil)
- 1999: *Rotoma* in the Poor Knights Islands (~7 tonnes of oily bilge discharge)
- 2000: *Sea Fresh* in the Chatham Islands (60 tonnes of diesel)
- 2002: *Jody F Millennium* in Gisborne (25 tonnes of fuel oil)
- 2011: *Rena* in Tauranga (~350 tonnes of heavy fuel oil)



Rena oil spill

The last big oil spill in New Zealand occurred in 2011, when the *M.V. Rena* struck Astrolabe Reef 12 nautical miles off Tauranga, and grounded. The vessel was carrying 1,368 shipping containers and 1,733 tonnes of heavy fuel oil at the time of the grounding. Around 350 tonnes of oil and 500 containers entered the marine environment. As a result of the disaster, a total of 2,410 dead birds were collected, of which 1,448 were oiled.⁴

Oil discharged into the marine area can have serious impacts on wildlife and habitats, depending on where the oil ends up. For example, oil spills in the open sea may cause little ecological damage as the oil is broken down naturally by micro-organisms (such as bacteria and fungi) into non-hazardous substances. Sunlight, wind and wave action also helps to break oil down. Conversely, major ecological impacts can occur if an oil spill moves onto the shoreline. For example, an oil spill in a sheltered environment such as an estuary means the oil can become trapped and seriously impact mangroves, worms, molluscs, juvenile fish and crustaceans, as well as contaminating shellfish beds.

Birds are particularly susceptible as oil damages the waterproofing on their feathers and, as a result, they can get cold and are prevented from swimming, floating or flying. The toxic components of oil can also cause serious or fatal illnesses and can contaminate the habitat and feeding grounds of coastal seabirds.

Dumping rubbish

Concerns regarding the impact of rubbish dumped in the marine environment have been increasing. A major threat to marine wildlife is plastic debris, mainly due to this being eaten or causing entanglement. It has been estimated that over a million birds and 100,000 marine animals, including mammals and turtles, die each year from plastic debris worldwide. A vast majority of plastic waste in the New Zealand marine environment comes from land; however rubbish discharged from vessels also has an impact. This issue is discussed further in Chapter 16: Marine tourism and recreation.

Harm to marine animals and birds

Ships moving through the water can hit marine mammals and fish and injure or kill them. Ship strike happens because dolphins and especially whales are moving much slower than vessels and spend considerable time on the surface, which can put them directly in the path of boats. The greatest number of marine mammal deaths from ship strike is caused by large ships travelling at more than 12 knots.

Bryde's whale ship strike

Ship strike is a particular problem in the Hauraki Gulf because it houses one of the busiest ports in the country and it is also important habitat for dolphins and the nationally critically threatened Bryde's whale. Between 1997 and 2007, 25 Bryde's whale deaths were recorded in the Gulf, of which 14 were found to have died from injuries consistent with collision with a large vessel.⁹ On average about two Bryde's whales a year are found dead from ship strike. The Hauraki Gulf Transit Protocol for Commercial Shipping is a voluntary measure agreed between the Ports of Auckland and the shipping industry. Vessels are asked to slow down to 10 knots while transiting the Gulf. If widely adopted, this should reduce the number of whale deaths caused by vessels. This is discussed further in the first case study below.

Vessels, including container ships, generate underwater noise, which can impact on the hearing system of marine mammals, both in enclosed waters and the open ocean. Human-produced noise has been shown to displace killer whales.¹⁰ Research has shown that the way marine mammals behave in response to noise varies considerably between the different species, as well as between individuals, sex and age classes. The context of the noise is also important.¹¹



Vessels can cause harm to marine mammals, such as the orca shown here in the Bay of Islands, through hull and propeller strike and the generation of underwater noise (Raewyn Peart)

Decision-making processes

Management of marine pollution in New Zealand is undertaken under two international conventions and several Acts and Regulations.

International conventions

The International Convention for the Prevention of Pollution from Ships (MARPOL) is the main international convention covering prevention of pollution in the marine environment by ships from operational or accidental causes. The Convention was adopted on 2 November 1973 and has been updated by amendments through the years. It includes regulations aimed at preventing and minimising pollution from ships – both accidental and that from routine operations. As well as addressing such matters as accidental oil spills from tankers, the Convention controls pollution caused by the discharge of chemicals, harmful substances, sewage and garbage. In addition, the Convention on the Prevention of Marine Pollution by Dumping of Wastes or Other Matter prohibits the dumping of waste at sea except for a restricted group of materials. Even then, dumping is only permitted if there are no practical opportunities for re-use or recycling, and the effects of dumping on the marine environment will be minor. It is important that New Zealand is synchronised with the international maritime regulations, including MARPOL.¹⁷

Resource Management Act

The RMA addresses the discharge of pollutants into the marine area within the territorial sea under section 15. It includes provisions which:

- Prohibit the dumping of radioactive matter from any ship or offshore installation, and the storage of any radioactive waste in the sea¹⁸
- Require a resource consent to authorise the dumping of any waste from a ship or offshore installation or the incineration of any waste within the coastal marine area¹⁹
- Require regulations, a rule in a plan or a resource consent to authorise the discharge of a harmful substance or a contaminant from a ship or offshore installation into water, land or air²⁰

Rena consent application

In May 2014, the *Rena*'s owner lodged resource consent applications under the RMA to leave sections of the wreck and associated debris in place on Astrolabe Reef, and to authorise any future discharges of contaminants that may arise as a result. The Bay of Plenty Regional Council publicly notified these applications and 153 submissions were received. A Council hearing of the application has been set down for June 2015.²¹

Rules in regional coastal plans can control shipping within the coastal marine area. This includes matters such as vessel speed, discharges, dumping waste and noise. For example, the Regional Plan: Coast for Otago recognises that a major source of noise in the coastal marine area is from the operation of ships, both commercial and recreational, and provides a policy to manage the effects from this noise.



The RMA controls the discharge of pollutants into the territorial sea, including from ships such as the Interislander ferry shown here (Raewyn Peart)

and masters on how to navigate inshore waters in an effort to reduce the likelihood of an incident and subsequent pollution. Recommendations include keeping at least 5 nautical miles from the land, charted danger, or any outlying islands until reaching the position where alteration is required to make port or depart the coast. Specific routing instructions are available for approaches to 14 of the main commercial ports throughout the country. The relatively low levels of shipping traffic around New Zealand make the implementation of a formal mandatory routing system unnecessary.²⁶

In sensitive marine areas, the voluntary code can be superseded by a mandatory exclusion which provides the marine environment with the best protection from potential pollution or damage posed by shipping operations.²⁷ 'Areas to be avoided' are implemented under the UNCLOS and are recognised under international law. Vessels meeting certain criteria are prohibited from entering the waters within these areas. Within New Zealand, there are two such designated areas: the Three Kings Islands (a wildlife sanctuary off the northern tip of the North Island) and area off the north-east coast including the Poor Knights Islands.²⁸

Areas to be avoided by shipping

The Three Kings Islands were declared as an area to be avoided in 1995, and all vessels exceeding 500 gross tonnes are prohibited. The area stretching between Bream Head and Cape Brett and extending out to the Poor Knights Islands was identified as an area to be avoided in 2003. After two incidents when oily bilge water was released into the sea between the Poor Knights and the Tutukākā coast, threatening the area's world-renowned wildlife and marine ecology, the Poor Knights Islands ship restriction was set in place. A review by the Maritime Safety Association highlighted that the area has a high probability of an oil spill²⁹ due to vessels transporting oil to and from the Marsden Point Oil Refinery, and sailing to and from the Port of Auckland.³⁰ Within this area, any vessel greater than 45 metres in length is now prohibited. The only commercial ships exempt from this rule are fishing vessels that are engaged in fishing operations and barges under tow, provided the cargo is not oil or other harmful substances.

Maritime New Zealand has responsibility for marine oil spill response planning. The Marine Oil Spill Response Strategy 2006 (which is currently being updated) sets out the overarching framework for how Maritime New Zealand and its partners will respond to a marine oil spill. This is funded by an Oil Pollution Levy payable by those sectors whose activities raise the risk of an oil spill. Maritime New Zealand maintains a response capability to counter an oil spill of 3,500 tonnes. Where a larger incident occurs arrangements are in place to secure overseas assistance.³¹

Oil spill preparedness is funded by an industry levy, the Oil Pollution Levy, which is required under the Maritime Transport Act. The Levy is collected to run New Zealand's maritime oil pollution preparedness and response system. The Oil Pollution Levy applies to all commercial vessels over 100 gross tonnes and 24 metres or more in length (except those operating in freshwater), offshore oil installations, exploration wells and oil pipelines.³² The intention behind Maritime New Zealand's oil pollution preparedness and response is that the polluter pays. In circumstances where the polluter is not identified, the Levy is used to pay expenses from reserves set aside for that purpose.³³

The following process is followed to develop a new marine protection rule, or to amend an existing rule:

- Maritime New Zealand prepares a draft rule and undertakes informal consultation
- An invitation to provide written comment on the rule is advertised widely to the maritime community and general public
- Maritime New Zealand considers the written comments and prepares a revised rule which may be circulated to key stakeholders for comment before finalisation
- The Minister of Transport approves the rule and it is notified in the *Gazette* (section 448)

Figure 15.2 Process to develop a marine protection rule

Bryde's whale voluntary protocol for shipping in the Hauraki Gulf

The Hauraki Gulf is home to a small population of Bryde's whales, estimated at less than 200 individuals, and is one of only a few places in the world that can boast of a semi-resident population of a critically endangered whale species. The shipping passage through the Gulf is one of the busiest in New Zealand, used by over 1,500 ships per year berthing at the Port of Auckland, the country's premier cruise destination and primary port. On average, two Bryde's whales are killed every year as a result of vessel strike within the area, threatening the populations' long term survival.³⁴

Scientific research undertaken by the University of Auckland found that Bryde's whales spend 90 per cent of their time within 12 metres of the surface, which is the very part of the water column which ships hulls penetrate, with most large ships drawing between 10 and 12 metres.



Bryde's whale in the Hauraki Gulf, where ship strike threatens the resident population (Raewyn Peart)

The problem of vessels hitting whales also occurs overseas. Vessel strike on the Northern Atlantic Right Whale off the eastern seaboard of the USA and Canada was successfully reduced through the establishment of traffic separation schemes, vessel avoidance areas and speed restrictions.³⁵ Scientists investigated the utility of applying these approaches in the Hauraki Gulf. Because Bryde's whales are distributed throughout the Gulf, with no apparent regular seasonal changes in distribution, a traffic separation scheme was found unlikely to work. The most effective means of reducing the risk to whales was identified as being reducing the speed of vessels transiting the Gulf to 10 knots or less.

A collaborative process involving the Ports of Auckland, the shipping industry, the Department of Conservation, the University of Auckland, the Environmental Defence Society, council representatives and others was commenced in March 2012 to find ways to address the issue. The process was successful, just 18 months later, in developing a voluntary protocol aimed at reducing the number and impact of collisions between Bryde's whales and large vessels in the Hauraki Gulf.

Pre-planning and speed reduction are the critical factors for preventing Bryde's whale mortalities. The protocol, which was announced on 20 September 2013, comprises three primary approaches to reducing ship strike:³⁶

- **Passage planning** - Vessels are encouraged to use recommended shipping routes to enter and depart the port. Narrowing the area of the Gulf transited by ships reduces the risk of encountering a whale. By travelling at a maximum speed of 10 knots whilst in the Gulf, risks associated with an accidental impact are significantly reduced. There is good scientific evidence to suggest that if a whale is hit when the vessel is travelling at 20 knots, the chance of survival is less than 10 per cent, whereas if the vessel slows down to 10 knots, the chances of a whale surviving an impact rises to 75 per cent.
- **Keep watch** - When entering or leaving the gulf, especially during daylight hours, vessels are asked to have a dedicated observer equipped with binoculars. If a whale is sighted, vessels are asked to alter course and keep a 1000m clearance.
- **Reporting** - In the event of a sighting, the vessels should call in the sighting to Ports of Auckland using a pre-prepared script. Harbour control will then broadcast the message, with co-ordinates, to all vessels in the area so that they may alter course accordingly. The sighting information is also used for further statistical analysis.

The voluntary protocol has been notified to all shipping companies, and pilots boarding incoming ships to Auckland ensure that crew are aware of the protocol. A baseline assessment of the speed of ships transiting the Gulf, based on automatic tracking information which is electronically sent by ships via satellite, was undertaken in early 2013. This indicated that the additional time taken to transit the Gulf at 10 knots or less would, for most vessels, be less than one hour. The assessment is being repeated a year later to determine if speeds are reducing as a result of the protocol.

Ongoing work to promote continued uptake of the new measures are continuing.³⁷ Notification of the voluntary protocol to the International Maritime Organization is being considered. Additional research is also underway to count and map whales and other marine mammals in the Hauraki Gulf to build up a better understanding of their distribution.

- Monitoring programme – the aim of this programme was to assess the long-term environmental effects of the *Rena* grounding and monitor recovery. A focus of the programme was to assess kaimoana species and their marine habitats. Results have shown that the oil and other contaminants released from the *Rena* have had few long-lasting effects on the Bay of Plenty's maritime habitats.
- Shoreline programme – this monitored recovery of beaches and rocky shorelines from oil and debris from the *Rena*. In conjunction with the monitoring programme, it assessed how quickly the environment recovered from the different clean-up techniques used. No significant amounts of oil have been reported since August 2013.
- Wildlife programme – this assessed the effect on wildlife, with a particular focus on sensitive ecosystems, where there were endangered species. This programme is ongoing and due for completion in June 2015.
- Biosecurity programme – this aimed to survey high-risk sites and habitats to assess if new unwanted aquatic organisms had entered the environment, particularly where barge and support vessels were working. The overall conclusion from the surveys was that no high-risk species were detected during the surveys.
- Cultural impacts programme – This work provided an opportunity for tangata whenua to assess how their cultural values had been affected. Five cultural impact assessment reports have been completed and recommendations made. All reports supported total removal of the *Rena* wreck from Astrolabe Reef as being the only way to restore the area's mauri (life force). Maketu (Te Arawa) is also assessing the *Rena* impacts using a mātauranga (traditional ecological knowledge) approach.
- Mātauranga programme – this is focused on researching the effects of the *Rena* grounding using a mātauranga Māori approach. The work includes an assessment of mauri and research which will be undertaken by Maketu (Te Arawa) over 18 months.

Figure 15.4 Summary of the '*Rena* Long-term Environmental Recovery Plan' projects⁴⁷



The crane barge *Smit Borneo* working near the *Rena* wreckage (LOC)

Endnotes

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Activities

There is a wide range of recreation and tourism activities that New Zealand's marine environment supports. Few locations now remain inaccessible due to the wide availability of new technologies.⁶ The 'Spectrum of Marine Recreation Opportunities',⁷ as shown in Figure 16.1, is a model that helps to categorise marine recreation and tourism experiences relative to their distance from shore and from human settlements. This model can be used to assist with understanding the range of experiences, and potential impacts that these activities are having or are likely to have on the environment.

Characteristics	Class 1: Easily Accessible	Class 2: Accessible	Class 3: Less Accessible	Class 4: Semi-remote	Class 5: Remote
Experiences	• Much social interaction • High level of services • Usually crowded • Noisy • Lots of activity	• Frequent contact with others • Some space/ times to escape from others	• Some contact with others • Locations/ times when no other people present • Quieter	• Infrequent contact with others • Peace and quiet • Close to nature	• Virtually no contact with others • Solitude • Tranquillity • Self-sufficiency
Environment	• Many human influences • Highly modified • Lower-quality natural environment	• Human structures and influences visible and close by • Environment quality variable	• Few human structures close by but some maybe visible • Higher environmental quality	• Limited evidence of human activity or structures • Very high-quality environment	• Isolated • Little to no evidence of human activity • Pristine environment
Locations	• Close to, or in, urban areas • Beaches, docks, piers, urban coastal parks and walkways/ cycleway • Close to roading, parking and mass transport options	• Intertidal zone and areas up to 100 metres offshore • Often requires walk of several hundred metres from car parking or mass transport	• 100 metres to 1 kilometre offshore • More than 20 kilometres from major urban area • More remote beaches, islands and coastal areas • Often requires boat for access or walk in >500 metres	• 1–50 kilometres offshore • Isolated coasts, islands or reefs difficult to access • Boats or overnight hiking needed to access	• >50 kilometres offshore • Coastal areas >100 kilometres from any significant human habitation
Examples of activities	• Sunbathing/ baking • People-watching • Swimming • Playing games • Eating • Social gatherings • Sightseeing • Special events (i.e. concerts)	• Swimming • Fishing • Boating • Surfing • Windsurfing/ kiteboarding	• Boating • Fishing • Snorkelling/ SCUBA • Nature-study • Surfing • Sailing • Sea-kayaking	• Coastal sailing • Remote coast hiking and camping • Live-aboard vessels (e.g. fishing, diving or surfing)	• Offshore blue water sailing • Live-aboard offshore vessels (e.g. for fishing or diving) • Remote coast/ reef sea-kayaking and surfing

Figure 16.1 Spectrum of marine recreation opportunities (Orams M B and M Lück, 2013)

Potential impact	Swimming	Board sports	Diving and snorkelling	Boating	Fishing	Marine mammal and seabird interactions	Special events and sporting contests	General coastal and beach activities	Holiday homes	Cruise ships
Wildlife disturbance	■	■	■	■	■	■	■	■	■	■
Ocean noise				■	■	■	■			■
Ship strike				■	■	■	■			■
Trampling	■	■	■				■	■	■	
Over harvesting			■	■	■					
Marine pests			■	■	■		■			■
Anchor damage			■	■	■					
Rubbish	■	■	■	■	■	■	■	■	■	■
Wastewater/sewage				■	■	■			■	■
Climate change		■	■	■	■	■	■			■
Conflict between users	■	■	■	■	■	■	■	■	■	■
Impacts on near-shore environments	■	■	■	■	■	■	■	■	■	
Impacts on remote environments			■	■	■	■				■

Figure 16.2 Potential environmental impacts of recreation and tourism activities

Swimming

Swimming at the beach only became a popular recreational activity in New Zealand at the end of the 19th century. Prior to that, the marine area was perceived as dangerous; few Europeans could swim, and many people drowned. In the late 1880s medical authorities in New Zealand championed the health benefits of swimming. By the early 20th century, sea bathers were forming organisations to improve the image of swimming and provide amenities such as changing rooms at the beach. It was during this time that the increasing popularity of beach swimming saw a rise in surf lifesaving as both a voluntary service and competitive sport. As a consequence, beach swimming became safer and even more popular.¹⁴ Today, the beach lifestyle is an integral part of the national way of life and highly valued by New Zealanders and visitors, particularly during the summer months.

In terms of environmental impacts, swimming is generally a benign activity. However, there is an increasing demand for infrastructure such as shops and toilets to enable people to swim at the beaches. While people are swimming they can also disturb the local wildlife, trample on seabed habitats, leave rubbish behind and cause conflicts with other users such as fishers and surfers. Swimming provides many people with a great deal of pleasure and enjoyment and as such delivers some valuable positive physiological benefits.

Board sports

Surfing is believed to have originated from Hawai'i. The well-known Hawai'ian surfer Duke Kahanamoku toured New Zealand in 1915 giving surfing demonstrations in Auckland, Wellington and Christchurch.¹⁵ Locals were inspired and surfing became established in those areas. Surfing as a widespread recreational activity only became established in New Zealand in the 1960s. It was during that decade that the growth of the surfing and hippy culture really gained momentum, as it spread from California to Australia and then to New Zealand. Several New Zealand coastal towns become strongly associated with surfing, and the related

The conflict between surfers and other users of the marine environment is an intense one that has played itself out in the Environment Court. One of the most well-known conflicts is the controversy over the Whangamatā Marina where local surfers were fighting to protect high-quality surf breaks from the damage that would be incurred by dredging to provide vessel access to a marina and safe harbour for recreational boaters and fishers.

There has also been much controversy over the artificial surf reefs which have been built along some parts of New Zealand's coast, for example:

- Mount Maunganui submerged reef, the first of its kind, is now set to be removed just a decade after it was constructed in the early 2000s. It was hoped that the reef would provide world-class surfing waves and attract new economic development into the area. But the reef has never functioned as it was intended and the Bay of Plenty Regional Council recently decided to remove the reef because it was altering coastal processes, changing currents and exacerbating risks to swimmers in the area¹⁸
- In 2006 construction began on an artificial reef 250 metres offshore from Ōpunake Beach in Taranaki. The work involved pumping sand into geotextile bags to create a wave-making reef. Work was never completed and in 2011 the South Taranaki District Council voted to restore the Ōpunake headland to its original state¹⁹

Diving and snorkelling

People have always had a keen interest in going beneath the sea. Ancient manuscripts contain depictions of early divers and centuries-old artefacts imply that people dove to gather materials for jewellery. In the 19th century people began to use diving bells, supplied with air from the surface, probably the first effective means of staying underwater for any length of time.²⁰ The invention of the SCUBA (self-contained underwater breathing apparatus) in 1943 changed the way we saw the underwater world, and increased the demand for ways that people could explore the marine environment. The Professional Association of Diving Instructors (PADI) is the world's largest recreational diver training organisation, with members teaching the vast majority of the world's recreational divers, issuing nearly one million certifications each year.²¹

New Zealand is a popular dive destination and is rated as having some of the top dive spots in the world. The protection offered by marine reserves has helped ensure that some key locations have remained attractive for diving. Jacques Cousteau, the famous marine explorer, rated the Poor Knights Islands (now a fully protected marine reserve) as one of the top 10 dive spots in the world. More recently, the marine reserve was named in CNN's top 50 dive sites, because it has some of the best subtropical underwater gardens on the planet. The *Rainbow Warrior*, a vessel deliberately sunk off the Northland coast to create an artificial reef and dive site, was named by the *Telegraph*, a major newspaper in the United Kingdom, as one of their top dive sites.²² The creation of artificial reefs, through the deliberate sinking of boats or other materials, is increasingly being used in New Zealand to create or enhance dive locations.

Snorkelling, where a mask and snorkel (usually also with fins for propulsion and safety) is used to view underwater, is also a very popular activity. It is an accessible and budget-friendly way for people to see and enjoy the marine environment and is commonly offered to a wide range of age groups at resorts and on charter boats throughout the world. Snorkelling in New Zealand is very popular, particularly over the summer months. With the country's long and varied coastline, there are many prime snorkelling spots offering people free and easy access. A number of marine reserves around the coast provide both diving and snorkelling opportunities. Examples include the Cape Rodney-Okakari Point (Goat Island) Marine Reserve, the Tāwharanui Marine Reserve and the Whanganui A Hei (Cathedral Cove) Marine Reserve.



Diving, such as that shown here near Leigh, can disturb marine life
(Raewyn Peart)

Yachting and motorboating has continued to develop in New Zealand, both as a competitive sport and a recreational activity. There is a large cruising boating community throughout the country. A New Zealand Marine Industry Association assessment of boat ownership in 2010 estimated the number of boats in New Zealand to be above 470,000 (including 5,000 charter and tour boats).²⁵ In 2011 Maritime New Zealand commissioned market research on recreational boat ownership which indicated that 19 per cent of New Zealanders owned at least one boat at the time of the survey.²⁶

The relatively protected and safe harbours around the upper North Island (such as the Bay of Islands, Hauraki Gulf) and in the South Island (such as the Marlborough Sounds) are a boating paradise. Increasingly the offshore islands and remote cruising destinations are becoming more popular as technology allows people to travel further, more comfortably and safely in their boats. In addition, New Zealand (particularly the north-eastern part of the North Island) has become a favoured destination for blue-water yacht cruisers. These international tourism arrivals, while low in terms of volume, have had significant impacts on the coastal communities such as Ōpua, Russell, Whangarei and Auckland.

Whilst it has a long history, particularly in the Arctic and Northern Canada, where its use spans at least 5,000 years, sea kayaking has only recently become a widespread marine recreational activity. This growth mirrors that of boating more generally, as kayaks began to be constructed out of durable materials such as polyethylene plastic, which were mass-produced from moulds and therefore could be sold at affordable prices. Tourism New Zealand promotes kayaking as one of the best ways to explore New Zealand's coastline and inland waterways. The activity is popular with both private recreational kayakers and commercial tour operators who offer sea kayaking tours in many places such as the Nelson Lakes National Park, the Hauraki Gulf Marine Park, the Bay of Islands Marine Park, the Marlborough Sounds and the Coromandel Peninsula.

The use of personal water craft (jetskis) has also become a popular marine activity, with significant growth in both high-powered, manoeuvrable single-person craft and those that are suitable for multiple riders. They are used for activities as diverse as fishing, waterskiing, wakeboarding, surf riding and tow-in surfing (where surfers are towed in to surf large waves by jetskis). In addition, a wide diversity of watercraft are being designed and utilised in the sea including radio-controlled yachts and powerboats, waka-ama, dragonboats and other paddle-powered craft, submersibles, semi-submersibles, specialist wakeboarding boats, offshore powerboats (for racing), large expedition-designed and -equipped cruising boats, jetboats, and superyachts.

Boating has a number of positive impacts. Improving people's well-being and quality of life is most certainly one and another would be the economic value of the boating industry, especially for local communities in smaller regions. However, as recreational and tourism boating continues to increase in popularity, it is important that the impact this activity can have on the marine area is well understood.

Hull fouling is one of the greatest contributors to the domestic spread of invasive marine pests once they arrive in the country (discussed further in Chapter 5: Marine biosecurity).

Wharves, jetties and marinas typically attract large numbers of people and boats that, in turn, have the potential for discharges into the marine area, including fuel spillage, sewage from boat toilets, rubbish and leaching of antifouling paints which contain toxic chemicals. The parking and boat washing areas found at marinas may also be a source of contaminated runoff into the sea if they are not appropriately managed. Elevated heavy metal concentrations have been found in seabed sediments in the vicinity of marinas.²⁷

Increased noise, both from engines and propellers, and the boats themselves can have impacts on marine life, and of increasing concern is the potential impact from fish-finders and sonar depth sounders. The impact of reliable and relatively cheap marine navigation systems makes favoured habitats easily locatable and shared with others, so scallop beds, schooling fish, habitats suitable for crayfish and other good fishing spots are now found easily.

Waste and sewage management is a big concern for the boating community, particularly as access to remote and pristine locations is becoming much easier. The way that boat owners dispose of rubbish and fish waste can be a contentious issue. Regional councils have created rules to control the disposal of sewage from boats, because it can



The use of personal water craft, as shown here in Marlborough, has become a popular marine activity (Raewyn Peart)

Before the 1960s, most land-based fishing spots were reached using bicycles and walking. Four-wheel motorbikes, trail bikes, 4WD vehicles and helicopters have now allowed people to access difficult-to-reach and remote spots. Rapid developments in boating, diving, navigation and fishing technology have also allowed recreational fishers to access more remote fishing grounds, further from shore, and in deeper areas.

Big game fishing is now a major part of the recreational fishing and boating industry, with charter boats offering week-long trips to some of New Zealand's most remote fishing spots. New Zealand is well known around the world for its billfish (e.g. striped marlin), tuna fisheries, as well as the opportunities to fish for species such as snapper, kahawai, hāpuka, bluenose and trevalley.

Recreational fishing brings a number of positive impacts, including improving people's well-being and quality of life as well as the economic value of the recreational fishing industry. However, there are negative impacts from this activity. Recreational fishing allowances are set as part of the total allowable catch, which is a component of the QMS. The Ministry for Primary Industries sets and enforces recreational fishing rules for key species, such as snapper and trevalley, including bag and catch limits, potting and netting methods, types of equipment that can be used and the restricted or closed areas. These are all intended to constrain the catch within the QMS allowance and manage impacts of fishing on associated species.

Controlling and estimating the recreational fishing catch is challenging for a number of reasons, including the country's huge coastline; sparse population; large number of species targeted; large number of recreational fishing boats; the methods used; and the small research budgets available. Fishery compliance officers work around the coastline and are the point of contact for any phone calls to the Ministry about people taking too much or an undersized catch.

Surveys are periodically conducted to estimate the quantity and type of species caught. For some fish stocks, the recreational take represents a large proportion of the total catch,³⁰ such as with the blue cod fishery in the Marlborough Sounds and the snapper fishery in the Hauraki Gulf. There is a paucity of information in New Zealand about the impact of recreational fishing on the marine environment. According to researchers in 2010, data extrapolated from interviews with fishers suggests that the accidental catch of seabirds and threatened marine mammals in New Zealand's non-commercial fisheries is a 'problem that deserves more focused attention in the context of species conservation'. The Minister intends to focus on this issue when developing the next Inshore Finfish Annual Operational Plan.

In New Zealand's recreational fisheries, seabirds have been recorded being caught during hook and line fishing and in set nets.³¹ In July 2002, Southern Seabird Solutions was formed in an effort to develop and promote sustainable and responsible fishing practices in both the commercial and recreational sector. This work is included as a case study in Chapter 9: Management of fisheries.

Scientific research suggests that when recreational fishers target larger fish, they may be having a disproportionately negative effect on the fish population. This is due to the removal of the larger and older fish, which can produce a far greater number of eggs than smaller fish, and which have a higher reproductive success rate. This ultimately decreases the



Recreational fishing has always been an important part of the national culture, and many New Zealanders are introduced to the sport at a young age, as shown here (Daniel Sharp)



Recreational fishing, such as shown here at Bethells Beach, Auckland, can improve well-being and quality of life but also impacts on marine wildlife (Raewyn Peart)

When the first marine mammal tourism permits were granted in the Bay of Islands, no one knew how many dolphins were actually present in the Bay, so it was not possible to make informed decisions about the level of pressure on them. The only information available was from sighting data provided by the dolphin-watching operators themselves.³⁶ More recently, research has provided clear evidence of a number of impacts from recreation and tourism on marine mammals, including:³⁷

- Disturbance to behaviour – there is a large amount of research demonstrating how tourist activity can disturb the behaviour of marine mammals. For example, the presence of tourist boats around dolphins in the Bay of Islands, the Hauraki Gulf, Mercury Bay, Bay of Plenty and Kaikōura, has been shown to reduce the time the dolphins spend foraging and resting. Instead, more time was spent milling and socialising. In addition, it took longer for the dolphins to return to a resting or foraging state, having been interrupted by tour boats, than if they were interrupted by other vessels.
- Negative impacts on successful rearing of young – there is some evidence to indicate that animals which are unable to forage and rest in the normal way may experience a reduced energy budget, which in turn makes it less likely that they will be able to breed successfully and rear healthy calves. Higher calf mortality can result. High calf mortality rates have been documented in bottlenose dolphins in the Bay of Islands and in Fiordland.³⁸
- Avoidance of habitat – the presence of tourist boats may cause some species to avoid the areas of their natural habitat where they are pursued. In general, dolphins will avoid boats either by diving vertically or by avoiding their path horizontally. A recent population study on bottlenose dolphins in the Bay of Islands has indicated that fewer dolphins are now present, possibly because they have begun to avoid the area which is frequented by large numbers of tour and recreational boats. In addition, fewer bottlenose dolphins are now present in Milford Sound during the busy summer tourism season.
- Varying degrees of impacts – some marine mammal species and populations are more susceptible to disruption by tourism than others. For example, the large pod sizes and night-time feeding behaviours of the dusky dolphin mean that the impacts of tourism are less than those on the smaller groups of bottlenose dolphins which forage during the day.

A lack of a robust scientific understanding about the size and range of marine mammal populations has contributed to the difficulty in managing the cumulative impacts of different tourism operations. For example, permits were granted for bottlenose dolphin interactions at multiple locations along the north-east coast of the North Island before it was fully understood that they were all targeting the same population of dolphins that ranged along the area. While there are similarities in the impact of marine mammal tourism across different species and locations, each scenario entails different challenges. In addition, in some locations there can be cumulative effects, when tourism impacts on marine mammals are combined with those created by other activities. For example, in Doubtful Sound, the decline in the bottlenose dolphin population is thought to be a result of a combination of factors including increasing boat tourism, historical commercial fishing activity and increased discharges of freshwater into the marine area from the Manapōuri Power Station outfall.³⁹

Ships moving through the water can hit marine mammals and fish and injure or kill them. Ship strike happens because dolphins and especially whales are moving much slower than vessels and spend considerable time on the surface, which can put them directly in the path of boats. The greatest number of whale deaths from ship strike is caused by large ships travelling at more than 10 knots. This is discussed further in Chapter 15: Vessels. The propellers of recreational boats can cause considerable damage when they strike dolphins, as evidenced by a number of animals which sport distinctive propeller scars on their fins and bodies.



Marine mammal tourism, as shown here in Akaroa Harbour, Banks Peninsula, can disturb the natural behaviours of the targeted animals (Raewyn Peart)

homes are now the norm and a number of coastal communities have become upmarket coastal resort-type developments, some complete with canal and marina front properties. Examples include Pauanui, Matarangi, Whangamatā, Ōmaha, Marsden Cove and Tutukākā.

What is clearly evident from this transition, and the financial inflation that has resulted, is how highly valued the coastal lifestyle is for New Zealanders. In a clear and almost uniform pattern, the monetary value of property increases the closer it is to the coast and waterfront, and properties with sea views attract premium prices. Proximity to the sea and all the opportunities it offers, both from a passive perspective (seeing, smelling and hearing the sea) and an active point of view (such as walking, swimming, fishing and boating), are hugely attractive to New Zealanders and highly valued by them.

The increase in the number of holiday homes around New Zealand's coast has resulted in a loss of natural character of the coastal environment.

This is particularly the case where infrastructure associated with holiday homes intrudes onto the coastal edge through such structures as seawalls, jetties and ramps (the impacts of such structures are discussed in Chapter 14: Major marine developments).



Holiday homes, such as those shown here at Tutukākā, have transformed the New Zealand coast (*Raewyn Peart*)



Holiday homes, such as these shown at Tongaporutu, can bring with them infrastructure that impacts on the natural character of the coastal edge (*Raewyn Peart*)

Cruise ships

New Zealand has seen a phenomenal growth in the cruise ship industry of over 250 per cent during the last five years, making cruise arrivals the third largest inbound market for New Zealand tourism.⁴² The New Zealand cruise season runs from September through to March. During the 2012–2013 cruise season, New Zealand received over 210,000 cruise passenger arrivals and more than 80,000 crew.⁴³ The cruise ship industry is an economic contributor to a number of New Zealand ports, particularly Auckland, Tauranga, Wellington, Akaroa, and Port Chalmers (Dunedin). Cruise ships also visit a number of remote New Zealand coastal locations such as White Island, Fiordland, Stewart Island (Rakiura) and New Zealand's subantarctic Islands.

Cruise passenger behaviour has been steadily shifting from organised tours to free and independent travellers. This change of behaviour means a shift in the demand for activities from these cruise passengers. Whilst most of these tend to be shore-based, increasingly there is demand for marine-based activities at stopover locations.

The NIWA risk assessment noted the increasing access of tourist ships into the subantarctic islands and the Ross Sea as remote cruising destinations. All vessels travelling to New Zealand (including pleasure craft) need to meet a number of requirements before and on arrival to ensure New Zealand's environment, economy and people are protected from imported pests and diseases.⁴⁴ These issues are discussed further in Chapter 15: Vessels.

Food scraps are either kept in bins on board cruise ships for discharge as quarantine waste ashore, under Ministry for Primary Industries supervision, or are burnt in incinerators whilst out at sea. All food-contaminated waste or plastics are discharged under Ministry supervision as quarantine waste for treatment. Where possible, cardboard, glass, waste cooking oil and scrap metal are sent

Discharge of plastics is prohibited, but other ship garbage (including food waste, paper, rags, glass, metal, bottles, and crockery) can be discharged if ground to a particle size of 25 millimetres or less and if the discharge occurs at least 3 nautical miles offshore. Discharge of oil or oily water is permitted from a ship only if the oil content does not exceed 15 parts per million.

Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

In general there are only a few recreational or tourism activities that occur in the EEZ because of its distance from the shoreline. Those that do take place there include cruise ships and offshore recreational boating. The NIWA risk assessment helped inform the development of the EEZ Act. This assessment concluded that there were no activities associated with offshore recreation and tourism that were considered to represent an extreme environmental risk and, as a consequence, should be prohibited.⁴⁷ However, the assessment did make a number of recommendations as to the status of certain tourism activities in the EEZ marine environment.

The following activities were considered to be of high environmental risk and were recommended to be categorised as discretionary:

- Tourist vessel activity impacting marine mammals (regulated under the Maritime Transport Act and the Marine Mammals Protection Act)
- The use of surface floodlights at night which may cause bird strikes (regulated under the Maritime Transport Act and the Wildlife Act)

In addition, surface activity and noise and subsurface activity were considered to be of moderate environmental risk and were recommended to be categorised as discretionary. The following activities were considered to be of low environmental risk and were recommended to be categorised as permitted:

- Operations near the seafloor likely to result in collisions with benthic fauna
- Subsurface lights and noise generated by camera systems, remotely operated vehicles and submersibles

The EEZ Act prohibits certain activities in the EEZ including structures, submarine pipelines, submarine cables, removal of non-living natural material, disturbance of the seabed or subsoil, and deposition of anything on the seabed, unless the activity is permitted by regulations or authorised by a specific marine consent.⁴⁸ The EEZ Act does not contain any explicitly permitted recreation/tourism activities and, as a consequence, all activities are considered discretionary and therefore require a specific marine consent.

Marine Mammals Protection Act

The Marine Mammals Protection Act protects all species of marine mammals within New Zealand waters. It is an offence to 'take' marine mammals, whether alive or dead, from their natural habitat or 'any other place' without a permit from the Department of Conservation. The term 'take' is interpreted to include any actions likely to harass, harm, injure, attract or to disturb a marine mammal.



Regulations under the Marine Mammals Protection Act require tourism operators, such as Dolphin Explorer shown here on the Hauraki Gulf, to obtain a permit from the Director-General of Conservation before offering marine mammal tours (Raewyn Peart)

Moratoria on the issue of marine mammal permits

Moratoria have been placed on marine mammal tourism permits for sperm whale, dusky dolphin and New Zealand fur seal tourism at Kaikōura, New Zealand fur seal tourism in Abel Tasman, southern right whale tourism in the Auckland Islands and bottlenose dolphin tourism in the Bay of Islands. However, the Department of Conservation only acts to implement moratoria when there is hard evidence of a problem, something which has been difficult for scientists to provide.⁵⁰ As a result, it can take years for a moratorium to be put in place after potential problems are identified. For example, a moratorium on new permits was first proposed in the Bay of Islands in 1997, but was not implemented until 12 years later in 2009.⁵¹

There are also a number of regulations that commercial operators and any people coming into contact with a marine mammal are required to comply with. These include abandoning the interaction if the marine mammal is showing signs of becoming disturbed or alarmed and not causing the separation of individuals of marine mammals from their group. It is also illegal to feed marine mammals.

The Regulations also place restrictions on the speed that vessels may go when in the vicinity of any marine mammal. Extra requirements apply to whales, including that no person in the water shall be less than 100 metres from a whale and no vessel shall approach within 50 metres of a whale, unless authorised by the Director-General. There are also additional requirements that apply to dolphins or seals, including that no vessel shall proceed through a pod of dolphins and there should be a limit of the number of vessels able to make contact with a group of dolphins or seals.

Marine Mammals Protection Regulations 1992, Part 3 – Behaviour around marine mammals

18 Conditions governing commercial operations and behaviour of all persons around any marine mammal

Every commercial operation, and every person coming into contact with any class of marine mammal, shall comply with the following conditions:

- (a) persons shall use their best endeavours to operate vessels, vehicles, and aircraft so as not to disrupt the normal movement or behaviour of any marine mammal:
- (b) contact with any marine mammal shall be abandoned at any stage if it becomes or shows signs of becoming disturbed or alarmed:
- (c) no person shall cause any marine mammal to be separated from a group of marine mammals or cause any members of such a group to be scattered:
- (d) no rubbish or food shall be thrown near or around any marine mammal:
- (e) no sudden or repeated change in the speed or direction of any vessel or aircraft shall be made except in the case of an emergency:
- (f) where a vessel stops to enable the passengers to watch any marine mammal, the engines shall be either placed in neutral or be switched off within a minute of the vessel stopping:
- (g) no aircraft engaged in a commercial aircraft operation shall be flown below 150 metres (500 feet) above sea level, unless taking off or landing:
- (h) when operating at an altitude of less than 600 metres (2,000 feet) above sea level, no aircraft shall be closer than 150 metres (500 feet) horizontally from a point directly above any marine mammal or such lesser or greater distance as may be approved by the Director-General, by notice in the *Gazette*, from time to time based on the best available scientific evidence:
- (i) no person shall disturb or harass any marine mammal:
- (j) vehicles must remain above the mean high water spring tide mark and shall not approach within 50 metres of a marine mammal unless in an official carpark or on a public or private slipway or on a public road:
- (k) no person, vehicle, or vessel shall cut off the path of a marine mammal or prevent a marine mammal from leaving the vicinity of any person, vehicle, or vessel:
- (l) subject to paragraph (m), the master of any vessel less than 300 metres from any marine mammal shall use his or her best endeavours to move the vessel at a constant slow speed no faster than the slowest marine mammal in the vicinity, or at idle or 'no wake' speed:
- (m) vessels departing from the vicinity of any marine mammal shall proceed slowly at idle or 'no wake' speed until the vessel is at least 300 metres from the nearest marine mammal, except that, in the case of dolphins, vessels may exceed idle or 'no wake' speed in order to outdistance the dolphins but must increase speed gradually, and shall not exceed 10 knots within 300 metres of any dolphin:
- (n) pilots of aircraft engaged in a commercial aircraft operation shall use their best endeavours to operate the aircraft in such a manner that, without compromising safety, the aircraft's shadow is not imposed directly on any marine mammal.

Local and regional bylaws

Local and regional councils may enact bylaws to manage behaviour and activities in the public interest. Regional councils have the jurisdiction for managing the coastal marine area within their regional boundaries and may designate specific zones or areas, restrict certain activities or require specific permits to conduct activities. Regional councils thus have the ability to manage and control a range of marine recreation and tourism activities. They do this via mechanisms such as:

- Designating ski lanes
- Providing boat mooring areas
- Closing areas to boating
- Restricting boat speed
- Requiring safety equipment (such as lifejackets)
- Temporary closures of areas to allow for special events
- Restricting dogs and other animals in certain areas or at certain times
- Installing signs and navigation aids
- Installing and maintaining boat launching ramps, floating docks, wharves and other structures (such as swimming platforms)
- Replenishing sand on beaches and the construction of coastal pathways
- Controlling cars and motorbikes driving on sand dunes and beaches

Elements of best practice

The Department of Conservation has identified a number of recommended techniques and practices for managing visitor impacts on marine reserves including:⁵²

- Modifying visitor behaviour through education
- Regulating types of activities (such as commercial operations and boating activities)
- Encouraging minimal impact uses (such as boardwalks, interpretation trails and moorings to avoid anchoring)
- Reducing the use of specific sites
- Restricting access to sensitive sites
- Promoting compliance with the marine reserve regulations
- Identifying areas where damage can be repaired and undertaking restoration

More information can be found at the Department of Conservation's website.⁵³ Figure 16.3 provides some considerations for best practice to help promote positive environmental outcomes for each activity discussed in this chapter.

Activity	Best practice considerations
Fishing	• Follow the Southern Seabird Solutions recommendations for mitigating impacts of recreational fishing on seabirds (discussed below) • Haul nets in as quickly as possible • Limit unnecessary use of any floodlights at night and ensure that those that are required are directed approximately vertically onto work surfaces • Reduce boat speed in areas where marine mammals are likely to be present and keep well clear of the animals • Reduce the speed of boats in areas susceptible to ecological impacts from wake • Identify public anchoring areas and install public mooring buoys in environments sensitive to anchor damage • Adopt the Clean Boating Programme recommendations while fishing, including the ways to reduce the spread of marine pests
Marine mammal and seabird interactions	• Ensure compliance with regulations governing interactions with marine mammals, so that the animals are not harassed • Educate the public about good behaviour around marine mammals • Restrict the number of marine mammal tourism trips to ensure that no significant adverse impacts on the populations occur • Provide rest times and exclusion areas for marine mammals to enable them to have a break and refuge from human interaction • Reduce boat speed in areas where marine mammals are likely to be present, keep a good look-out, and keep propellers well away from the animals • Exclude vehicles and members of the public from areas of the beach used by nesting seabirds
Cruise ships	• Exchange ballast water in mid-ocean or operate a type-approved ballast water management system • Reduce ships speed in areas where whales are likely to be present to less than 10 knots • Reduce the speed of vessels in areas susceptible to ecological impacts from wake • Reduce the impacts on vulnerable species by rotating any visiting cruise ships around a number of sites • Restrict the number of visitors able to access sensitive environments and habitats • Minimise the use of night illumination to reduce impacts on seabirds including limiting unnecessary use of any floodlights at night and ensuring that those that are required are directed approximately vertically onto work surfaces
Special events and sporting contests	• Identify areas for special events and sporting contests that are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets and rubbish bins) appropriate for the level of activity to ensure the impacts from the event are reduced • Provide educational resources on registration that support good environmental outcomes
General coastal and beach activities	• Identify areas for public swimming which are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets and rubbish bins) appropriate for the level of activities • Provide public education around caring for the environment
Holiday homes	• Concentrate development in areas which have already been impacted and avoid development in places with high natural character or landscape values • Where coastal areas are to be developed, design the development in a way that maintains natural character and landscape values, such as through setting buildings back from the coastal edge and away from sensitive habitats such as dunes and wetlands • Incorporate restoration of the coastal environment into the development

Figure 16.3 Best practice considerations for marine recreation and tourism

Dolphin Encounter Kaikōura

In the 1980s, Kaikōura was an economically depressed town with high unemployment levels. At this time a whale watch business called Nature Watch Charters was set up which was soon followed by a second enterprise known as Kaikōura Tours (subsequently renamed Whale Watch Kaikōura).⁵⁹ The interest in whale watching at Kaikōura rapidly increased. At the same time, Dolphin Mary Charters (subsequently named Dolphin Encounter) persuaded the Department of Conservation that viewing and swimming with dusky dolphins was a viable business, and it issued them a permit. This enabled many people to interact with dolphins for the first time. Previously, dolphins had only been widely accessible in captivity. With increasing tourist interest in the wild dolphins, the Department placed a moratorium on the issue of further permits until research into the impacts of tourism on the dolphins could be undertaken.

Research undertaken during the late 1990s found that the dolphins were accompanied by boats for much of the day (over 70 per cent of daylight hours), and that the boats changed dolphin behaviour. However, it was not possible to determine whether the changes in behaviour observed had a significant impact on the welfare of the dolphin population overall. The researchers suggested that a cautionary approach should be applied to any increase in dolphin tourism activity.⁶⁰

In 1999, as result of the research, the Department put in place a moratorium on the issue of any more dolphin swimming permits for a 10 year period. In addition, Dolphin Encounter and the other permit holders agreed to give the dolphins a two-hour 'time off' period during the middle of the day over the summer months. In anticipation of the moratorium expiring in 2009, the Department commissioned a three-year investigation into the effects of tourism activity on the dolphins.

Following this further research, the moratorium on the issue of additional permits was extended for another five years. The two-hour rest period for the dolphins during the summer months was made mandatory. A limit was placed on the number of swims which could be attempted on each trip and a prohibition was placed on dropping swimmers in front of approaching dolphin pods.⁶¹

The dolphin and whale encounters have transformed Kaikōura into a vibrant community bristling with new tourist accommodation, restaurants and souvenir shops. Marine mammals now underpin the town's economy.



Dolphin Encounter, operating in Kaikōura, initiated dolphin tourism in New Zealand (Raewyn Peart)

Environment Southland's Deed of Agreement for cruise ships

In 1997 a study into the environmental impacts of tourism on the inland waters of the Southland region identified cruise ships as presenting the greatest threat to the area. A key area of concern was the pollution resulting from the discharge of wastes into both water and air.⁶⁶ Environment Southland (Southland Regional Council) responded by introducing the Deed of Agreement which is a contract between the council and individual cruise ships. The Deed of Agreement for cruise ships wishing to enter the fiords was introduced through the Southland Regional Coastal Plan in 2001. Section 13 of the Plan deals with cruise ships operating in the internal waters of the Southland region, and cruise ships either require a resource consent or are required to sign the Deed of Agreement with Environment Southland.⁶⁷

The agreement places environmental obligations on all vessels visiting the fiords, including requirements for waste discharge.⁶⁸ Whilst within the Southland coastal management area, a 'zero discharge' policy applies to discharges to water covering both solid (food waste and packaging) and liquid wastes (sewage, grey water, bilge water and hazardous waste). Air discharges must be minimised, with a goal to reduce visible smoke emissions in the fiords. A suggested way of achieving this is to substitute marine gas oil for heavy fuel oils,⁶⁹ but ultimately it is up to the vessels to determine how they will meet the requirement. The taking on or exchanging of ballast water whilst in internal waters is strictly prohibited.⁷⁰

The Environment Southland Marine Fee was set up as part of the Agreement, where each ship entering the fiords is charged based on the vessel's gross tonnage. The fee enables the council to offset any costs arising from risks to the environment posed by the industry and to alleviate financial pressure on the council's coastal expenditure. Introduced in time for the 2001/2002 cruising season, the fee generated \$260,000 from 23 visits and the 2012/2013 season was estimated to generate \$1.9 million. By the end of the 2011/2012 season it was estimated that the cumulative total of fees taken was \$10 million.⁷¹ Part of the Marine Fee helps pay for a wide range of the council's coastal-based functions, including its marine biosecurity programme, supporting shoreline clean-up and coastal water quality science reporting and monitoring, all of which are impacted by the vessels entering Doubtful Sound.



Environment Southland has introduced a Deed of Agreement for cruise ships wishing to enter the fiords, including Doubtful Sound shown here (Raewyn Peart)

- 10 Cessford G et al., 2012
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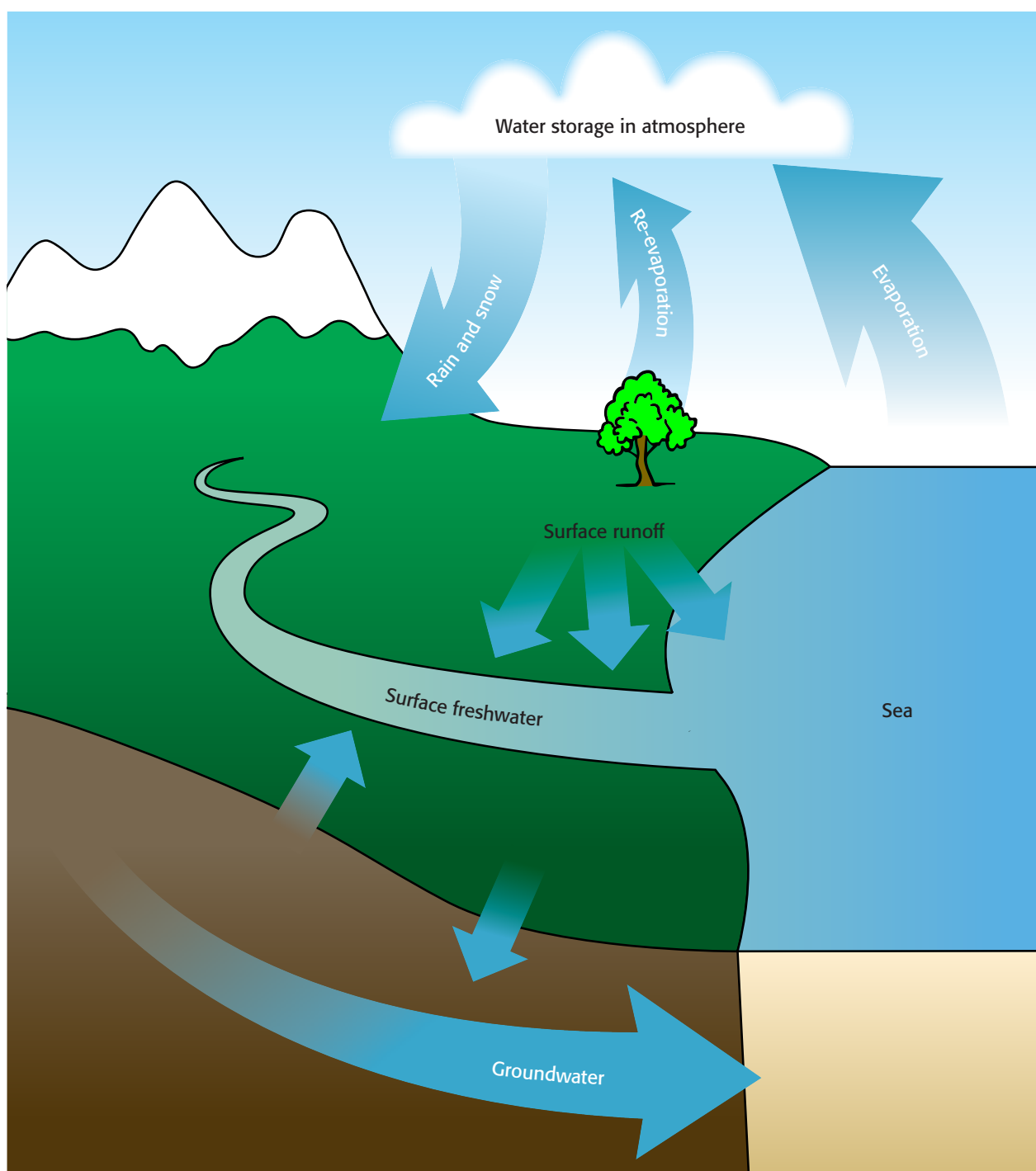


Figure 17.1 The water cycle

Catchment-sourced discharges can be divided into two types: point source and non-point source. Point source discharges refer to contaminants that enter the catchment from a single defined source such as a sewage treatment plant or a milking shed. Non-point source pollution refers to contaminants that enter the catchment from multiple diffuse sources, such as contaminant in runoff from pasture or urban areas. Point source and non-point source pollution are dealt with separately below.

Discharges into the coastal environment are also covered in depth in *'Caring for Our Coast: An EDS Guide to Managing Coastal Development'*, including an assessment of implementing the national policy on this issue and examples of good practice.

Nitrogen-nitrate and phosphate pollution

Nitrogen can be a significant pollutant in both freshwater and marine environments. Phosphorus is naturally more common in marine environments so systems there tend to be nitrogen limited. Phosphorous is often more of a concern in freshwater, but the addition of both substances to the marine environment can cause problems.

Nitrogen, and to a slightly lesser extent, phosphorus, are some of the most significant by-products of agricultural activity. Nitrogen and phosphorus are essential parts of amino acids and nucleic acids, and consequently they are critical building blocks for all life on earth. Unfortunately when they appear in the marine environment in excess, they can be very harmful because they accelerate natural processes of growth and decay.

Along with sediments and microbes, nitrogen-nitrates (nitrogen dissolved in water) and phosphorus-phosphates (phosphorus dissolved in water) produced by agricultural activity find their way into the marine environment. These nutrients can come from a number of sources, the principal of which is animal waste. Nitrogen from livestock manure contributes around five times the amount of nitrogen to the land as nitrogenous fertilisers.⁶ The breakdown of organic nitrate and phosphate in faeces, and the oxidation of ammonia in urine, release large amounts of nutrients which frequently enter water bodies.

Nitrogen and phosphorus pollution is also derived from fertilisers, which are applied to agricultural pasture in order to promote grass growth. Total fertiliser use significantly increased in New Zealand between 1985 and 2004. The amount of nitrogen fertiliser used in New Zealand has increased about ten-fold since 1985 and has doubled since the mid-1990s.⁷ Furthermore, soil in agricultural pasture often has high natural levels of nitrogen due to the presence of clover. This is sown by farmers as a method of fixing nitrogen into the soil, and superphosphate is often applied to promote clover growth. As farming becomes more intensive, more nitrogen fertiliser is often used to increase yields, more is excreted in dung and urine, and consequently more nitrogen finds its way into the catchment.

Forestry

Although the effects of forestry on a catchment are generally short-term and localised, they can be significant. Tree-felling and track construction can increase the amount of sediment and nutrients (in particular nitrogen, together with potassium, phosphorus, magnesium and others) that are washed into the water because without the trees and roots to provide a barrier there is little to stop rainwater washing the soil into the waterways. The extent of this depends on the particular area, the type of soil and the amount of rain normally experienced. The worst effects are generally experienced for the first three to five years after felling. The impacts increase if large areas of forest are clear-felled at one time or in succession.

The negative effects of forestry can be mitigated by selective felling, clear-felling of only small areas within a catchment at a time and the retention of riparian planting, which protects the waterways from the majority of the runoff. In addition, careful track construction with cut-off drains and the like can help reduce runoff. Land in pasture produces greater levels of sedimentation and nutrient runoff than mature forest, so afforestation of an area previously used as pasture can return stream flows and sediment loads to normal levels within six to eight years of planting, and reduce runoff by 30 to 50 per cent within five to ten years.⁸

While water quality may benefit from forestation, there is a potential downside to reduced sediment loads in rivers, which can result in increased potential for coastal erosion because of less sediment reaching the coast.

Urban stormwater

Approximately 86 per cent of New Zealand's population lives in urban centres of more than 1,000 people.⁹ As a result of land clearance and urban development, most of these urban areas are covered in impermeable surfaces. For example, in the Auckland region only 19 per cent of the 10,000 kilometre-long stream network passes through native forest.¹⁰ In these paved and developed areas, rainwater cannot be absorbed by vegetation and soil, as it would be in rural areas. Therefore, to reduce the risk of flooding, towns and cities have stormwater systems that channel rainwater into gutters and pipes and eventually out into streams, lakes and coastal waters. Because a large number of New Zealand's urban centres are sited on estuaries and harbours, the stormwater many towns produce discharges directly to the coast.

Stormwater systems are often a significant source of pollutants into the catchment and coastal areas because the water that enters the pipe network picks up and carries with it a wide range of contaminants from the streets, industrial and construction sites, roofs and other surfaces. These substances include heavy metals (especially zinc, copper and lead – lead contamination has been falling

Mining

In the past, mining had a significant impact on water quality. Mines can discharge huge amounts of silt and sediment into the catchment, as well as large amounts of processing chemicals such as cyanide and heavy metals from the rock. Nowadays, mines have relatively effective treatment systems, which minimise the amount of silt and sediment being transferred into the catchment (e.g. by holding wastewater in pools to allow sediment to settle before releasing it into the river).

Gold mining gives rise to particular risks because when the mineral ore is exposed to air and water the metals can react chemically, dissolving and entering groundwater and streams. This can cause the catchment to become more acidic. The industry has methods to minimise the risk of this by transporting the rock to a treatment plant in a slurry, separating off the water, and then treating it to remove the heavy metals before discharge.

Mining often changes groundwater flows, and water flowing through exposed ore remnants or mine waste can acquire high levels of heavy metals, with such contaminated flows often continuing long after the mining has ended.

Wastewater treatment plants

Approximately 1.5 billion litres of domestic wastewater is discharged into the environment in New Zealand every day.¹⁶ Human effluent contains a number of substances that can be extremely damaging to the marine environment if released into the coastal waters untreated. The most significant pollutants are rotting organic matter, disease-causing micro-organisms, excess nutrients, ammonia and suspended solids.

Most of the household sewage in New Zealand is piped through treatment plants to remove pollutants before discharge: most towns of 5,000 or more residents have a reticulated sewage system which pumps everybody's wastewater into a common treatment plant and discharge point. There are more than 250 treatment plants in New Zealand, and although 24 of them discharge waste onto land, the rest discharge either into the ocean, rivers, streams or drains.

The plants vary in age and technological capability. Treatment systems use a number of stages in the treatment process, depending on how advanced they are. 'Primary' treatment means that suspended solids are filtered out of the effluent before it is discharged into the catchment, and 'secondary' treatment refers to the use of micro-organisms to remove organic matter and some nutrients, which are then decomposed in process facilities or oxidation ponds. Tertiary treatment further reduces nutrients and organic matter using oxidation ponds and/or various types of organic or chemical treatment. Ultra-violet treatment, and other technologies, are used by the most advanced plants in New Zealand to kill the pathogens that remain in the water.

Treatment plants are particularly designed for human wastes and effectively reduce organic pollution from this source. They are not generally designed for a broad range of industrial pollutants and may be quite ineffective in removing these. Industrial effluent is much more difficult to manage in treatment plants because it can be highly variable in loading and volume.

The solid material from treatment plants (biosolids) needs careful management. It may be suitable for use as a soil additive (but this may add to pollution risks) or it may need to be disposed of in a landfill site.



The Mangere Wastewater Treatment Plant, shown here, is one of the most advanced plants in the country, undertaking tertiary and ultra-violet treatment along with using other technologies (Raewyn Peart)

Some regions use combined sewer and stormwater drains – this is common in old networks which have not yet been replaced with more modern systems that minimise the risk of pollution. In these areas pumping stations can overflow when there is heavy rain that quickly reduces their holding capacity. In other areas with separated sewers, stormwater infiltration through damaged pipes and poorly arranged or illegal connections can also result in inflows in excess of capacity. When this happens, raw sewage can be discharged into the marine area, although it will be diluted to some extent by the large amount of stormwater.

Stormwater infiltration can also cause major problems in sewage treatment plants by increasing the volume of liquid that they have to deal with, reducing the effectiveness of treatment, and increasing the risk that treatment systems will flood and allow effluent to escape. This is actually quite common as a result of faulty pipes, illegal cross-connections (where private properties illegally attach sewage pipes to the public stormwater drain), and old connections that have not been cut off.

Another way in which untreated urban sewage can enter the catchment is where sewer systems overflow into the stormwater network (exfiltration) and the natural environment. This generally happens as a result of failures in the pipes such as cracking followed by tree root invasion. It can also occur as a result of heavy rainfall. Pumped sewerage systems are becoming more common, and are resistant to infiltration, but failures can result in spills.

Auckland Central Interceptor Project

The Auckland Isthmus has had ongoing issues with major wastewater overflows into the marine environment resulting from a wastewater network in need of upgrading. Water quality at a number of Auckland beaches exceeds national guidelines for the concentration of enterococci bacteria. These bacteria indicate the presence of faecal material and disease-causing micro-organisms in coastal waters, high levels of which pose a significant risk to public health. In November 2011, more than a third of the region's most popular beaches failed water quality tests, and signs recommending 'no swimming' were erected for at least 10 beaches.¹⁷ There are now permanent health warnings in place at Cox's Bay, Meola Reef, Weymouth Beach, Little Oneroa Lagoon and Wairau Outlet.¹⁸

The growth and development predicted for the Auckland region, and the ageing infrastructure, means there is much concern about the capacity of the network in the future. Watercare Services is now focusing on delivering an \$800 million Central Interceptor Project which will see a large wastewater pipeline constructed to improve the wastewater network service for western and central Auckland. The interceptor, which could be up to 24 kilometres long, will reduce wastewater overflows, provide for population growth and mitigate the risk of pipe failures. The interceptor will be constructed in three phases between 2016 and 2030 to transport wastewater from central and west Auckland to the Mangere Wastewater Treatment Plant.¹⁹

Potential environmental effects

The amount of pollutants that are transferred from land into the oceans, and the effect that they will have on coastal marine ecosystems, varies depending not only on the amount of waste actually released and the type of waste, but also on the characteristics of the marine environment affected. As one would expect, transfer rates are much worse at times of more extreme wet weather, when more water is moving from the land to the sea.

Whether or not land-sourced pollutants remain in the coastal marine area depends on a number of factors, in particular the geographical makeup of the area and the winds and currents experienced by the locality. In harbours and estuaries, the amount of mixing that occurs and the time the inflowing water is retained in the area are important. Shallow inlets are more prone to retain sediment and pollutants than deeper ones, and estuaries and narrow inlets are more prone to retaining material than wide exposed beaches, which are subject to stronger currents that can carry inflowing substances out to sea. Many river estuaries have natural sea barriers broken only in high flows. Sediment and pollution retention behind these barriers is high.

New Zealand's particular geographic make-up means that there are large stretches of coastline that do tend to trap sediment and pollutants. As the Ministry for the Environment's 1997 State of the Environment report notes: 'Much of the coastline is made up of river-fed estuaries whose wide, shallow waters are permanently protected from ocean waves by sand or shingle bars or offshore islands'.²⁰ Where there are high river flows, the freshwater may pass along an open coast, floating over seawater for some distance before it is mixed with the seawater.

Sediment can bring about changes in vegetation, be detrimental to fish habitats, degrade spawning areas and deplete invertebrate populations. These impacts can flow through the food web, particularly if a particular life stage of a species is affected. As most sediment is deposited in intertidal zones, it is the species that live in these areas that are often most affected by excess sedimentation.²³ This is particularly significant, because healthy intertidal zones support a particularly high diversity of species in comparison with other parts of the coastal area. For example, the Kaipara Harbour is the biggest estuary in the Southern Hemisphere, and is the place where most of the snapper on the west coast of the North Island originate. Sediment from the catchment threatens to overwhelm horse mussel beds and seagrass meadows, which are nursery grounds for snapper.²⁴

Some of the main environmental impacts from sediment suspended in the water or settling on the seafloor are identified in Figure 17.2 below.

Type	Impact
Direct	- Decreases oxygen concentrations and changes sediment properties such as grain size, chlorophyll and organic matter content²⁵ - Clogs the gills of fish, making it harder for them to breathe - Interferes with food intake and potentially affects growth and condition of fish and shellfish - Reduces visibility, making it harder for some fish to find food - Negatively affects microphytes or benthic primary producers, an important food source for many macrofaunal species - Smothers shellfish beds and cause cockles, pipi, mussels and scallops to die - Causes seafloor plants to die off - Reduces the hard surface area for kelp spores or baby paua to attach to
Indirect	- Reduces the food that is available for rocky reef ecosystems and larger animals - Reduces the growth and extent of kelp forests - Damages the entire productivity of the rocky reef ecosystem - Damages sensitive habitats that are important for part of a species life stage, such as nursery or juvenile areas - Reduces the abundance of many biologically important fish species - Reduces the stocks of recreationally and commercially important fisheries species (such as paua and crayfish)

Figure 17.2 Environmental impacts resulting from increased sediment in the marine environment²⁶

Sediment impacts in Tasman Bay

Tasman Bay in the South Island regularly suffers from murky seabed conditions which are believed to have come from sediment carried into the marine area by floods of the Motueka River, which then gets resuspended in the water by tidal currents in the bay. In addition, ongoing trawling and dredging resuspends the sediment. Research has found that scallops on the seafloor temporarily stop feeding when in highly turbid water, whereas those in clearer water in the same vicinity continue to feed. In addition, the growth of scallops was shown to be adversely affected by high levels of suspended sediments over the long term.²⁷

There are now concerns that coral-like bryozoan colonies further north of the Motueka River, which are important snapper and tarakihi nursery habitats, are under increasing stress from the sedimentation. With the seabed in the area becoming largely covered in sediment, there are few hard surfaces for the bryozoans to settle on. Ultimately this is reducing the overall productivity of the tarakihi and snapper fisheries as well as decreasing the area's biodiversity.²⁸

Heavy metals and other dissolved pollutants

Many heavy metals and other pollutants find their way into the marine area from roads, from industrial and agricultural activity, and from domestic households. Such toxic substances have also been used directly in the marine environment. In the 1950s, 'booster' agents such as DDT (dichloro-diphenyl-trichloroethane), organomercurial, organolead and arsenical compounds were used on underwater surfaces to discourage antifouling organisms.

In particular, heavy metals, chemicals, hormones, dioxins, organochlorines, PCBs (polychlorinated biphenyls) and DDT cause problems. The problem may not just be mortality of marine organisms, but disruption of growth or reproduction, which may occur

Excess nutrients

Nutrients, in particular nitrogen and phosphorus, occur naturally in aquatic ecosystems and are essential elements for growth. This is why they are used in large quantities in fertiliser to promote grass growth on agricultural pasture. However, where excess amounts of nutrients reach the catchment they can upset the balance of the ecosystem and cause serious problems.

When there are excess nutrients in the water, they serve to fertilise microscopic algae, leading to abnormal increases in their growth and subsequent decay. Their decay stimulates aerobic (oxygen-consuming) bacteria, which in turn causes severe reductions in water clarity and oxygen levels, potentially leading to the deaths of fish and other marine life. When the plants decay, the nutrients are released back into the water, stimulating algal growth and allowing the cycle to start again. This phenomenon is known as eutrophication. The consequent reduction in oxygen levels can result in 'dead zones', such as occur at Chesapeake Bay in the USA during mid-summer periods.³⁶

The Firth of Thames' nitrogen overload

In 2012, the Waikato Regional Council published a report into the water quality of 17 rivers in the Hauraki region using data gathered between 2000 and 2009. This showed that the Piako, Waitoa and Waihou Rivers were in poor to intermediate condition, being oxygen-depleted and murky. The report showed these rivers had concentrations of nitrogen and phosphorus five to seven times above guidelines.³⁷ As a consequence, the Firth of Thames is considered to be probably more productive today than it was in previous times, but water quality (in terms of nutrient levels, colour and clarity) is likely to have diminished.³⁸ More recently long-term monitoring has shown evidence of oxygen depletion as well as nutrient runoff-driven acidification of the seawater in the Firth of Thames.³⁹

The 'State of the Hauraki Gulf 2014' report confirmed that riverine inputs of nitrogen to the Firth of Thames now exceed loads from oceanic sources, and that the nitrogen loads entering the Firth from Hauraki Plains rivers are significant. The report noted that this is consistent with the Plains having one of the highest stocking rates for cows in the country. Long-term studies indicate that nitrogen inputs cause 'seasonal elevation in carbon dioxide, corresponding sags in oxygen and increasing acidification'.⁴⁰

Bacteria and viruses

Animal and human faecal matter contains large numbers of bacteria, which are produced by the gut. Although many are harmless, some of these bacteria can cause serious health problems in humans if they reach water used for recreation and shellfish gathering. Shellfish contaminated with sewage may contain pathogens such as norovirus, hepatitis A, shigella, vibrio and salmonella. These can cause dysentery, gastro-enteritis or other illnesses within a few hours or days.⁴¹

Coliform bacteria are common in human and animal faecal waste – but can come from other sources. The amount of coliforms per hundred millilitres of water is used to indicate the likely presence of faecal matter in water. A more precise test for animal and human waste is that for a particular coliform, *Escherichia coli*, which comes from warm-blooded animals alone. Viruses can also be found in human and animal faecal matter and the main class is the enteric viruses, which cause illnesses such as polio, meningitis and hepatitis.

Where levels of viruses or bacteria in coastal waters are relatively high, they can pose a risk to swimmers and other contact water users. They can also concentrate in shellfish, which eat by filtering food through the water, and can therefore be dangerous to humans who consume the shellfish.



Cox's Bay in Auckland is permanently closed to swimming and shellfish collecting due to a high level contamination from sewage (Raewyn Peart)

Decision-making process

The RMA provides the framework for the control of activities that result in land-sourced pollutants entering the marine area. The RMA adopts a catchment-based approach to environmental management, with the boundaries of regional councils largely coinciding with the boundaries of water catchments and including the coastal marine area. This is based on the understanding that many different activities within a catchment will cumulatively impact on the quality of lakes and rivers and ultimately on what is discharged into the sea. Effectively managing these impacts requires an integrated and catchment-focused approach.

Under section 15 of the RMA, the discharge of contaminants into water (or onto land where there is a possibility that the contaminants will run off into water) is prohibited unless the activity is expressly allowed by a national environmental standard, a rule in a regional plan as well as a rule in any proposed regional plan, a resource consent or regulations.

National Policy Statement for Freshwater Management

The National Policy Statement for Freshwater Management (Freshwater NPS) sets out national policy for freshwater management within New Zealand under the RMA.

Freshwater NPS water quality objectives

- *Objective A1: To safeguard: a) the life-supporting capacity, ecosystem processes and indigenous species including their associated ecosystems of fresh water; and b) the health of people and communities, at least as affected by secondary contact with freshwater, in sustainably managing the use and development of land, and of discharges of contaminants.*
- *Objective A2: The overall quality of fresh water within a region is maintained or improved while: a) protecting the significant values of outstanding freshwater bodies b) protecting the significant values of wetlands and c) improving the quality of freshwater in water bodies that have been degraded by human activities to the point of being over-allocated.*

The Freshwater NPS provides a framework for achieving its objectives. Regional councils are required to develop freshwater objectives (the intended environmental outcome), to set limits (the maximum amount of resource use which will allow a freshwater objective to be met), and then to establish methods to avoid overallocation (resource allocation beyond a limit or resource use where a freshwater objective is not being met) for all freshwater management units (a water body, multiple water bodies or a part of a water body determined by the regional council as the appropriate scale for freshwater management).

National freshwater objectives set out the intended environmental outcome for a freshwater management unit. Part CA – National Objectives Framework sets out the approach for establishing freshwater objectives. Regional councils must:

- Identify the values for each freshwater management unit, including two compulsory values ‘ecosystem health’ and ‘human health for recreation’ as well as any other values
- Identify the attributes applicable to each value, including the attributes listed in Appendix 2 as well as other attributes
- For those attributes listed in Appendix 2 (currently phytoplankton, total nitrogen, total phosphorus, periphyton, nitrate toxicity, ammonia toxicity, dissolved oxygen, *E. coli* and cyanobacteria), assign an attribute state at or above the minimum acceptable state
- Formulate freshwater objectives, by reference to the relevant numeric attribute state for those attributes listed in Appendix 2 or, for other attributes, in numeric terms where practicable or narrative terms

Appendix 1 describes national values and uses for freshwater, including the two compulsory values ‘ecosystem health’ and ‘human health for recreation’. Appendix 2 identifies some ‘attributes’ relevant to the two compulsory values. It contains attribute tables which set out for each attribute the unit in which it is to be measured, four attribute states, numeric and narrative descriptions of those states and the national bottom line. Appendix 2 is currently incomplete. For example, there are a number of attributes relevant to ecosystem health that are not yet specified, including dissolved inorganic nitrogen, dissolved reactive phosphorus, dissolved oxygen, temperature, macro-invertebrate community index, and water clarity. Regional councils will need to continue to set objectives for these attributes for each freshwater management unit.

Freshwater limits specify the maximum amount of resource use available which will allow the objectives to be met. For example, nitrogen-leaching limits for agricultural land use may be included to achieve a dissolved inorganic nitrogen objective.

New Zealand Coastal Policy Statement

The NZCPS applies within the coastal environment – this includes the coastal marine area and all land with a coastal influence (generally defined by the first ridgeline but can include coastal systems such as wetlands and dunes extending further inland). Regional policy statements, regional plans and district plans must ‘give effect’ to the objectives and policies contained within the NZCPS.

A number of policies relate to catchment-based activities that may impact on the marine area, in particular Policy 21 (set out in Chapter 2: Marine management), 22 and 23.



Estuaries, such as the one shown here at Ngunguru, Northland, are not included in the Freshwater NPS, but regional councils are required to provide for the integrated management of the use of land and freshwater on coastal waters (*Raewyn Peart*)

NZCPS, Policies 21, 22 and 23: Water quality

Policy 21: Enhancement of water quality

Where the quality of water in the coastal environment has deteriorated so that it is having a significant adverse effect on ecosystems, natural habitats, or water-based recreational activities, or is restricting existing uses, such as aquaculture, shellfish gathering, and cultural activities, give priority to improving that quality by:

- a. identifying such areas of coastal water and water bodies and including them in plans;
- b. including provisions in plans to address improving water quality in the areas identified above;
- c. where practicable, restoring water quality to at least a state that can support such activities and ecosystems and natural habitats;
- d. requiring that stock are excluded from the coastal marine area, adjoining intertidal areas and other water bodies and riparian margins in the coastal environment, within a prescribed time frame; and
- e. engaging with tangata whenua to identify areas of coastal waters where they have particular interest, for example in cultural sites, wāhi tapu, other taonga, and values such as mauri, and remedying, or, where remediation is not practicable, mitigating adverse effects on these areas and values.

Policy 22: Sedimentation

1. Assess and monitor sedimentation levels and impacts on the coastal environment.
2. Require that subdivision, use, or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water.
3. Control the impacts of vegetation removal on sedimentation including the impacts of harvesting plantation forestry.
4. Reduce sediment loadings in runoff and in stormwater systems through controls on land use activities.

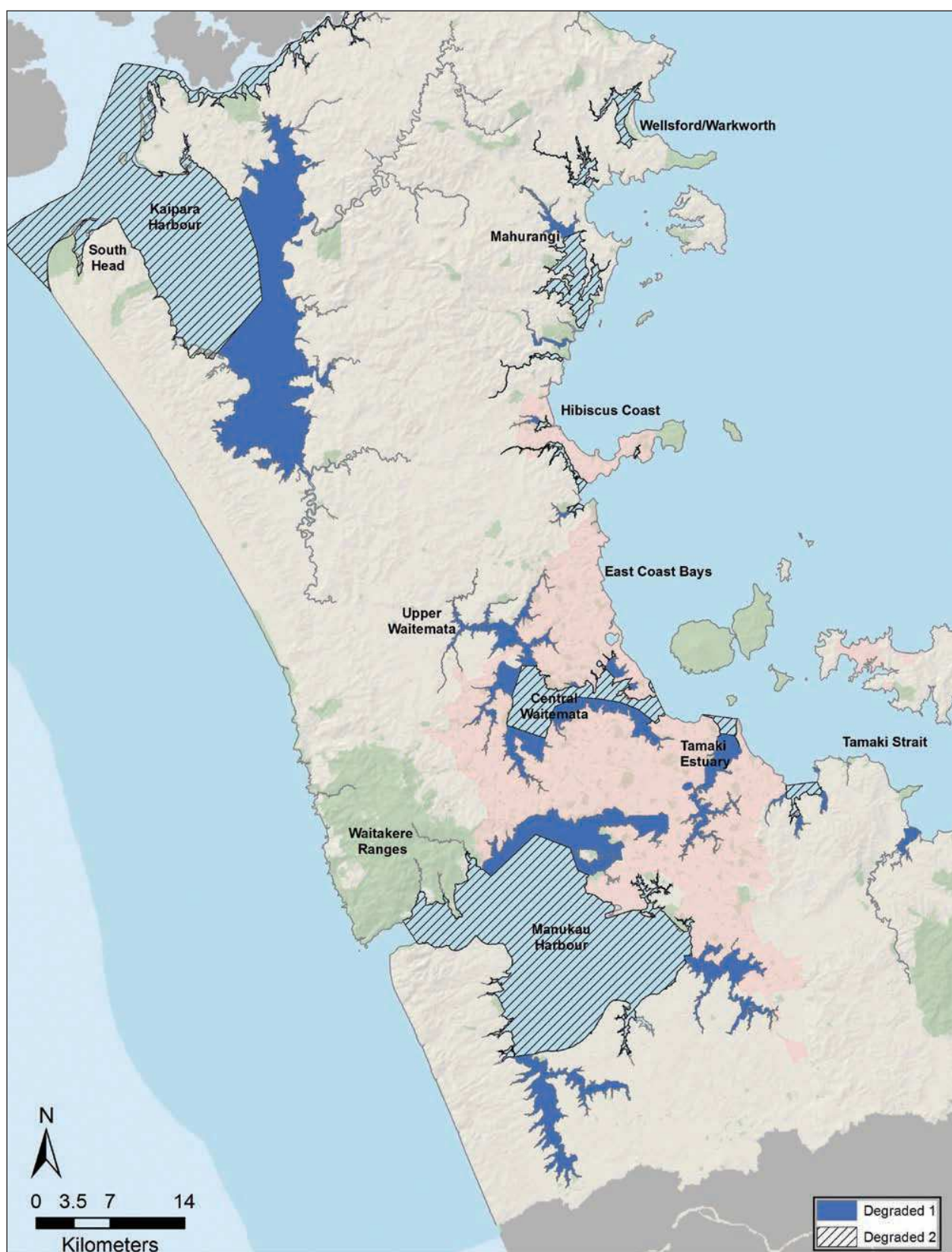


Figure 17.3 Map showing the marine degraded areas in Auckland (*Auckland Council*)

- Another method is to focus more on the land use activity itself and apply methods to control or regulate these types of activities. For example, if a rural catchment is identified as having a negative impact on the ecological health of the marine area because of the high levels of sedimentation, and the main causes are identified as forestry and farming activity, the regional plan might include rules to mitigate these effects, such as requiring stock to be kept out of waterways and preventing harvesting in riparian areas.

These methods can be combined – such as those relating to land uses being linked to water quality standards.

Non-point source pollution poses more of a challenge for regulators because it is not caused by isolated and identifiable acts which are generally more easily controlled by the consent process. Regional councils are responsible for identifying the environmental effects of both urban and agricultural runoff, and implementing regional solutions and management. To discharge this duty, councils may produce technical guidelines for the management of stormwater systems and sediment control programmes. They may also set up incentive programmes and form community-based groups to encourage farmers, forestry companies and other users to alter the way they use their land to minimise the effects of runoff. There are opportunities to undertake stronger actions on these issues, including through new regional rules prohibiting intensification in some areas, requiring farm environmental management plans, and requiring planting/harvesting plans for forestry.



Regional councils are responsible for identifying the effects of runoff from forestry, farming and urban activities, such as those shown here at Kawakawa Bay, Auckland, and implementing solutions (Raewyn Peart)

Integrated catchment management plans

Catchment management plans are a tool to integrate the management of natural resources and the protection of community values. Such a management approach is an established concept used around the world. However, in the past they have tended to focus on the planning and management of a catchment. In recent years the focus has widened to include the coastal-receiving environment for that catchment.⁵⁰

Harbour and catchment management planning focuses on managing land and water together to achieve long-term outcomes which are beneficial to the wider environment. Such a tool integrates multiple stakeholder concerns, issues and objectives into one overarching plan, as highlighted by Figure 17.4. Such types of planning documents are non-statutory but have the capacity to inform and support statutory documents such as district and regional plans.

Topic	Issues addressed
Allocation of scarce water resources among competing land and instream uses	• How do local land uses affect the availability of ground and surface water over the entire catchment? • What is the most defensible way to plan for the allocation of water resources among competing alternative uses? • How much water is required to maintain instream values? • How do economic returns affect landowners' and recreational uses of water?
Managing land uses in harmony with freshwater resources	• How is sediment mobilised to reach rivers, and what impacts does it have? • What information is needed to best manage river gravel allocation? • Why has the Motueka catchment trout population declined then partially recovered? • How can water quality be maintained or improved with intensifying land use? • Are there some fundamental solutions to water quality contamination? • Is riparian revegetation the 'silver bullet' for improving water quality?
Managing land and freshwater resources to protect and manage marine resources	• What are the risks to marine farming from activities on land? • What are the factors that increase or decrease the production and values of fish and shellfish? • Where are the faecal bacteria affecting marine farms coming from? • What are the relative effects of river flows (especially floods) on scallop and mussel production compared to activities like dredging and marine farming?
Integrative tools and processes for managing cumulative effects	• What are preferred development pathways to ensure continued sustainable management at catchment scale? • What are the opportunities for using modelling to predict cumulative effects of land use scenarios? • How can models help decision-makers balance environmental impacts alongside social, economic and cultural benefits when planning for further development?
Building human capital and facilitating community action	• What methods would best motivate environmental stewardship by catchment and community groups? • What methods can we use to promote effective interaction between scientists, resource managers, and the community? • How can iwi build their capability in the resource management process? • What are some 'off the wall' examples of innovative and enduring community engagement that we could use to facilitate community action?

Figure 17.5 Research issues addressed by the Integrated Catchment Management for the Motueka River
(Fenemor A, 2013)

Territorial authorities

Under section 31 of the RMA, territorial authorities are responsible for controlling the effects of land use outside the coastal marine area, and this includes downstream effects on the marine environment. Territorial authorities can prepare district plans to address these issues. Whereas regional councils tend to focus on catchment-wide issues, territorial authorities focus more on site development issues such as earthworks and the construction of impervious areas.

District plans can indicate the amount of impervious area (building or paving) that a property can have, and property owners who wish to exceed this limit must apply to the territorial authority for consent to do so. Territorial authorities and their local network operators also have responsibility for stormwater infrastructure in their area, such as maintenance and replacement of stormwater pipes, maintaining stormwater pumping systems and outfalls, construction of silt traps and detention ponds.



Constructed wetlands, as shown here in the Whisper Cove development at Snells Beach, Auckland, can help reduce the contaminant load in stormwater before it reaches the marine environment (Raewyn Peart)

Potential environmental effects	Elements of best practice
Urban stormwater	
In periods of heavy rain stormwater increases the volume of sewage so that wastewater treatment plants cannot treat it all	• Replace combined sewage/stormwater systems with separate networks • Monitor and maintain sewage pipes to prevent leakage
Earthworks generate increased sediment	• Prepare catchment management plans to protect natural channels, restore wetlands and provide stormwater retention ponds • Require an erosion and sediment control plan as part of building consent applications and include strict conditions in resource consents to require adequate sediment detention methods to be used. These may include chemical sediment control (flocculation), silt fences, decanting earth bunds or sediment retention ponds. • Undertake major building works at times of the year when rainfall is unlikely to be heavy • Retain existing vegetation wherever possible and effect revegetation as soon as possible • Stabilise disturbed areas and minimise the size of the area disturbed as much as possible • When considering zoning for plans, pay particular attention to the potential for the cumulative effects of earthworks in an area to cause significant sedimentation of the receiving marine environment
Urban sewage	
Wastewater treatment works fail to remove organic matter, nutrients and micro-organisms from the effluent before it is discharged into the catchment	• Use the best available technology (tertiary with ultra-violet pathogen treatment or similar) in wastewater treatment plants to ensure that the water discharged into the coastal marine area contains minimal amounts of organic matter, micro-organisms and nutrients • Plan the construction of wastewater treatment works so that they have the capacity to deal effectively with all the waste generated by the community, and with future or seasonal growth in the area. • Ensure sufficient capacity to avoid overflows of raw sewerage into the marine environment
Septic tanks	• Construct septic tanks only in areas where the soil is suitable • Construct septic tanks only where there is an adequate amount of land surrounding the property to cope with the amount of effluent generated • Adequately maintain septic tanks and soakage fields to ensure that they do not fail • Use the latest technology in designing effective septic tank systems
Sewage enters the stormwater system and from there enters the catchment untreated	• Prevent illegal connections • Include stormwater control measures such as permeable paving in new developments and modifications to existing urban areas • Adequately maintain piping • Use swales and planted areas as buffers to limit and filter stormwater runoff • Restore old wetlands to filter and limit the flow of stormwater to the sea
Industrial effluent	
Industrial processes result in the discharge of effluent into the catchment	• Use the best available technology to minimise pollution, and instigate programmes for the reduction of waste • Use constructed or restored wetlands as an alternative treatment and disposal system; they can be effective at removing nutrients as well as organic material
Heavy metal contamination in stormwater	• Phase in a prohibition on the use of unpainted zinc roofs • Reduce vehicle use through good urban design and provision of public transport, biking trails and pedestrian friendly routes • Require copper-free brake linings for vehicles and more chemical-friendly tyres
Figure 17.6 Best practice considerations for catchment activities	

Tairua Harbour and Catchment Management Plan

Sedimentation of Tairua Harbour, on the east coast of the Coromandel Peninsula, has long been a concern for the local community and management agencies. A report by the Hauraki Catchment Board in 1977 stated that it seemed 'fairly certain that the tidal compartment is under attack' from the greatly increased sediment flow that had occurred over the previous 100 years. It continued, 'there is an urgent need for a comprehensive catchment scheme aimed ... at a plan of land use which will conserve native forest, and will create an appropriate compromise between 'production' and 'protection' for the purpose of controlling the entry of sediment into the river system'.⁵⁷

The Waikato Regional Council worked closely with the local community and stakeholders during the late 2000s to identify issues related to the health and well-being of the Tairua catchment and harbour and the preservation of environmental and community values. It was agreed that an integrated Tairua Harbour and Catchment Management Plan was needed to provide an assessment of current pressures and issues in the Tairua Harbour and catchment and to provide a practical strategy to alleviate these.⁵⁸ In addition, relevant matters identified in the Coromandel Peninsula Blueprint (a regional strategy for managing growth and development) would be addressed within the Plan. More detailed site-specific plans are to be developed with landowners and land managers relating to specific works and issues on private properties.

During the drafting of the Plan, consultation was undertaken with a range of stakeholder groups, agencies and landowners in an effort to ensure the issues, outcomes and works areas focus on what is needed by the community. Key current and potential issues for the Tairua Harbour and catchment identified by the Plan are:⁵⁹

- Declining aquatic and terrestrial habitat
- Evidence of a general decline in water quality, particularly in peak summer period
- The perceived and actual impact of forestry activities
- Potential for pine to pasture land conversion
- Mangrove expansion impacting on recreational use and other significant habitats
- Salt water paspalum expansion and impact on habitat and hydrology
- Impact of plant and animal pests
- Sedimentation rates in the harbour and surrounding streams
- Decline in fish numbers and species diversity in the harbour

Figure 17.8 provides an example of the way that issues are identified, methods to respond to these and the expected outcomes.



The Waikato Regional Council is preparing a harbour and catchment plan for Tairua to address sedimentation and decline in habitats within the harbour (Raewyn Peart)

Ōkahu Catchment Ecological Restoration Plan

The Ōkahu Bay catchment lies within the Tamaki Estuary of the Auckland Harbour. It is a small embayment that is now home to a diverse group of communities and is an important tourism destination within Auckland City. Historically it was the tribal base or papakainga of Ngāti Whātua Ōrākei,⁶¹ and remains an important location for the hapū.

More recently, Ōkahu Bay has suffered from a number of environmental impacts, including:⁶²

- Severely degraded ecological values through human colonisation and past grazing practices
- Highly modified urban ecology following lengthy human occupation
- Contamination from being the primary receiving environment for stormwater runoff from the wider catchment
- Halting of the natural coastal processes interacting between land and sea from the construction of Tāmaki Drive, including frequent flooding episodes
- Impacts of tidal influences on groundwater and resultant impacts on groundwater levels and Ōkahu Bay urupā and burials

In an effort to manage some of these impacts, Ngāti Whātua Ōrākei prepared the 'Ōkahu Bay Ecological Restoration Plan', which in addition implements the 'Whenua Rangatira Reserves Management Plan' and the 'Ngāti Whātua Ōrākei Heritage and Resource Management Kaupapa, Strategy and Policy 2010–2011'.⁶³

Visionary Statement

'Waters fit to swim in at all times, with thriving marine eco-systems that provide sustainable kaimoana resources to a Ngāti Whātua Ōrākei community who have strong daily presence in and on the bay as users and kaitiaki.'

The Plan has been designed in a highly practical manner to promote, develop and enhance Ōkahu Bay as the public face or gateway to the whenua rangatira while respecting its existing cultural and spiritual value to the tangata whenua and enhancing its relationship with the Waitematā.⁶⁴ Priorities include brokering strategic relationships with agencies, expanding the work of Ōkahu Rākau Bush Care and extending Ko Te Pūkākī (ecological restoration project) throughout the catchment.

Both quantitative measures (testing the effect of stress associated with anthropogenic activities on ecology) and qualitative research (where the Ngāti Whātua Ōrākei and the Ōrākei Community define ecological health and social indicators) are being used to monitor the outcomes from the Plan.⁶⁵ Their work has included the first hapū-based mussel reef restoration programme in New Zealand.



Ōkahu Bay is a popular recreational destination for a wide range of activities (Raewyn Peart)

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Conclusion

New Zealand is blessed with a very large, varied and productive marine environment. With its estuaries, fiords, reefs, soft sediments, seamounts and hydrothermal vents, it is home to a huge variety of marine life and significant mineral and petrochemical resources. The marine area is of enormous value to New Zealanders in terms of economic, social and cultural values. It is of particular importance to Māori who have a long and particularly close historical association with the sea and harvesting its bounty. It is therefore critical that we manage it well.

Decisions affecting the country's marine environment are made within a complex and overlapping legislative framework which has developed over the last 50 years. The most recent addition is the EEZ Act, passed into law in 2012, which applies an environmental management framework to a range of deep-sea activities. This Guide has sought to demystify this framework by clearly describing how decisions are made and by whom. It has also identified how members of the public and other parties can become involved in the various decision-making processes.

Many industries are dependent on accessing the marine environment and its resources and these have been described in detail in Part 2 of the Guide. These activities can potentially cause adverse effects on the marine environment, but in many cases these can be addressed through the application of best practice approaches. The Guide identifies a number of ways in which this can be achieved.

With the implementation of best practice in respect of marine and catchment activities, supported by proactive marine spatial planning and a comprehensive network of marine protected areas, New Zealanders should be able to continue to enjoy their oceans for generations to come.



Blue maomao, Poor Knights Islands, Northland (Daniel Sharp)

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New Zealand has jurisdiction over a very large, varied and productive marine environment. It is of enormous importance to New Zealanders in terms of economic, social and cultural values. With increasing interest in developing marine resources it is critical that we manage this marine area well.

Sustainable Seas demystifies a complex and overlapping legislative framework by clearly describing how decisions affecting the marine environment are made and by whom. It identifies how users of the marine environment, members of the public and others can participate constructively and effectively in these decision-making processes. Many activities can potentially cause adverse effects on the marine environment, but these can often be addressed through the application of best practice approaches. By bringing together environmental and development interests in oceans management this publication identifies a number of ways in which this can be achieved.

Sustainable Seas identifies the potential benefits of marine spatial planning and marine protection and describes their application in practice. With the implementation of such approaches, coupled with best practice applied to marine and catchment activities, New Zealanders should be able to enjoy their oceans for generations to come.

Sustainable Seas is an invaluable resource for all those who have an interest in New Zealand's marine environment including members of the fishing, aquaculture, shipping and mining sectors, tourism operators, marine developers, recreational users, conservationists, catchment and marine managers, scientists and students.

